

Unified Communications Release 15 and On- Premises Update CST

November 9, 2023

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Edge

04 CUBE

Edge

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Hardware Endpoints

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CST On-Premises Update: Presenters and Q&A Mgrs



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Introduction

Unified Communications Release 15 and On-
Premises Update CST

Sudhir Parsuram – Product Manager

Connected calling



Powering all workers



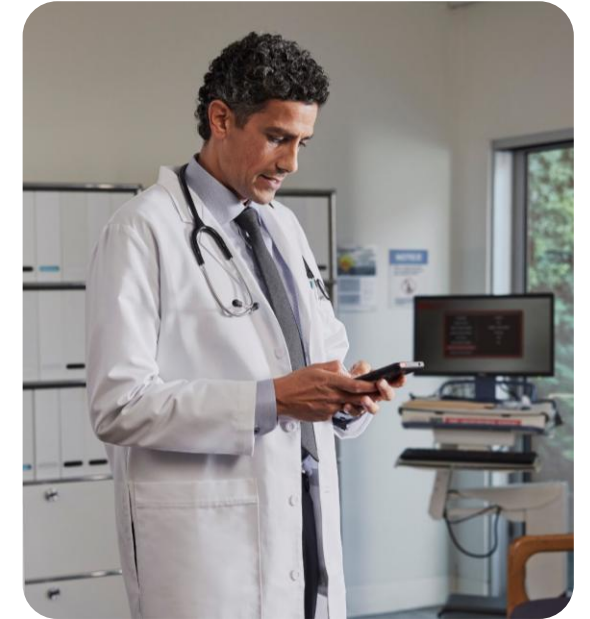
Knowledge workers



Frontline workers



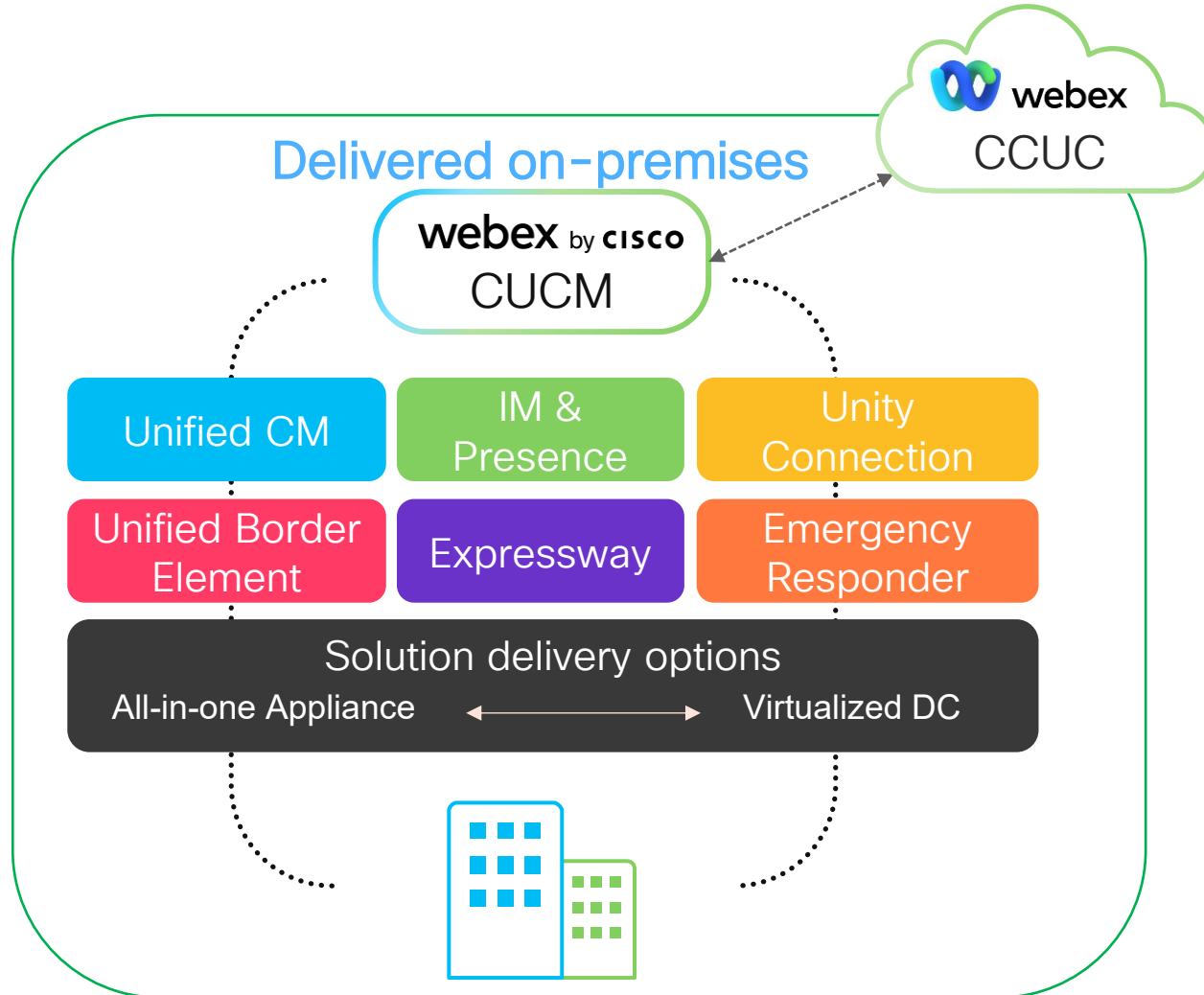
Support workers



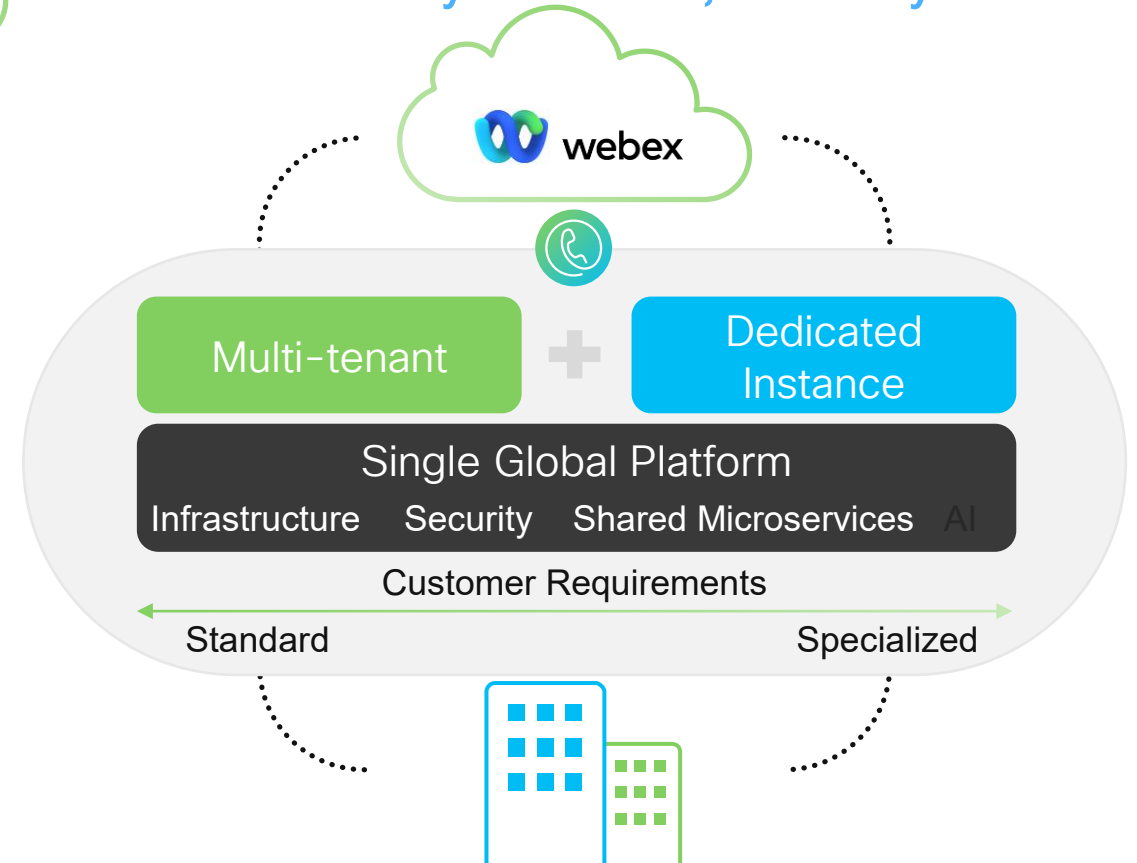
Regulated workers

This powerful experience delivered through flexible Cisco Webex Calling suite

The most complete and flexible enterprise-grade calling experience is now part of the Webex Suite



Delivered fully on Cloud, or as Hybrid



On-Premises Calling Portfolio Strategy

- Help our customers transform to unified and integrated Webex experience across Webex Suite, Webex Calling, Webex App and Webex devices, with flexible journey options
- Maintain highest product quality and supportability of the overall UC solution across various deployment models
- Deliver best in class security, certification and compliance needs ensuring business continuity for customers
- Enhance compatibility with evolving ecosystem to provide flexible deployment choices helping customers modernize

CUCM Solution Updates – Last 24+ months

Keeping pace with Innovation, Security and Compliance

	Simplified Administration	Security / Compliance	Better User Experience	Connected to Cloud
14	- Simple Phone refresh - Higher cluster scale - Fresh Install w/ Data import	- E911 compliance - Simple Phone Security (oAuth) and revocation - Simple certificate mgmt.	- MRA HA – Expressway & LTE/Wifi failover - Jabber zero downtime - Headset based EM login	- Cloud-Connected UC – Analytics - IM&P Open Federation - Unified Messaging w/ Google and O365
14 SU1	- Serviceability for Jabber/ MRA, Persistent chat mgmt - TFTP large file download - BAT support for VG 420	- E911 for remote worker - PLR Support for licensing - SHA-2 for SCCP analog gw and conf bridge	- Wx App Self provisioning - IPv6 over MRA - Opus transcoding	CCUC – Jabber App Migration, Self Enable, Analytics, Certificate Management. Operations Dashboard, WebRTMT
14 SU2	- Expressway Smart Licensing (w/ PLR/SLR) - Cluster s/w location for simple upgrades	- National suicide prevention Hotline (988) - Redsky E911 loc service - EST for Cert enrollment - Authenticated SMTP	- Borderless CTI for MRA - Accessibility enhancements	CCUC – Borderless CTI, Troubleshooting, Touchless provisioning of Webex App, Migrations,
14 SU3	- SRST and Webex Calling SGW colocation. - CUC Scale > 100K	- DTMF interworking for Secure Media MTP - HA for oAuth Refresh Token renewal - CRL enhancement - ExpWay – ECDSA - ExpWay – Select >2500 - CUBE – Secure media forking for cloud speech services	- iOS Local Push Notification - Presence, DND sync w/ Wx App - VDI Support for Webex app - Webex Calling Dedicated Instance – Remote Survivability Node	- CCUC – Presence Synchronization, Azure AD support for UCM/CUC, DND Sync - MS Teams integration – Call History, Presence, Voicemail
Quality – Automation, Library/Platform updates Cisco Product Security Baseline, Federal Certification				

CUCM 15 & Roadmap

User Experience

- Improved Webex App experience with persistent CUCM login flow
- *Local Push Notification**
- CUC – Better noise cancellation for VM deposit via Babble Labs integration
- CUBE - NAT Traversal using RTP keepalives
- **Expressway - 88XX and 78XX Series Phone Failover**

Admin Simplicity

- Core Linux Transition
 - Python 3
 - IDS
 - Kernel crypto
 - FIPS Toolkit
 - 64-bit Support
- Support for VG410
- *PUSH Notification Service enhancements for CUCM**
- CER – New device support for Switch Port tracking
- **CER – IPv6 Subnet Tracking**
- **Megacluster Survivability via Higher Node Count.**
- **Higher scale/density (via 64-bit-app)**

Security & Compliance

- Certificate based Multi Factor Authentication for RTMT
- FIP 140-2 certification
- USGv6 R1 certification*
- V6 Ready Logo*
- **TLS 1.3**
- **DOD-IN APL Certification**
- **Secure Communications Interoperability Protocol (SCIP) support – UCM & CUBE**
- **FIPS 140-3 support**
- **Common Criteria Certification – UCM & Expressway**
- **IPv6 only for UCM & CSR apps**

Bridges to the Cloud

- *Microsoft Teams Integration (via CCUC) – Call History, Voicemail, Bi-directional Presence, Call Forwarding Status and more*

Webex Cloud-Connected UC

- **Directory Services (Azure AD support) – near real-time delta sync based on Webex CI Notification services**

<https://gobeta.webex.com/key/cucm>

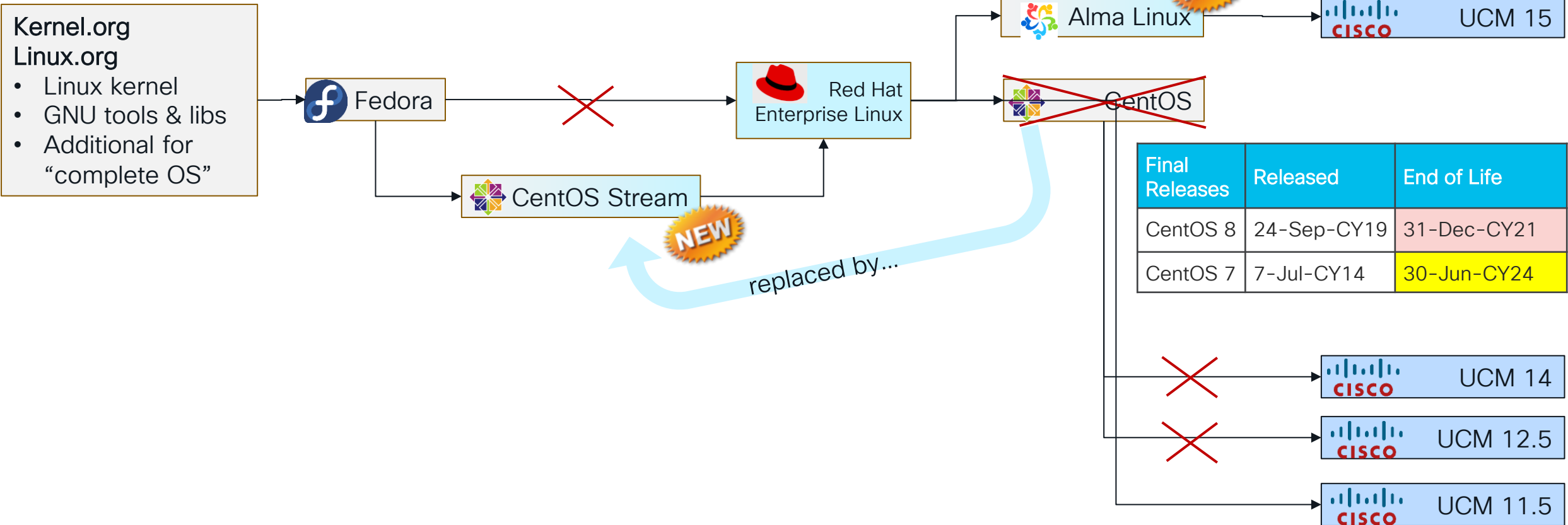
Available beyond initial 15 release

Certification Time Frames

- FIPS 140-2 : UCM 15 Nov'23
- USGv6 R1 : UCM 15 Q1CY24
- DoDIN APL TLS 1.3 : UCM 15SU1 CY24
- Common Criteria : UCM 15SU3 1HCY25

End of Support for CentOS Linux distro

(Industry/Macro Event)

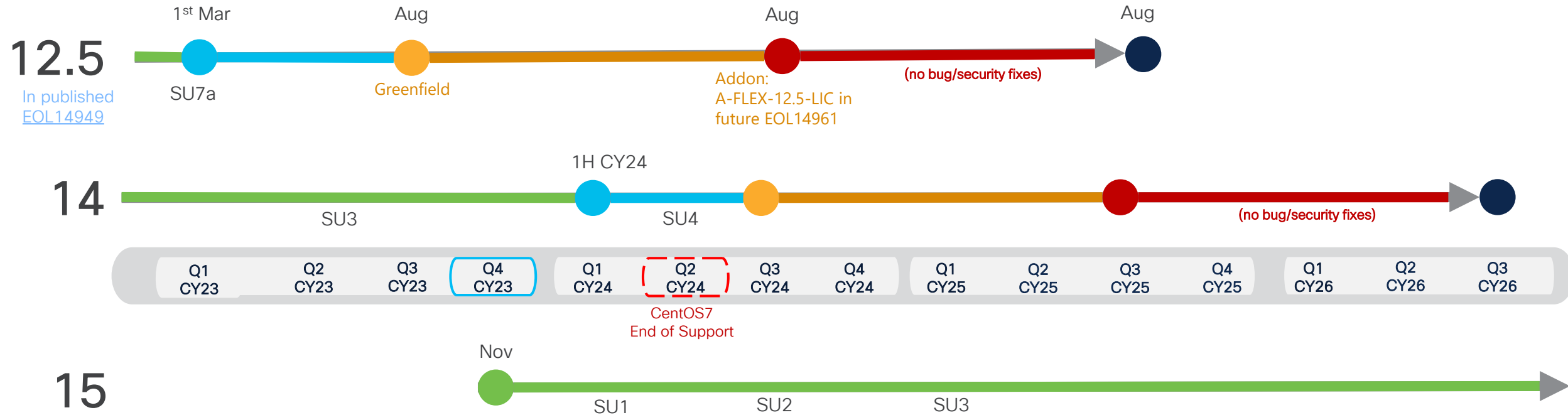


UCM 12.5 release EOled

Plan to Upgrade to UCM 15 or Migrate to Webex Calling.

- Macro industry event: End of Support for CentOS Linux distribution. **CentOS7 goes end of support June CY24.**
- Cisco is developing **UCM release 15** based on alternative Linux, for long term customer supportability.
- End of support for CentOS7 forces **End of Life of UCM releases based on CentOS7.**
- **12.5 release EOL** bulletin published on **1-March-CY23.**
- For best long-term solution, customers should either...
 - Upgrade to UCM release 15
 - Migrate to Webex Calling

Release FCS/EOL – CUCM 15, 14, 12.5 (Cisco dates **subject to change**)



Products that this timing applies to:

- UCM – Unified Communications Manager
- SME – Session Management Edition
- IM&P – Instant Messaging & Presence service
- CUC – Unity Connection
- CER – Cisco® Emergency Responder
- PCD – Prime Collaboration Deployment
- Elements of Webex Calling dedicated instances, UCMC-G, Webex Contact Center Enterprise

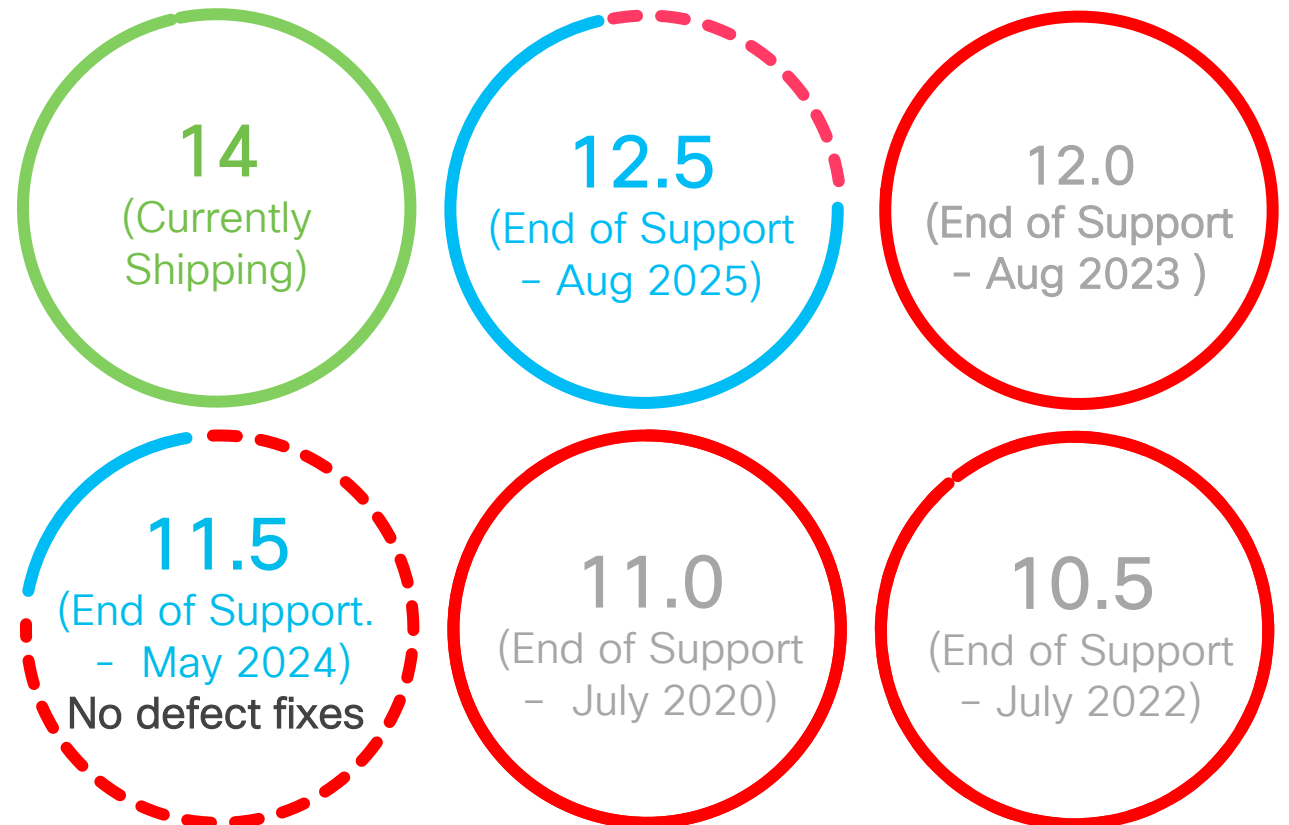
Supported UCM Versions

Available releases for

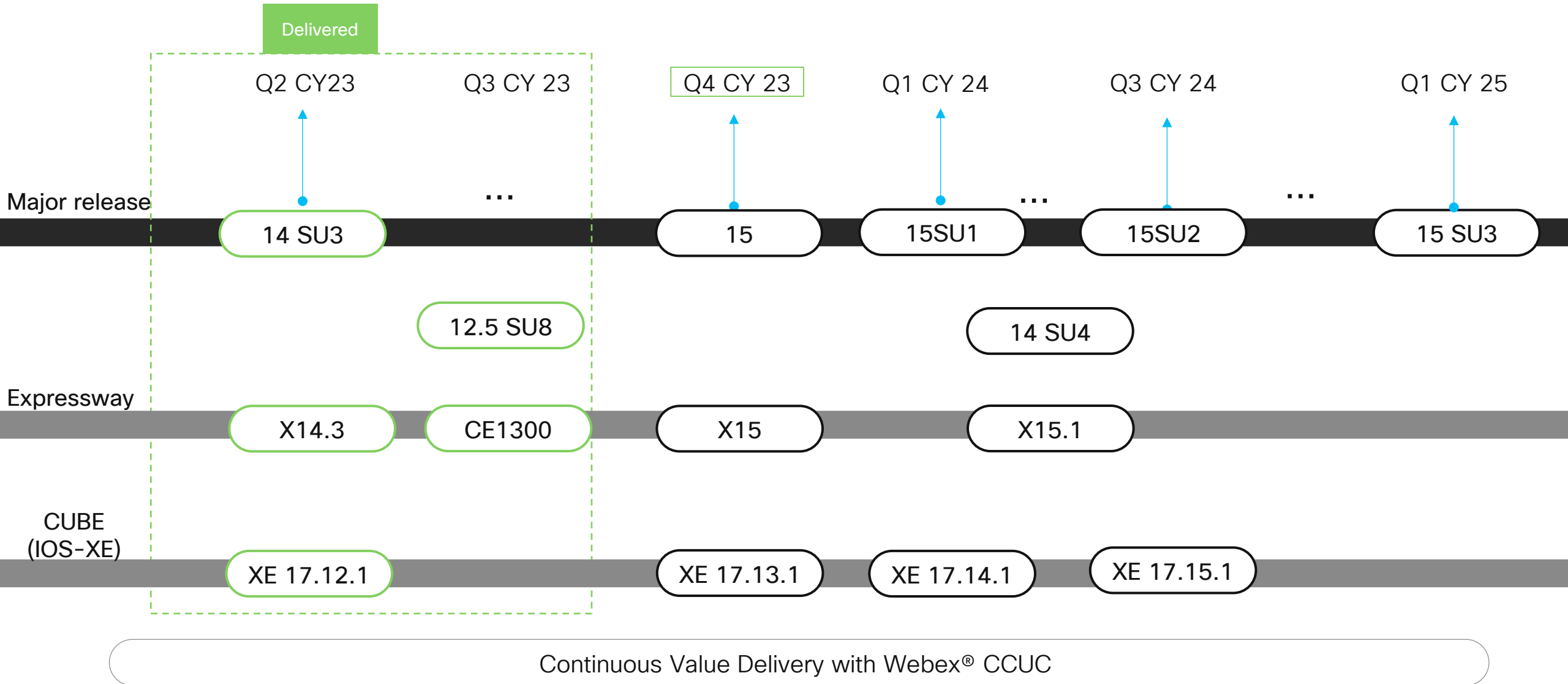
- UCM – Unified Communications Manager

To which these applications and cloud services also align:

- SME – Session Management Edition
- IM&P – Instant Messaging & Presence service
- CUC – Unity Connection
- CER – Cisco® Emergency Responder
- PCD – Prime Collaboration Deployment
- And parts of
 - UCCX – Contact Center Express
 - P/UCCE – Packaged/Unified Contact Center Enterprise
 - Webex Calling dedicated instances
 - Webex Contact Center Enterprise



Cisco Unified Communications Manager Release Timeline



Plan as of Nov 2023 - Subject to change without notice

What should I do?

For best long-term solution, customers should either:

1. Migrate to Webex Calling (Multi Tenant) ... last migration
2. Migrate to Webex Calling (dedicated instance) ... last migration
3. Upgrade to UCM 15

For more information / details...

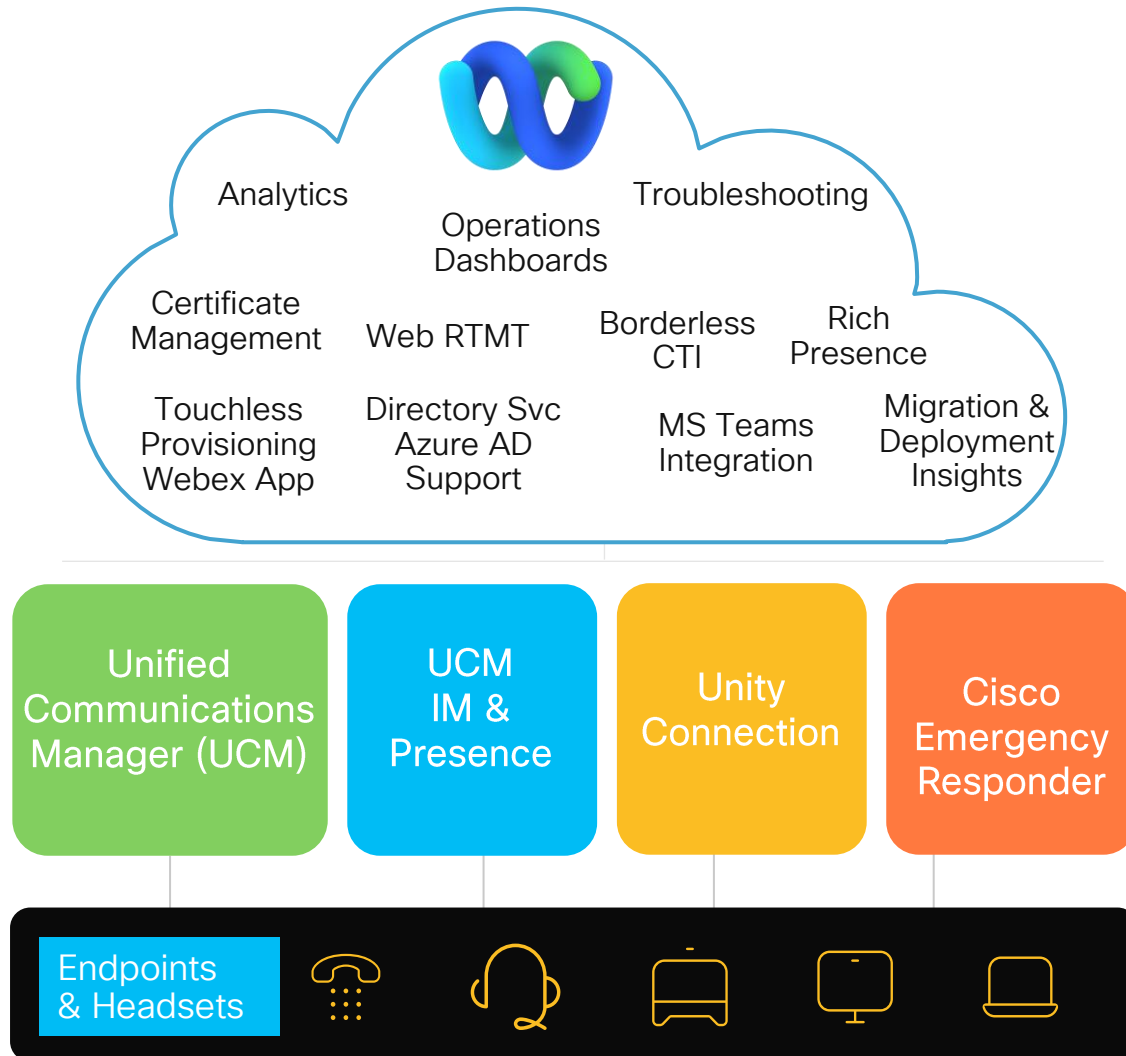
What...	Where...
PM alias	ask-ucm-release-eol@cisco.com ask-ucm-release-eol-partner@cisco.com
PM Webex space 	Ask-UCM-Release-EOL - Link to space: webex teams://im?space=2fbda8c0-b38e-11ed-b874-abbe62d26526 - Join link: https://eurl.io/#EBoSUJrld
Partner webinar <i>Cisco UCM 15 Update - Get Current or Go Cloud Now</i>	http://cs.co/cvoe (8-Nov-CY23)
UCM Release EOL bulletins	12.5: EOL14949 12.0: EOL13303 11.5: EOL13693 11.0: EOL11446 10.5: EOL12642

CALLING, MESSAGING

Unified Communications Release 15

Release 15 Application Updates

Webex Cloud-Connected UC (CCUC) – Bridge to Cloud



1. *Analytics*
2. *Operational Metrics*
 - a. *Operations Dashboard*
 - b. *Troubleshooting*
 - c. *Certificate Management*
 - d. *Web RTMT*
3. *Migrations & Deployment Insights*
4. *Rich Presence for Webex App*
5. *Borderless CTI*
6. *Touchless Provisioning for Webex App*
7. *CCUC Directory Service - Azure AD Support*
8. *Microsoft Teams Integration*
 - a. *Call History*
 - b. *Presence Sync – Webex <-> Microsoft*
 - c. *Voicemail (CUC)*

Webex Calling for Microsoft Teams Integration

Call Tab

Access dial pad, contact search and speed dials

Dialpad

Access to dialpad to key in numbers

Add Speed dial

Click to add up to 25 speed dial cards

Contact Search

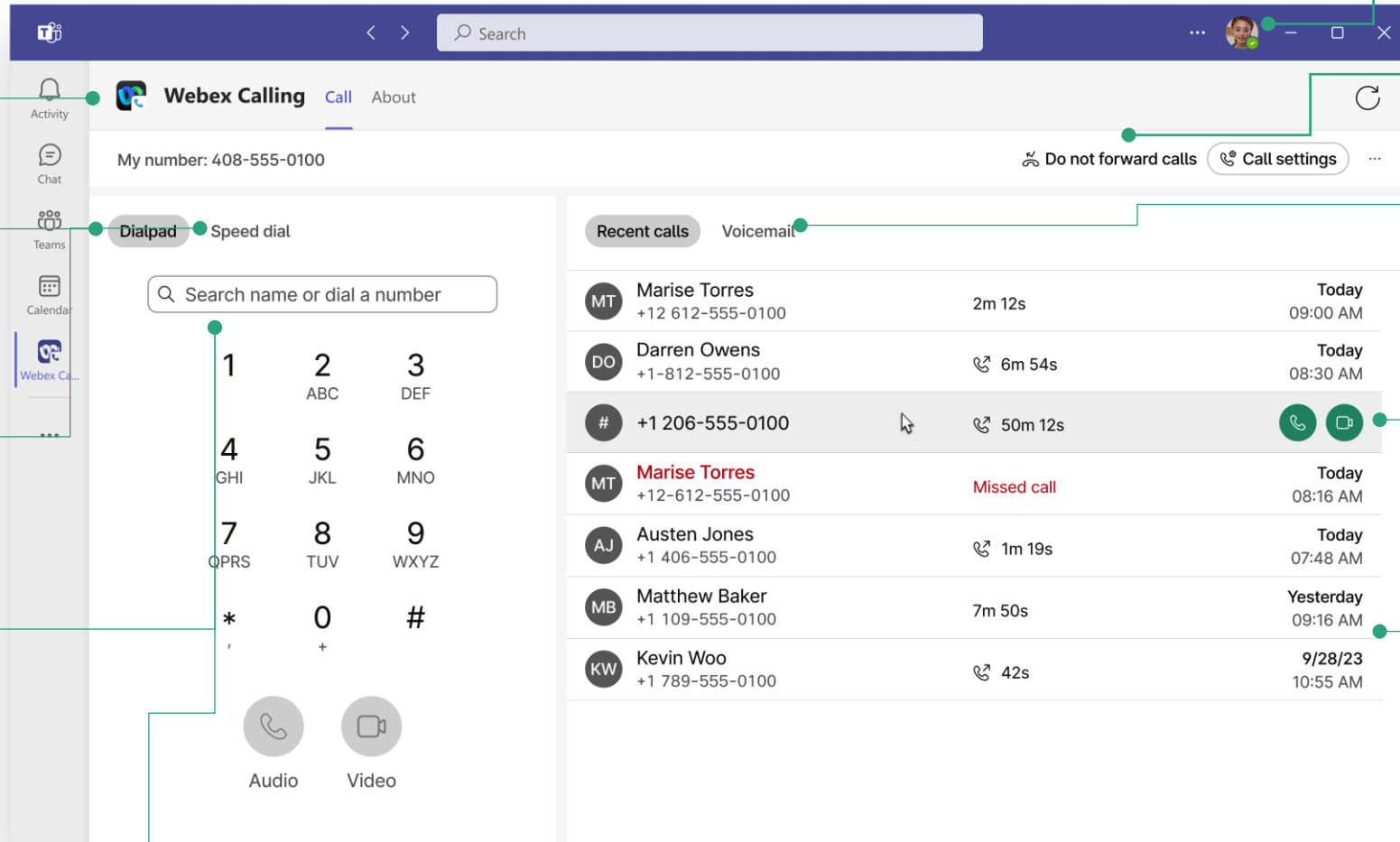
Search for personal and corporate contacts

Azure AD
Contacts

Outlook 365
Contacts

Contact Resolution

Resolves the incoming call identifier to Azure AD



Call/Search Input

Enter a number, SIP address or type a name to search for Azure AD/Outlook contact

Bi-directional Presence

Display your on UCM call presence to your contacts and see their call presence too

Call Forwarding

See your call forwarding status at a single glance

Voicemail

Access and play back the voicemail

Call

Select number and call contact

Call History

See history of incoming/outgoing/missed calls

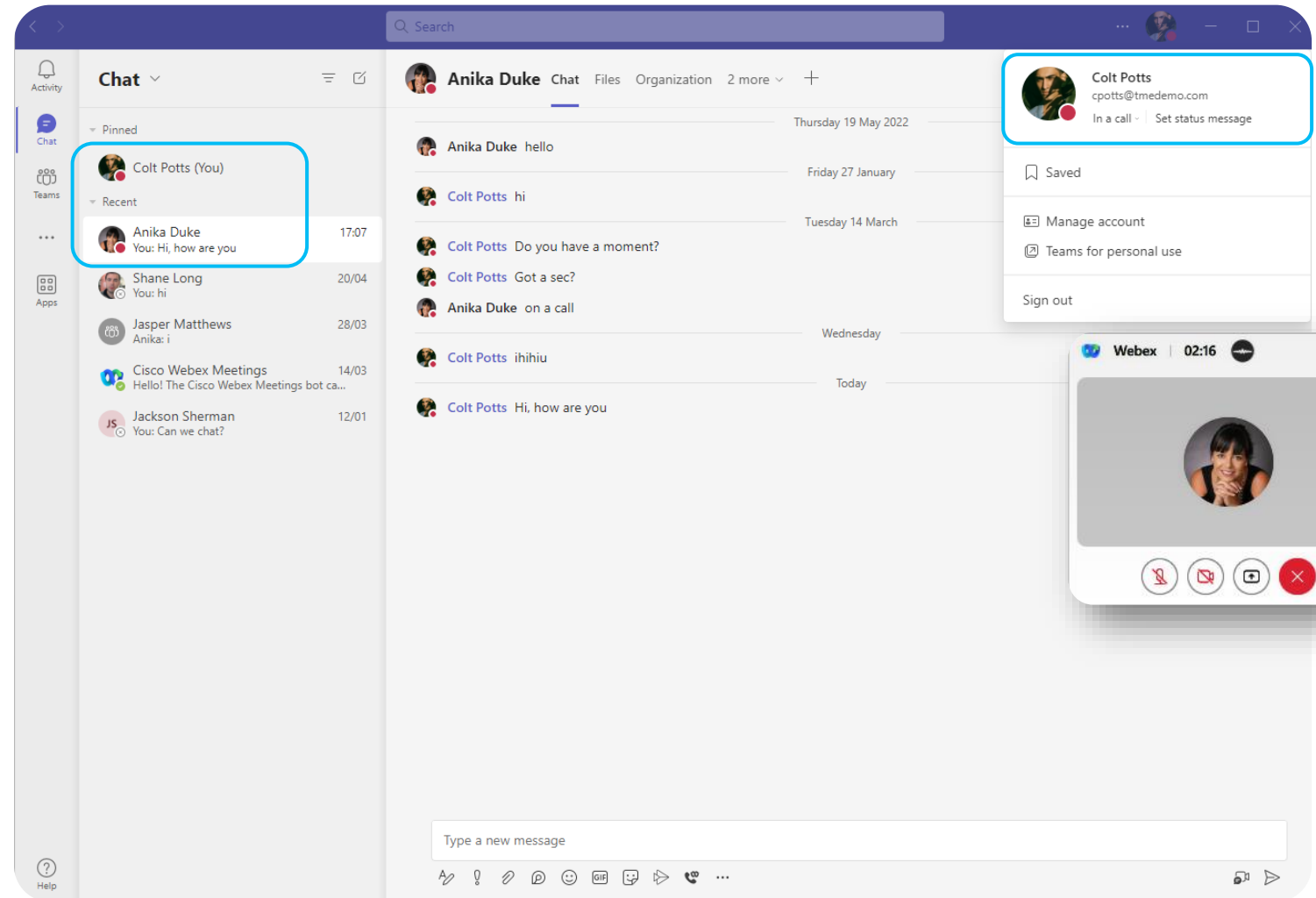
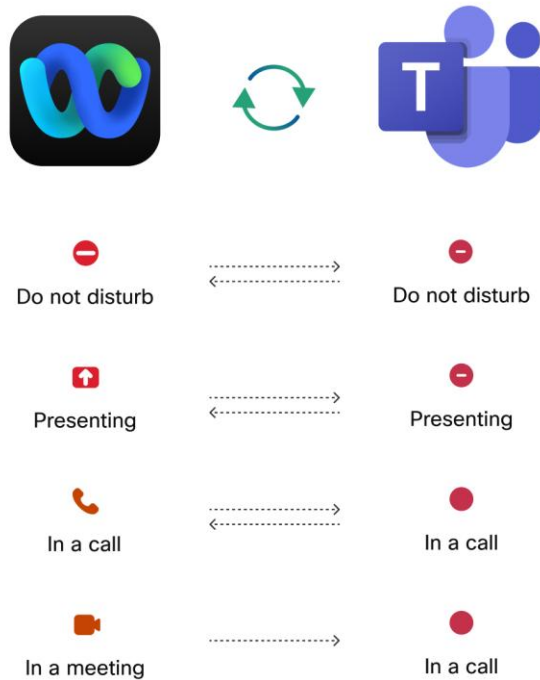
Webex Cloud-Connected UC (CCUC) required

Supported Releases

- UCM – 15, 12.5 SU7+ , 14 SU3
- CUC – 15, 12.5, 14 SU3

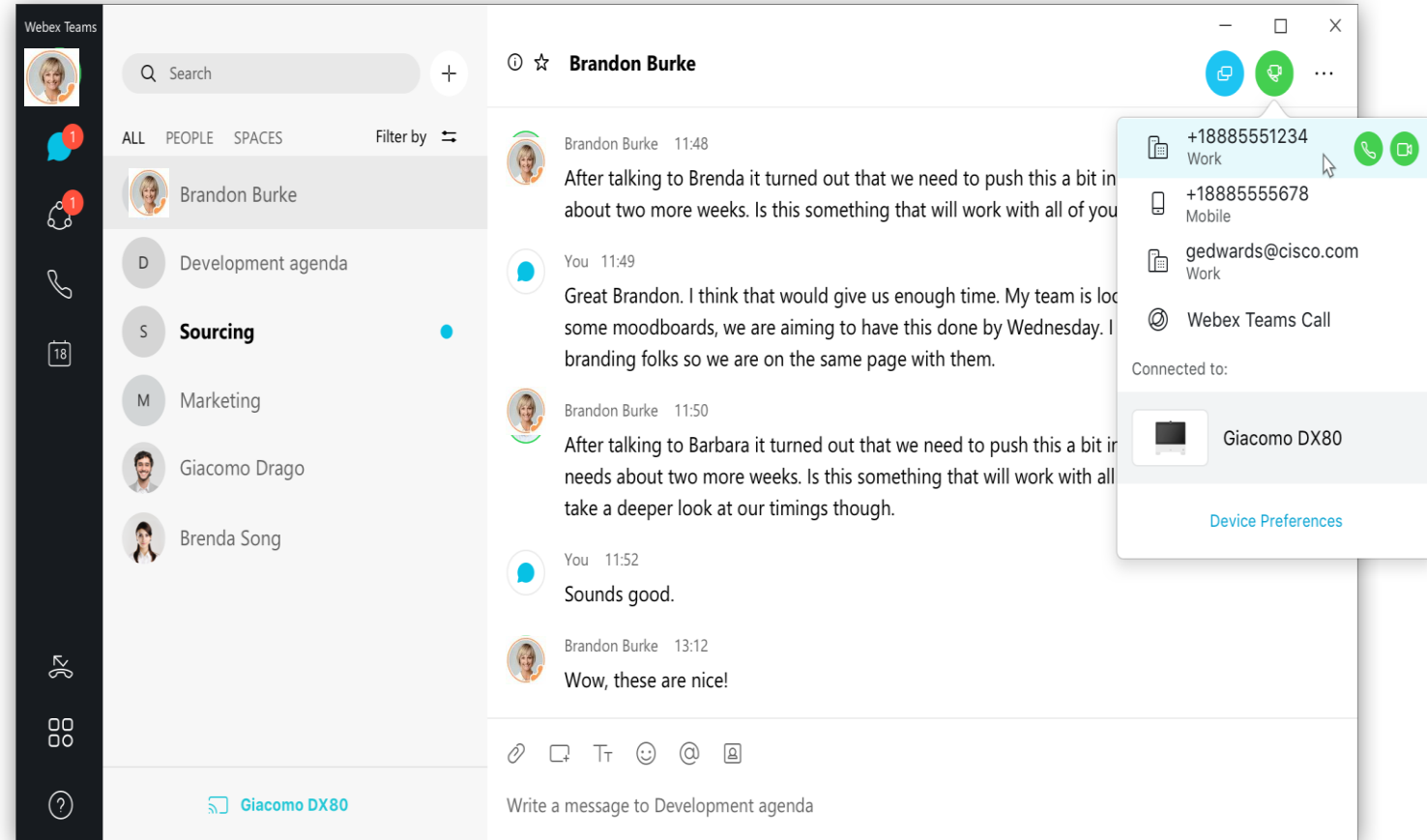
MS Teams: Bi-directional Presence Sync

User's presence status on either application (MS Teams / Webex App) will now be kept in sync with the other application for the below



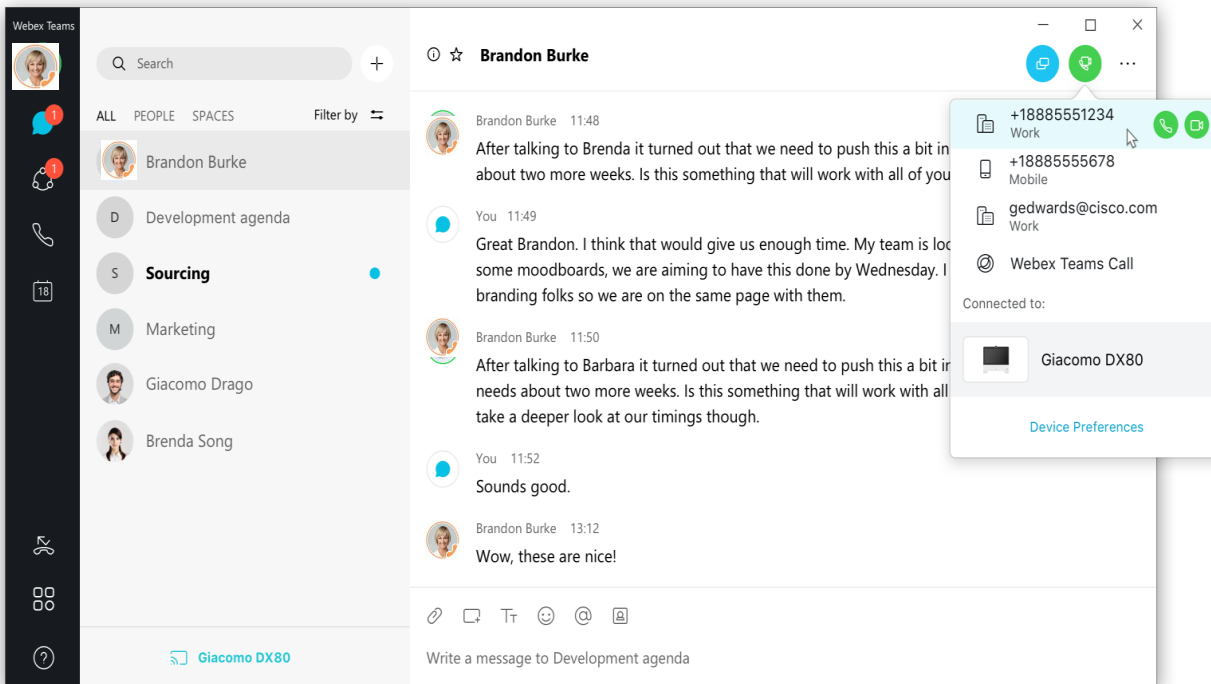
Rich Presence: User's Phone Presence in Webex App

- Integrate User's IP phone off-hook/on-hook status with Webex app presence
- Accurate real-time presence information for contacts
- Enables users to determine best communication mode
- Webex Cloud-Connected UC (CCUC) required
- Supported Releases :
 - Cisco UCM: 15, 12.5 SU7+, 14SU3+
 - Webex app: all releases



Presence Sync

DND from Webex app to IP Phone and vice-versa



- Sync User's IP phone DND status with Webex app and vice-versa
- DND status is cleared on all devices when DND expires
- Supported on CTI mode and Shared lines
- Webex Cloud-Connected UC (CCUC) required
- Supported Releases:
 - Cisco UCM: 15, 12.5 SU7+, 14SU3+
 - Webex app: all releases

Borderless CTI

End user experience

Benefits

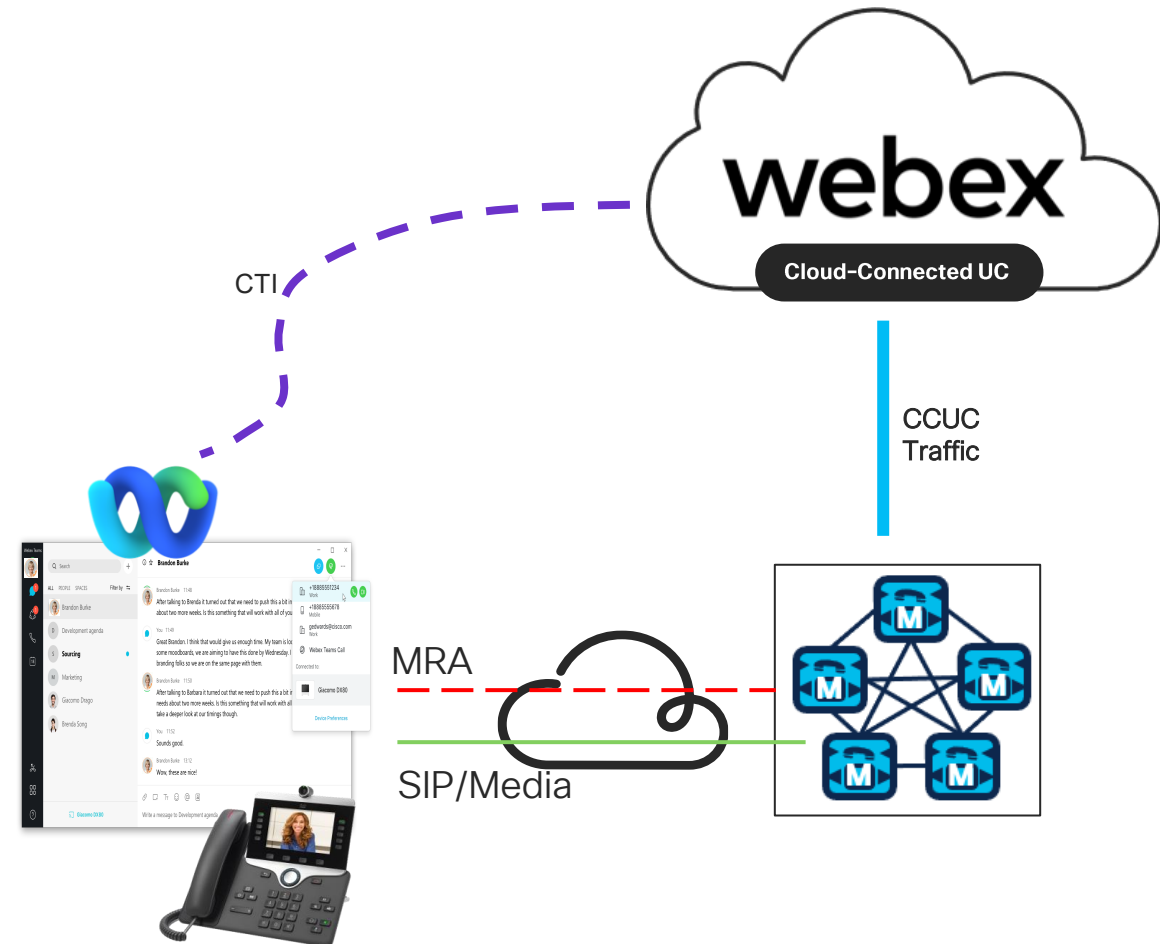
- Webex App CTI application functionalities available remotely, no VPN connection required

Key Features

- Desk phone control of 88xx/7xx, CE, DX from Webex App remotely
- CTI applications such as receive incoming enterprise calls, making/disconnect/Hold/Retrieve calls among other applications.
- Webex Cloud-Connected UC (CCUC) required

Supported Releases:

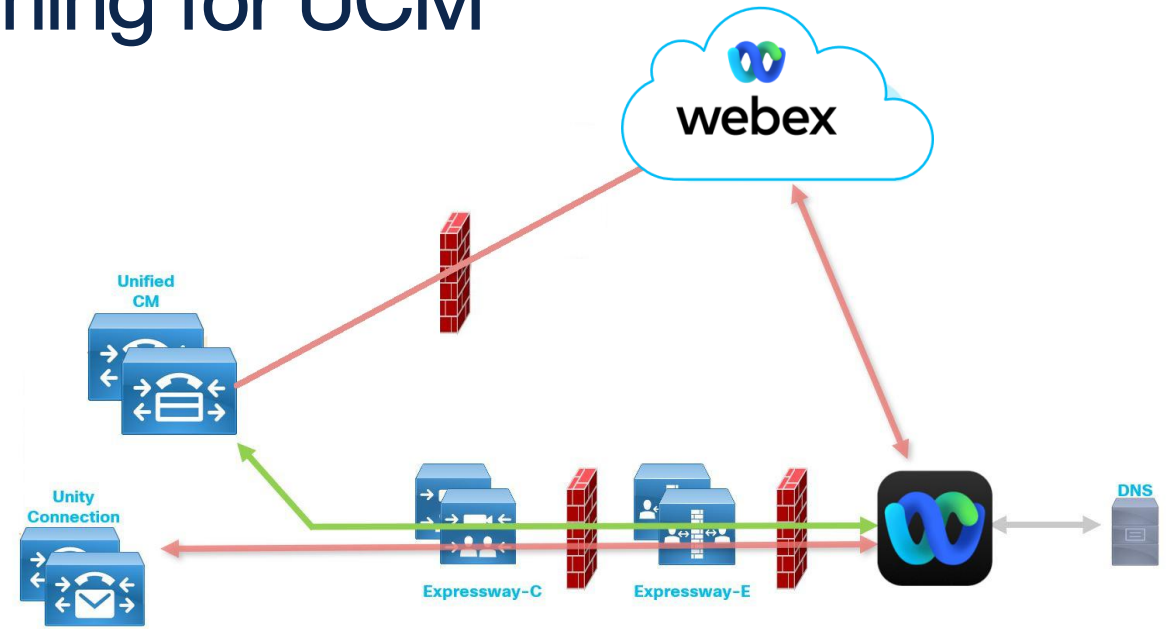
- Cisco UCM: 15, 14SU2 onwards
- Webex App Jun 22



Touchless Webex App Provisioning for UCM

User can self-provision the Webex® App in UCM

- Avoids over provisioning of devices for Webex App
- Minimizes impact on cluster scale & licensing
- Control Hub based policy control
- Enable Jabber® to Webex App migration
- Webex Cloud-Connected UC (CCUC) required
- No admin intervention required
- Supported Releases:
 - CUCM 15, 11.5+, 12.5+ and 14+
 - Expressway X14.0.2+
 - Webex app 41.10 and higher



MANAGEMENT - Users - Workspaces - Devices Apps - Account - Organization settings	Unified CM Settings	Allow Unified CM registration without trusted certificate When enabled, the security of the Unified CM connection is not authenticated. We recommend that you only use this setting in a lab environment. ☒ Allow auto-provisioning for webex Turn on for automatic device provisioning when a Unified Communications Manager user signs in with their registered email address. You can also change this through individual user settings. ☒ Allowed device types Allow auto-provisioning Webex app for desktop, mobile, or both. ☒ Desktop and mobile ☐ Desktop only ☐ Mobile only Delete inactive auto-provisioned devices ☒ Don't delete inactive devices ☐ Delete after 30 days
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CCUC Directory Service: Azure AD Integration

Synchronize users from cloud-based directory such as Azure AD into on-premise UC infrastructure

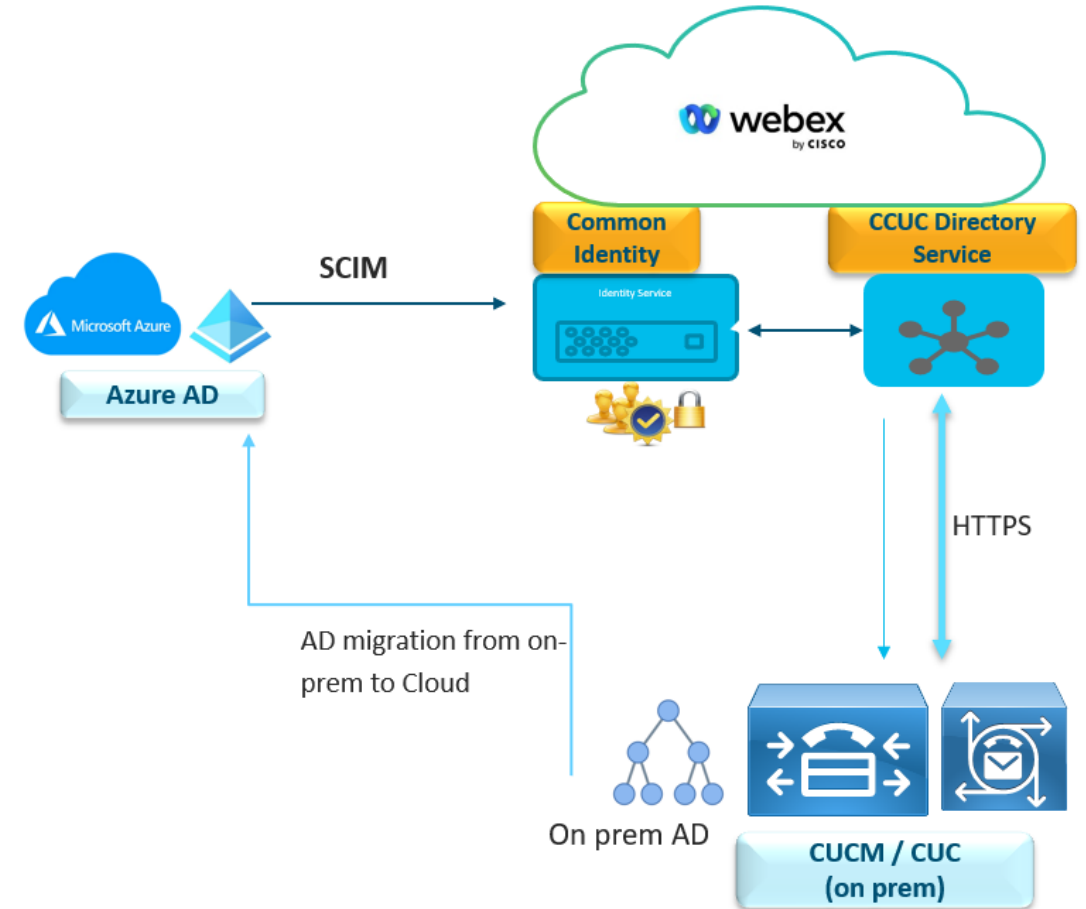
- Webex CI integrates with Azure AD using SCIM to sync the users
- CCUC Directory Service will sync users from Webex CI into CUCM and Unity Connection
- Webex Cloud-Connected UC (CCUC) Required

Benefits

- Ubiquitous Identity and Access management while moving directory to Cloud
- Reduction in TCO
- Enables migration options – incremental to full cloud native

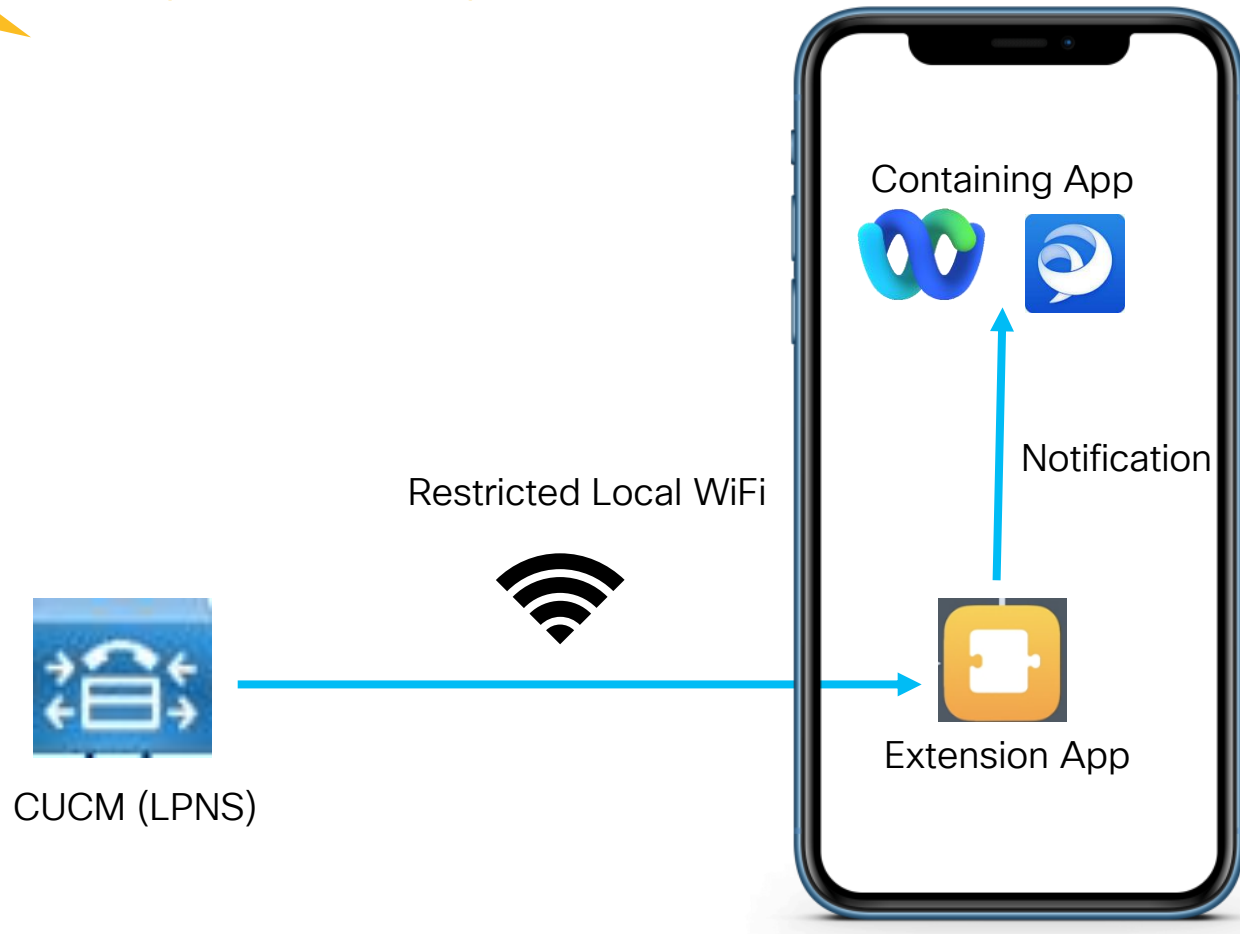
Supported Releases:

- UCM 12.5(1)SU4+ and 14 or later



iOS Local Push Connectivity for calls (LPNS)

★ Industry first and only solution



Benefits

- **Reliable & Secure** way to notify Webex App / Jabber users on iOS devices of incoming VoIP call and messages when iOS devices operate in a WiFi constrained network when there is **no internet connection** and doesn't have access to APNs

Key Features

- To the user, LPNS and APNS notifications will be identical
- Mission critical / Time critical Local notifications get delivered when App is suspended and running in background
- Persistent connectivity to LPNS service

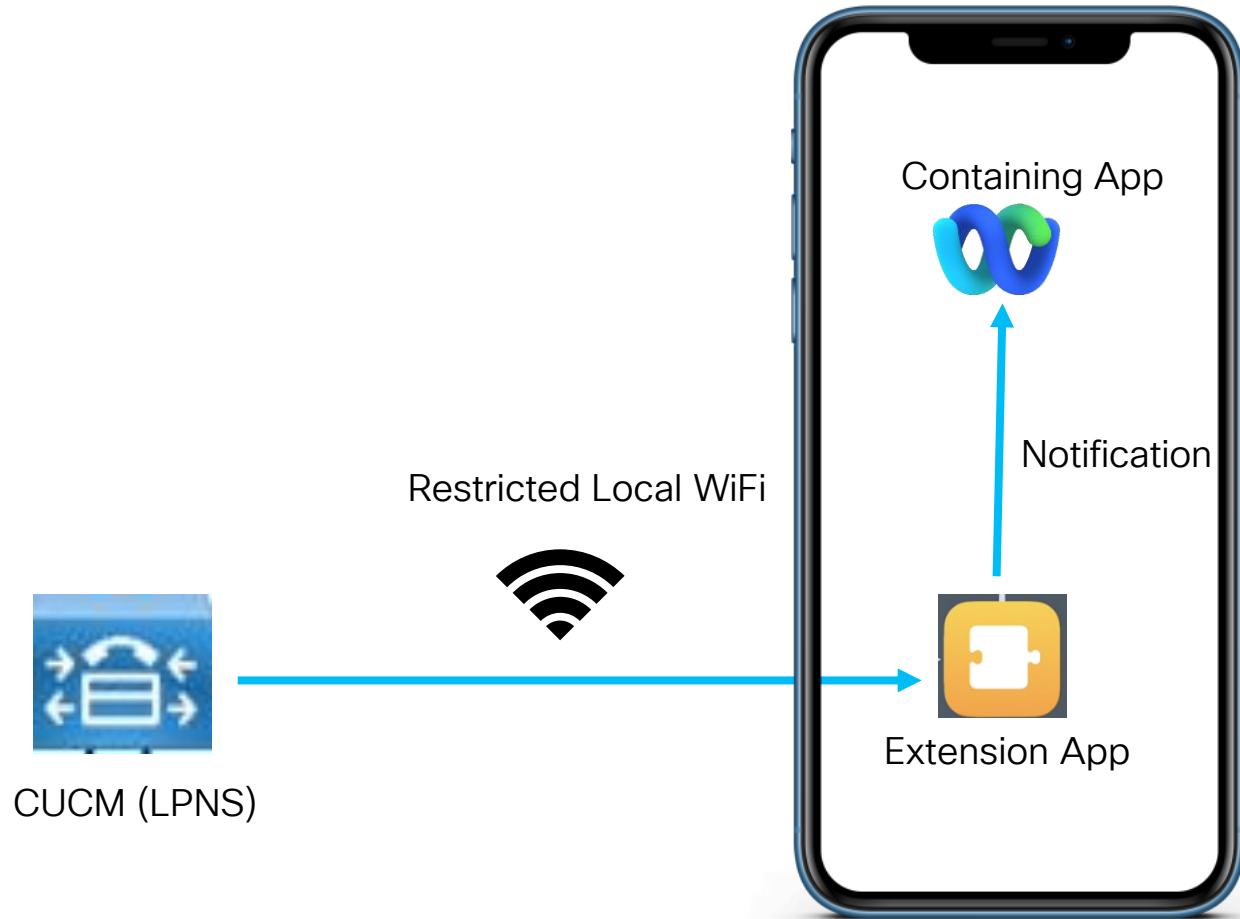
Supported Releases:

- Cisco UCM: 15, 14SU3 onwards
- Webex App 43.6
- Jabber: 14.2

Expected delivery dates are subject to change



iOS Local Push Connectivity for calls (LPNS)



- App Extension can directly communicate directly with your CUCM (LPNS) on a local WiFi network
- Web Socket Protocol between CUCM (LPNS) and App Extension to deliver the notifications
- App must specify WiFi networks where you wish to enable local push connectivity
- When device joins specified WiFi network App Extension is responsible for maintaining network connection with CUCM(LPNS) and receiving the notifications
- App Extension continues to run in background as long as device is associated to the specified Wifi network System stops all extension when devices loses connectivity to specified wifi network



CUC : Supporting 160K users

Feature:

Improvement in Directory Size from 100K to 160K in an HTTPS network

Benefits

- Customers can have a larger network with 160K voice mailboxes.
- All 160K users can perform Directory Search, leave Voice messages, Cross Box Sign-in and Transfer in the network
- Users can send Voice messages to Distribution Lists across the larger network
- Ease of Administration with more efficient topology utilizing larger OVAs

Supported Releases:

- CUC 15, 14 SU3

Smart Receiver as method of transport for Smart Licensing in UCM Calling

Roadmap



Offline



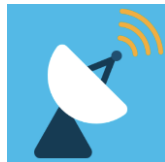
CUCM



Direct Access



Cisco Smart
Software Manager (CSSM)



On-Premises

Benefits

- Smart Call Home (SCH) as method of transport is EoL and replaced by Smart Receiver based transport
- Smart Receiver is HTTPS based

Key Features

- New fresh install will be supported with Smart Receiver only
- Upgraded or Migrated system will be supported with Smart Receiver provided system is able to connect to endpoint URL of Smart Receiver automatically, If the system is not able to connect to Smart Receiver endpoint URL , then admin would need to manually configure and migrate to Smart Receiver based transport
- Pre and Post upgrade checks will be provided

Minimum release support:

- Cisco UCM : 14 SU4, 15 SU2
- CER : 15 SU2
- CUC : 15 SU2

USGV6 R1 Certification for CUCM



As per [Federal Acquisition Regulation \(FAR\) - Case 2005-041](#), IPv6 compliance is mandatory for all US Government (USG) agencies.

To meet this requirement the [National Institute of Standards and Technology](#) (NIST) created the USGv6 certification program.

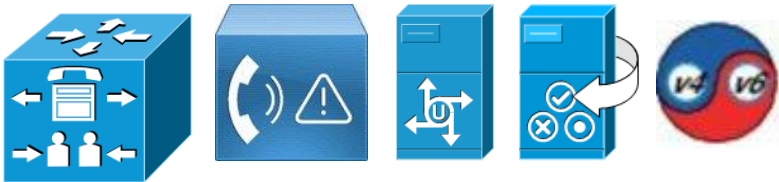
IPv6 Ready Logo Program is a conformance and interoperability testing program intended to increase user confidence by demonstrating that IPv6 is available now and is ready to be used.

UCM 15 will be USGV6 R1 certified ~Q1CY24

IPv6 only support on CUCM

Future

Current



Problem Statement

- IPv4 address exhaustion
- OMB Mandate : 20% by FY 2023 , 50% by FY 2024, 80% by FY 2025

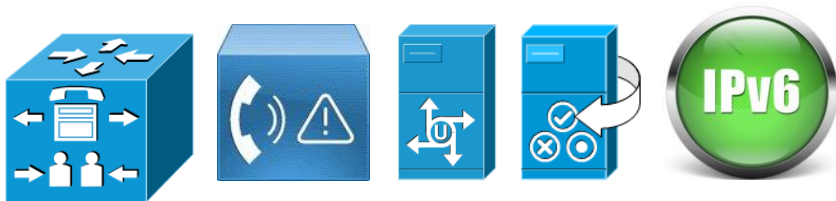
Current Recommendation:

- CUCM, IMP, CER, CUC in Dual Stack
- Endpoints , SIP GW in IPv6 only
- Network Infrastructure: IPv4 LDAP Dir Sync (IPv6 compliant), IPv6 NTP, IPv4 CDR / CMR/ Trace/ Backup server (IPv6 aware) , IPv4 Syslog server, IPv6 DNS, IPv6 DHCP
- The same CUCM / IM&P will be able to handle all phone services for IPv4 only endpoints as well

Future Goal:

- CUCM, IMP, CER, CUC in IPv6 only
- Endpoints , SIP GW in IPv6 only
- Network Infrastructure : IPv6 LDAP Dir Sync (IPv6 compliant), IPv6 CDR / CMR/ Trace/ Backup server (IPv6 aware) , IPv6 Syslog server

Future



Improved Webex App experience with persistent CUCM login flow

- Webex app registered to CUCM
Registered prompts user to login every 60 days to maintain phone service.
- User may cancel login for phone service and still have access to messaging, meetings, and internal calls as the account is still authenticated with IdP on Cisco Webex cloud
- If the user on Webex app is not authenticated by CUCM:
 - Phone service disconnects which results in Missed calls
 - Mixed calling User Experience. Internal calls may work as connectivity to Cisco Webex cloud is maintained, but PSTN calls don't.

Key Feature

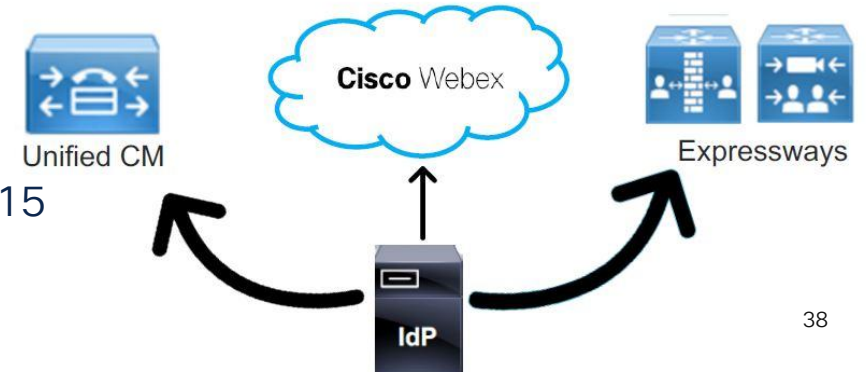
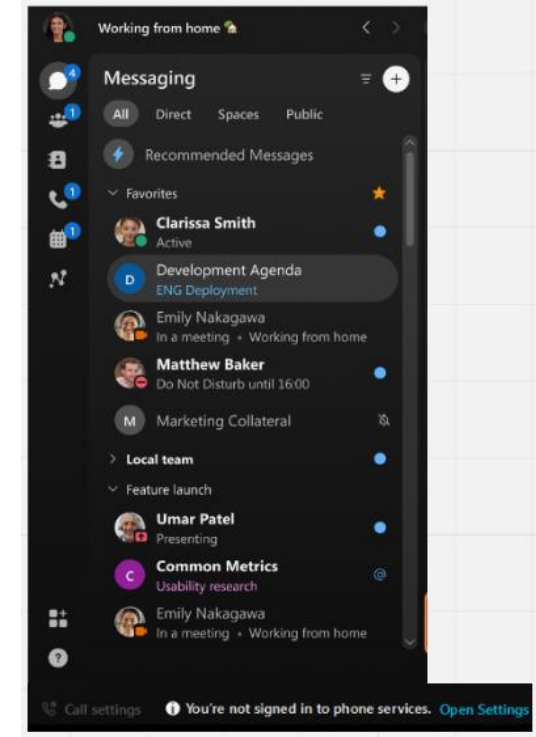
- The refresh token in Webex App is automatically renewed without user intervention so that the user does not get logged out of the phone service.
- Expressway will be able to pass through the auto-renew parameter for refresh token so that the clients will renew the tokens automatically

Benefits

- Improved Webex App user experience resulting in end users not missing calls due to phone service disconnects

Supported Releases:

- UCM 15, Expressway X15
- Webex App 43.8



TLS 1.3

Roadmap

Why TLS 1.3 ?

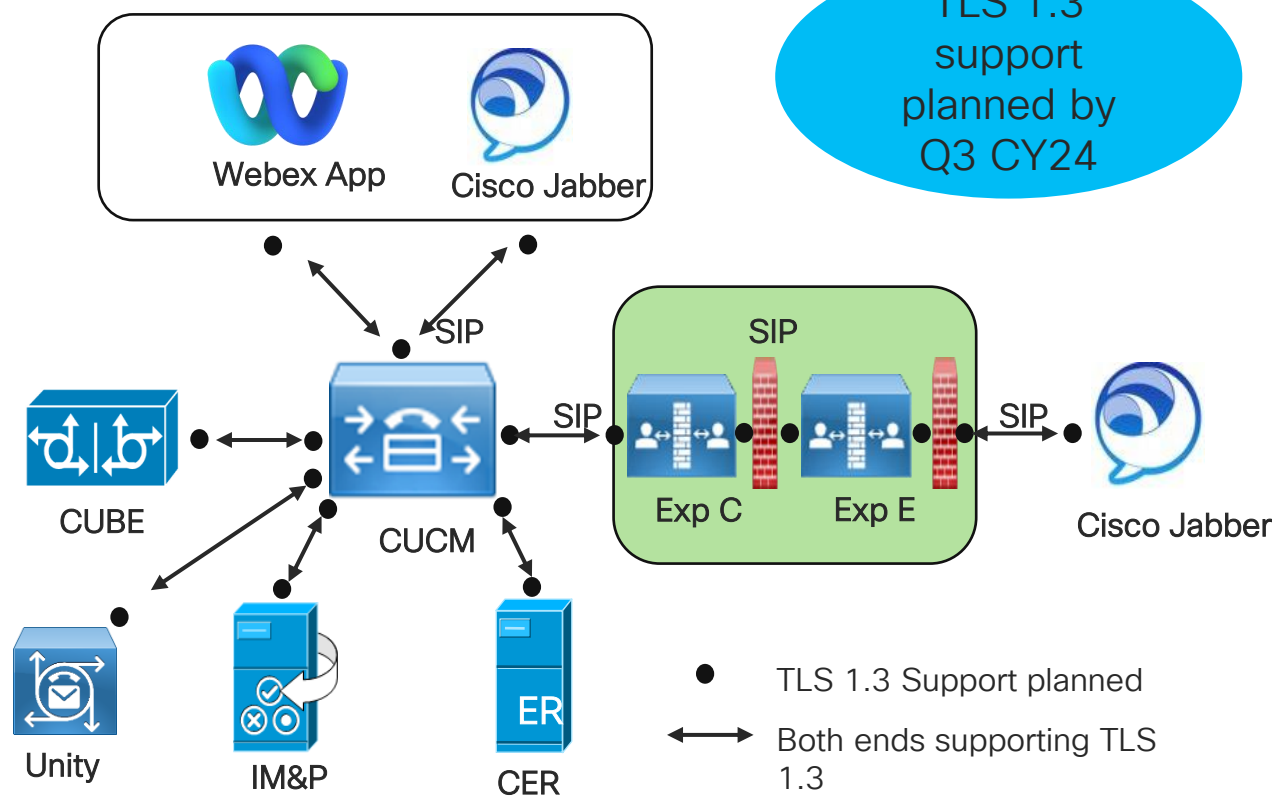
- Improved Security – TLS 1.3 more secured than TLS 1.0, 1.1 and 1.2
- Improves performance – Shorter handshake mechanism
- Simplified Cipher Suites – Use of simple and stronger Cipher suites

Requirements

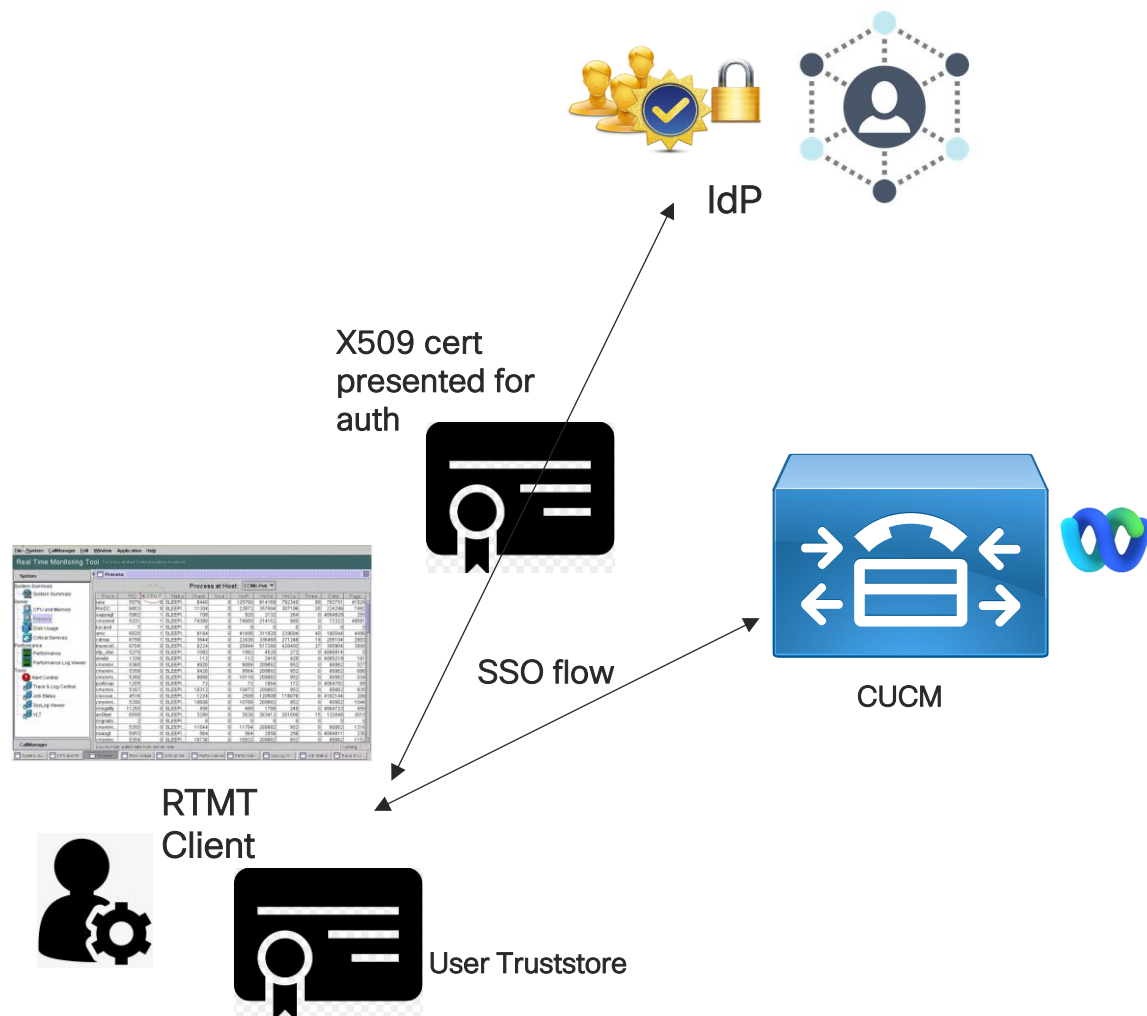
Support TLS 1.3 end-to-end at solution level

Support for both inbound and outbound connections

TLS 1.3 handshake should auto-fallback to supported lesser version if client does not support TLS 1.3



Certificate based Multi Factor Authentication for RTMT



Problem Summary:

UCM supports Certificate based Multifactor Authentication along with SSO from the regular web interface, but this is not available for RTMT.

Benefits

- Phishing resistant cryptographic technique avoiding use of passwords for authentication and access.
- Protects against password breach / compromise.
- Suitable for use in classified systems

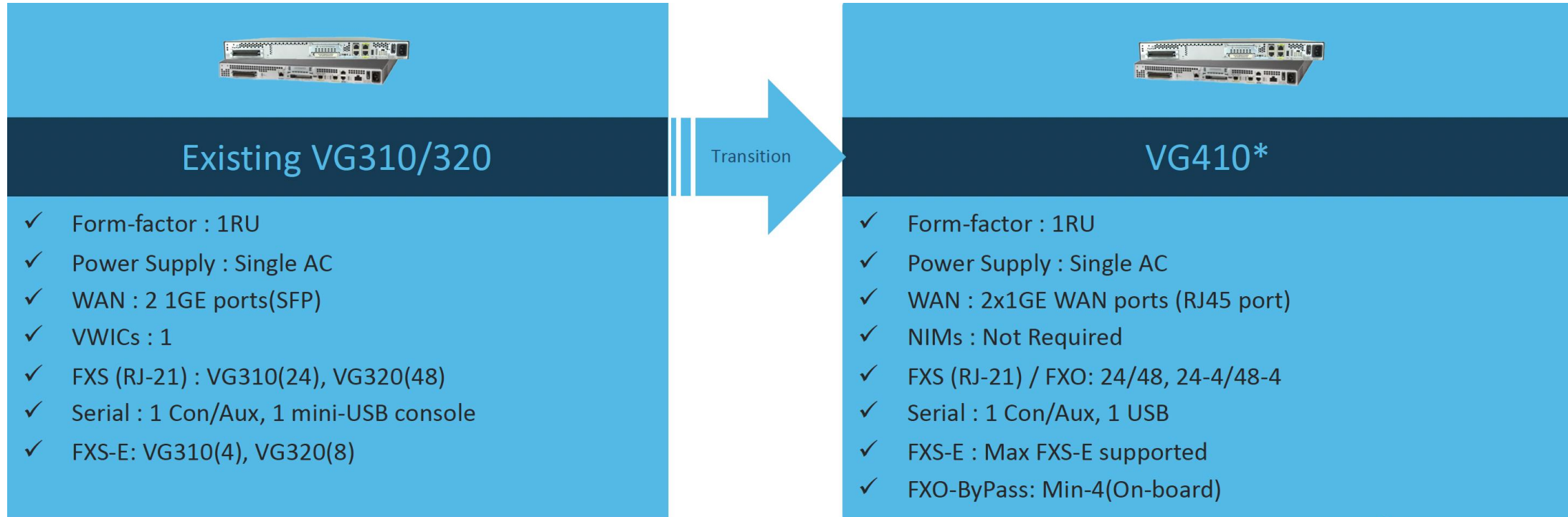
Key Features

- Leverages SSO flows
- User certificates are selected from RTMT Client machine truststore and presented to IdP for authentication
- May be used in conjunction with username / password based auth (MFA)

Supported Releases:

- Cisco UCM 15

CUCM support of VG 410



Supported Releases:

- UCM 15 (adds BAT support)
- UCM 14 SU3 (basic support)

CER: New device support for switch port tracking

Switch Port Tracking:

For the most accurate location tracking, Switch Port based phone tracking can be utilized. Emergency Responder queries Cisco Unified CM for a list of phones registered with the cluster. Emergency Responder then queries the switches (using SNMP) on the network to determine the port used by the phones (using CDP/MAC-Table). Emergency Responder can then send emergency calls to the correct PSAP (Public Safety Answering Point) based on the ERL assigned to the switch.

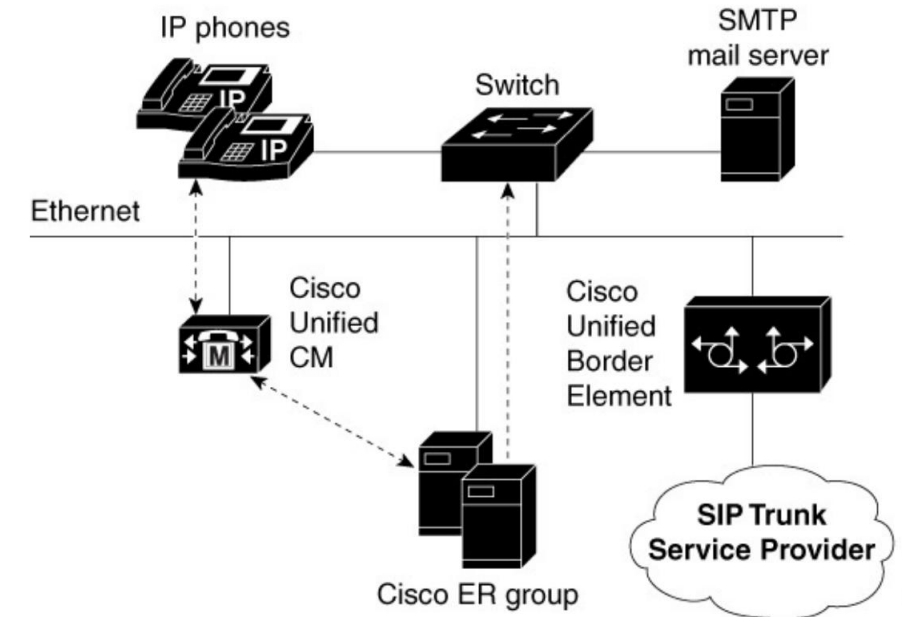
Feature:

CER is constantly adding new switches and routers to the list of supported devices. The following are the most recently added based on customer requests:

- Cisco Integrated Services Routers
 - Cisco C1111-8P
 - Cisco C1111-8PWA
 - Cisco C1111-8PWB
 - Cisco C1111-8PWE
 - Cisco C891F-K9
- Cisco Catalyst Switches
 - WS-C2960L-8PS-LL

CER now supports:

- 326 Cisco Catalyst Switches
- 60 Cisco Industrial Switches
- 53 Cisco Meraki Switches
- 29 Cisco Integrated Services Router



Supported Release: CER 15

For an up-to-date list of supported network devices (and their SNMP OIDs) refer to the CER release notes

CER: IPv6 Subnet based phone/client location tracking

Problem Summary: CER does not support IPv6 subnet based phone/client location tracking making it difficult to manage endpoints in an IPv6 network.

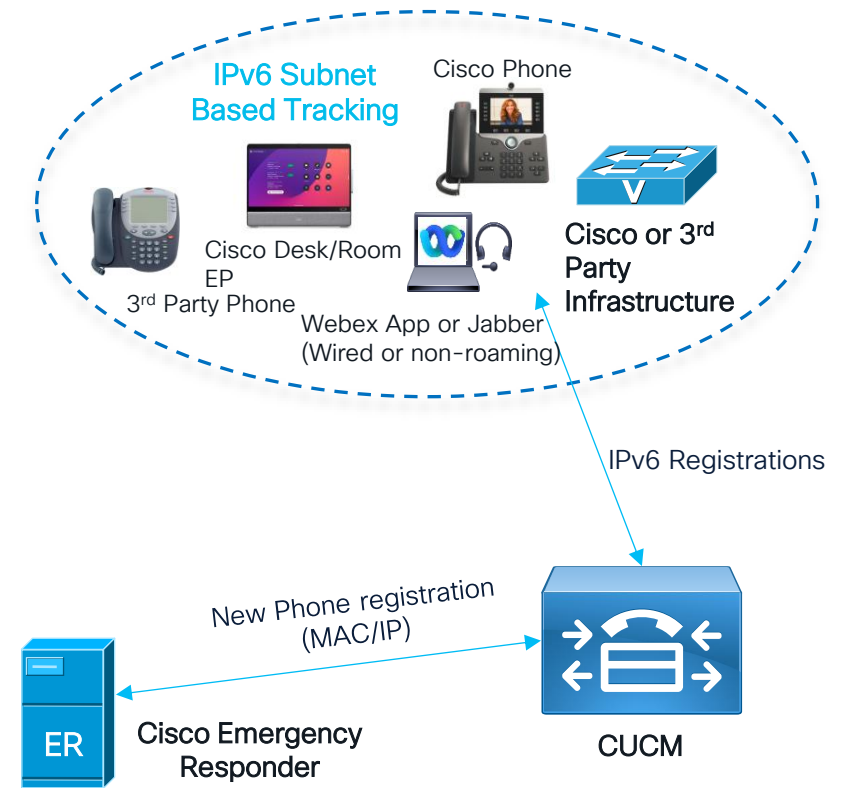
Solution: CER is now enhanced to support IPv6 subnet tracking.

Benefits: IPv6 subnet tracking support on CER will now allow to

- Track device locations using IPv6 Subnets
- Map physical location to IPv6 Subnet
- Define a combination of IPv4 & IPv6 Subnets in CER
- Support IPv6 registrations to CUCM from IP Phones, Desk/Room EPs, and wired soft-client support
- Support IPv6 On-Prem ERL
- Support IPv6 Off-Prem ERL (e.g. VPN Subnet or Expressway IP)
- Support Cisco and 3rd party phone

Supported Release: CER 15 SU1

Roadmap



The screenshot shows the 'Cisco Emergency Responder Administration' web interface. The 'Configure IP Subnet' page is active, displaying a form to add a new IP subnet. The form includes fields for Subnet ID, Prefix, Location, and ERL Name. The Subnet ID field is populated with '2001:db8:0:1:256:2bff:fe05:9797'. The Prefix field is set to '128 (1 address)'. The Location field is set to 'MH1'. The ERL Name field is set to 'erl1'. There are 'Insert' and 'Cancel Changes' buttons at the bottom of the form. A note at the bottom states: '* Indicates required item. Note: Please enter a valid IPv6 format in Subnet ID: X₁:X₂:X₃:X₄:X₅:X₆:X₇:X₈ where X₁₋₈ represents a four digit hexadecimal value.'

Product Bulletins / Deprecation Notice

Feature	Impacted Date	Action / Details
Data Center Migration for Push REST Service	24th November 2023	Add the new IP address: <u>144.196.130.125</u> for Push REST service in Firewall's Access Control List or any other Allow List before 24th November 2023 Production Cutover Date Product Bulletin - https://www.cisco.com/c/en/us/products/collateral/unified-communications/unified-communications-manager-callmanager/uni-cm-im-presence-service-bulletin.html
Deprecation of Remote Call Control with Microsoft Lync Server for IM&P	Release 15	• Microsoft Lync Server 2013 is past Extended End of support Date of Apr 11, 2023 • Customers are encouraged to migrate to Cisco Webex App users to control their Cisco Unified IP Phone

Release 15 Platform Updates



Release Set (emerging) for version 15

Calling



CUCM
15



SME
15



CUC
15



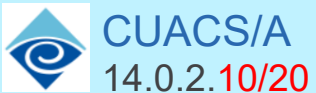
CER
15



Expwy
X15



CPS
15



CUACS/A
14.0.2.10/20



PCD
15



CUBE
TBD

Messaging/ Teams



IMP
15

Meetings



CMS
3.9

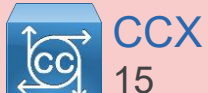


VMN
TBD



TMS
TBD

Contact Center



CCX
15



CCE
15

Endpoints & Cloud Interop



Webex App
TBD



Jabber
TBD



Webex CCUC
TBD



IP Phones
FW TBD
15 beta uses from 12.5su8



Headsets
FW TBD



Video Endpoints
CE/FW TBD

TSP?

- For “everything on 15”...
 - Known versions are “so far”.
 - Version TBD for products not impacted by CentOS EOL.
- **Backwards compatibility**
 - Calling & CC on subsequent slides.
 - Meetings expecting low-risk (UCM NOT changing SIP interface when interacting with CMS as a conf bridge).

Release Set (cont'd)

Planned Calling/CC compatibility

(pending testing, **subject to change**)



1. if UCM version is...

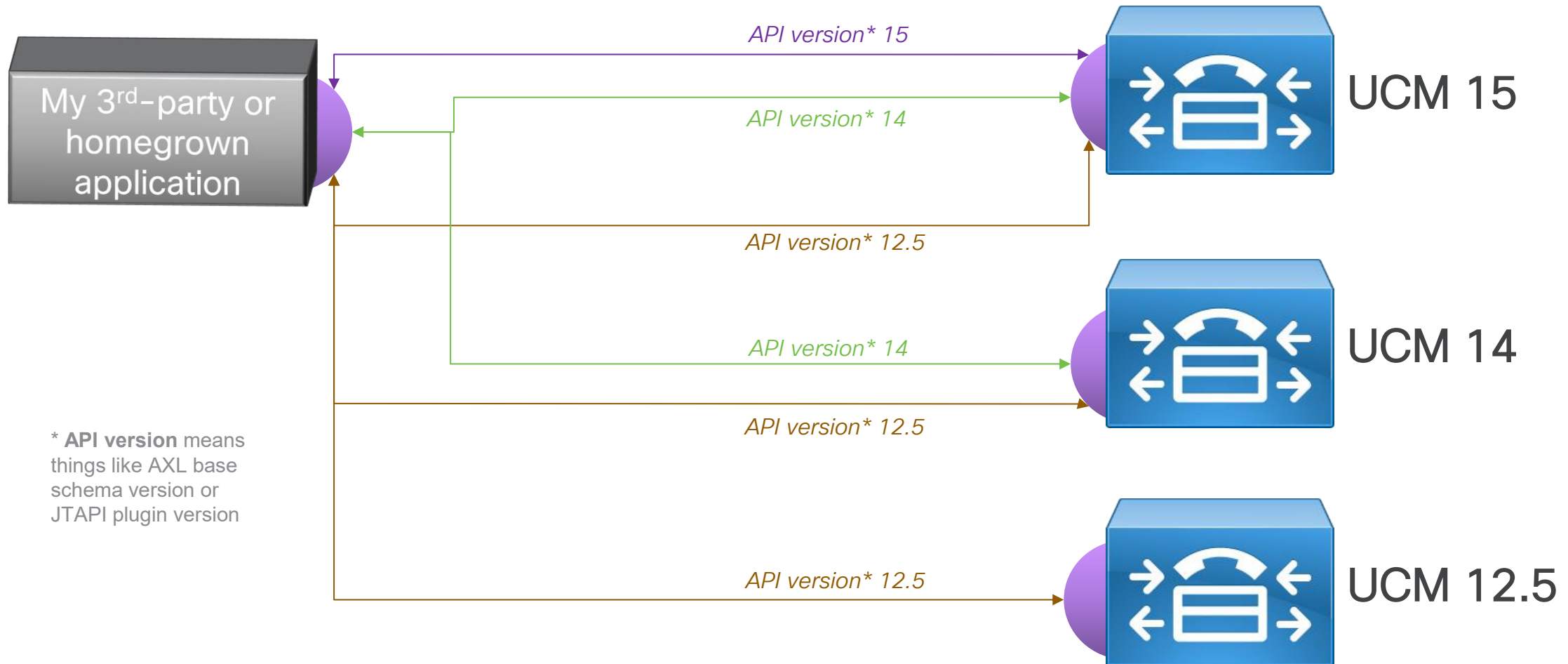
2. ...then other app can be version(s) ...



	UCM 12.5	UCM 14	UCM 15
SME	12.5	14 / 12.5	15 / 14 / 12.5
IMP	12.5	14	15
CUC	15 / 14 / 12.5	15 / 14	15
CER	15 / 14 / 12.5	15 / 14	15
PCD	15 / 14 / 12.6	15 / 14	15
Expressway	X14.3	X15 / X14.3	X15 / X14.3
CPS	14.4.x	14.4.x	15
CUAC Std/Adv	14.0.2.10+	14.0.2.10+	14.0.2.10/20
UCCE suite	15 / 12.x	15 / 12.x	15 / 12.x
UCCX suite	15 / 12.x	15 / 12.x	15 / 12.x

UCM 15 API Compatibility

- In general, UCM R supports APIs R, R-1, R-2.
- v15 is not changing external API definitions.
- 3rd-party validation “Gold/Silver/Bronze” for v15.



UCM 15 endpoint support vs. deprecation

- UCM 15 will NOT deprecate any additional endpoints from 14/older.
- UCM 15 WILL inherit deprecation from prior releases.
 - UCM 14+12.5 did NOT deprecate any additional endpoints from 12.0/older.
 - UCM 12.0+11.5 DID deprecate some endpoints, and these carry over to 14+15.
 - Summary here:
https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/trouble/14_0_1/fieldNotices/cucm_b_d_eprecated-phones-14.html
- Usual caveats apply if using endpoint that is “not deprecated, but end of support”.
 - Can’t repair/replace phone/endpoint hardware or buy spares.
 - Risk of unfixable security issues (e.g. insufficient phone memory to hold newer/bigger certs/keys)
 - Risk of other unfixable software issues (e.g. bug in old firmware, or inability to load new firmware on the old phone).
- Softclient details still being worked out for Jabber vs. Wx App.

Phone Firmware Bundling Strategy : 15 onwards

Description	Change for 15
Supported Phone Firmware	No change , continue to be included in CUCM ISO
End of Support Phone firmware	• Bundled separately in Device Pack • Impacts fresh install of 15 • Does not impact upgrade from previous version to 15 deprecated phone firmware is already installed in previous version of CUCM

Compatibility & support planning

On-prem Calling vs. ESXi vs. Hardware

	CSR 10.6	CSR 11.6	CSR 12.8			CSR 14		CSR 15	
UCM 10.5 eoSupp now 11.5 eoSupp MayCY24 12.5 EOL announce MarCY23 14 Pre-announce MarCY23	10.5	11.5	12.5 SU2+			14		15 (pre-FCS)	
VMware ESXi 6.0/older eoSupp now 6.5/6.7 eoSupp NovCY23 7.0 eoGS AprCY25	6.5	6.7 / 6.5	6.7 / 6.5	7.0	6.7	7.0	8.0	7.0	8.0
UCS M7 SapphireRapids (+Genoa select apps)									
UCS M6 Ice Lake (+Milan select apps)									
UCS M5 CascadeLake / Skylake BE6K/7K M5 eoSupp CY28									
UCS M4 Broadwell / Haswell BE6K/7K M4 eoSupp JunCY23									
UCS M3 Sandy Bridge / Ivy Bridge BE6K/7K M3/older eoSupp now C220/C240 M3 eoSupp now									

Good to go...

Complications...

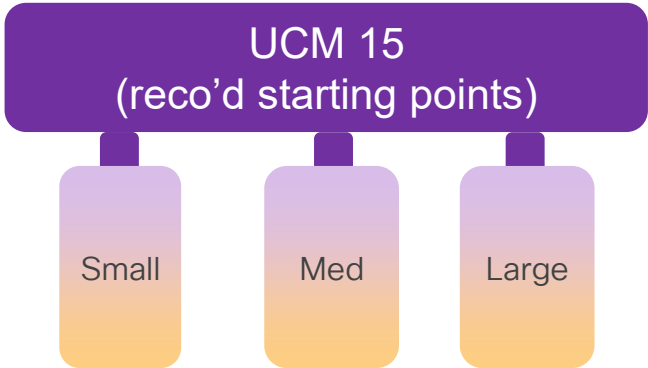
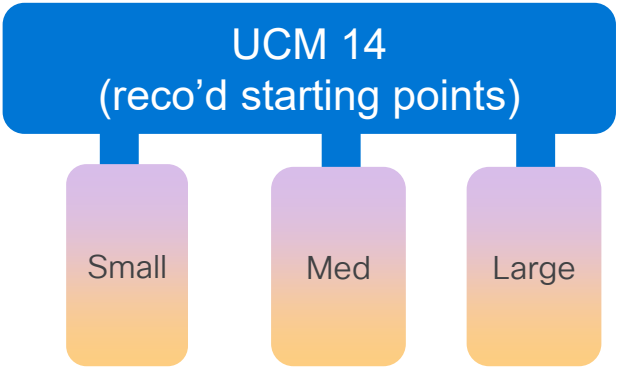
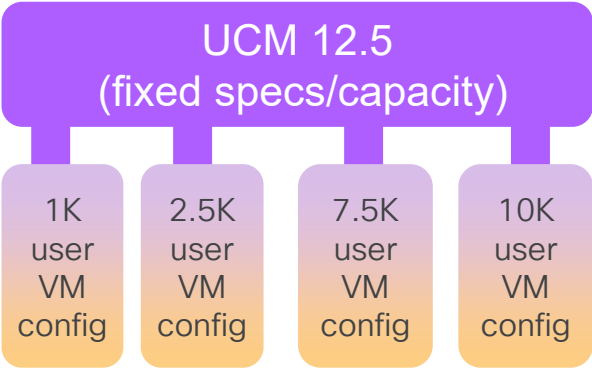
Something’s NOT supported!

- Unsupported combos
- Anything on esxi 6.0 / 5.x
 - No 10.5 on esxi 6.7+
 - No 11.5 on esxi 7.0+
 - No 12.5 on esxi 8.0+
 - No 14 on esxi 6.5
 - No 15 on esxi 6.7

UCM 15

VM Configurations / Specs / Footprints

3. Derive VM Count+Specs



vcpu		2	1	2	4
Base Freq (GHz), full support		2.0+	2.5+	2.5+	2.5+
vram (GB)		6	6	8	8
Vdisk (GB), fs, partition ver	1 st FI as 11.5+	1x80 ext4, 11.5		1x110 ext4, 11.5	
	1 st FI as 8.0-11.0	1x80 ext3, 8.0-11.0		2x80 ext3/4, 8.0-11.0	

Cluster built out of 2-22 VMs.
Each VM same config for full cluster scale & node density.

2	2	4
2.0+	2.0-2.5 (run CST)	
6	8	8
1x80 ext4, 11.5	1x110 ext4, 11.5	
1x80 ext3, 8.0-11.0	2x80 ext3/4, 8.0-11.0	

Higher density/scale for same VM count.

2	2	4
2.0+	2.0-2.5 (run CST)	
10	12	14
1x 110 ext4, 15	1x 110 ext4, 15	
N/A	N/A	

15 min vmv / vmx 17 (13 in 12.5/14)
Guest OS type = "Other 4.x or later Linux (64-bit)"

CAN direct upgrade to 15 or 15 SU.

Can NOT direct upgrade to 15 or 15 SU.
See next slide for alternatives.

15 SU2+ might
reduce VM count.

15 SU2+ might support higher node counts at same
cluster/node capacities.

SME 15

VM Configurations / Specs / Footprints

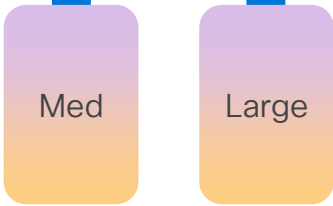
SME 12.5 (fixed specs/capacity)



vcpu		2	4
Base Freq (GHz), full support		2.5+	2.5+
vram (GB)		8	8
Vdisk (GB), fs, partition ver	1 st FI as 11.5+	1x110 ext4, 11.5	
	1 st FI as 8.0-11.0	2x80GB ext3/4, 8.0-11.0	

Cluster built out of 2 to 22 VMs.
Each VM same config for full cluster scale & node density.

SME 14 (fixed specs/capacity)



2	4
2.5+	
8	8
1x110 ext4, 11.5	
2x80GB ext3/4, 8.0-11.0	

SME 15 (fixed specs/capacity)



2	4
2.5+	
12	14
1x 110 ext4, 15	
N/A	

Same capacity as SME 14.

CAN direct upgrade to 15 or 15 SU.

Can NOT direct upgrade to 15 or 15 SU.
Same alternatives as for UCM.

SME 15

VM Configurations / Specs / Footprints

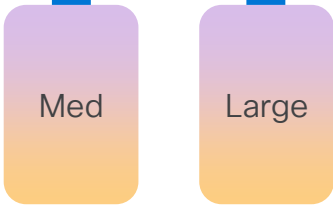
SME 12.5 (fixed specs/capacity)



vcpu		2	4
Base Freq (GHz), full support		2.5+	2.5+
vram (GB)		8	8
Vdisk (GB), fs, partition ver	1 st FI as 11.5+	1x110 ext4, 11.5	
	1 st FI as 8.0-11.0	2x80GB ext3/4, 8.0-11.0	

Cluster built out of 2 to 22 VMs.
Each VM same config for full cluster scale & node density.

SME 14 (fixed specs/capacity)



2	4
2.5+	
8	8
1x110 ext4, 11.5	
2x80GB ext3/4, 8.0-11.0	

SME 15 (fixed specs/capacity)



2	4
2.5+	
12	14
1x 110 ext4, 15	
N/A	

Same capacity as SME 14.

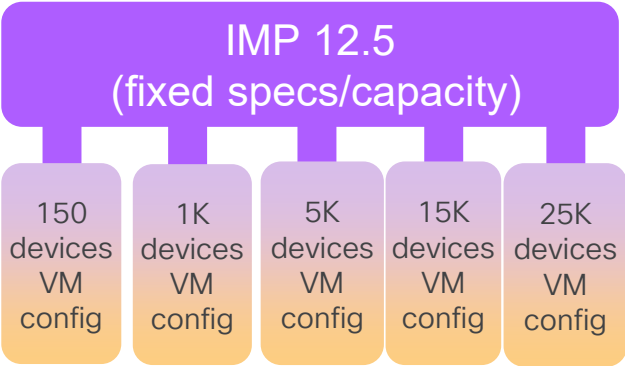
CAN direct upgrade to 15 or 15 SU.

Can NOT direct upgrade to 15 or 15 SU.
Same alternatives as for UCM.

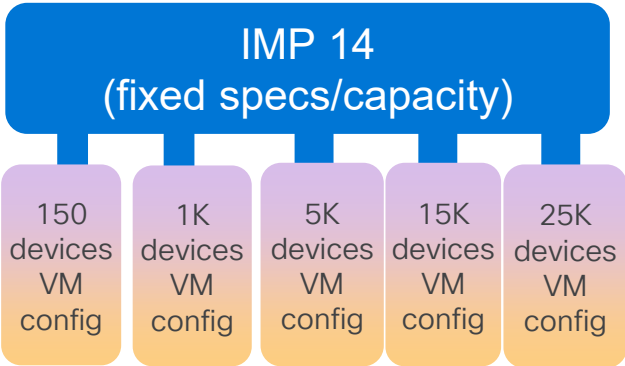
IMP 15

VM Configurations / Specs / Footprints

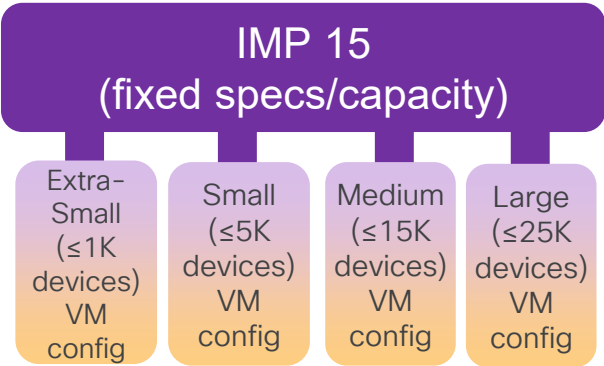
3. Derive VM Count+Specs



vcpu	1	1	2	4	6
Base Freq (GHz), full support	2.0+	2.0+	2.5+	2.5+	2.5+
vram (GB)	2	4	4	8	16
vdisk (GB)	1x80	1x80	2x80	2x80	2x80



1	1	2	4	6
2.0+	2.0+	2.5+	2.5+	2.5+
2	4	4	8	16
1x80	1x80	2x80	2x80	2x80



1	2	4	6
2.0	2.5	2.5+	2.5+
6	10	14	18
1x110	2x80	2x80	2x80

Cluster built out of 2-6 VMs.
Each VM same config for full cluster scale & node density.

CAN direct upgrade to 15 or 15 SU.

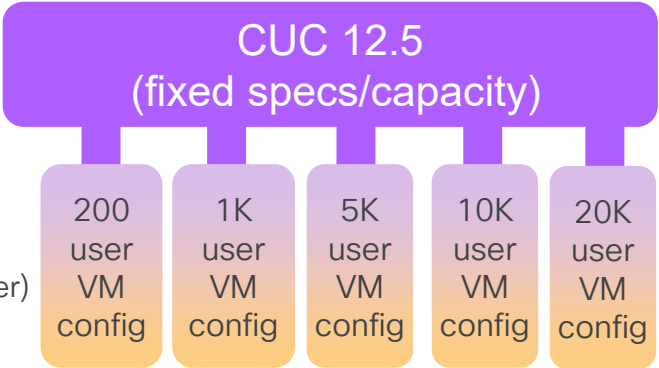
Can NOT direct upgrade to 15 or 15 SU.
Move to another VM config.

CUC 15 VM Configurations / Specs / Footprints

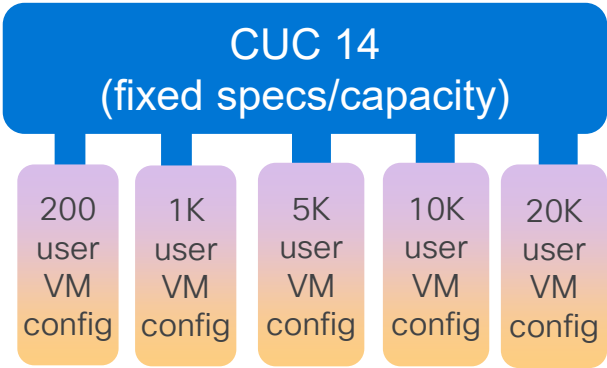
* vcpu 1 (+1 if UM)
Min GHz 2.0 (1.8 if UM)

** 2x146GB, 300GB or 500GB

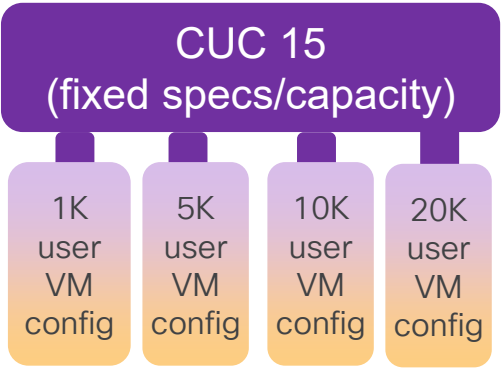
*** vram 8GB (+2GB w/ Secure Jabber)
vdisk 2x300GB or 500GB



vcpu	1	*	2	4	7
Base Freq (GHz), full support	1.8+	*	2.5+	2.5+	2.5+
vram (GB)	4	4	6	6	***
vdisk (GB)	160	160	200	**	***



1	*	2	4	7
1.8+	*	2.5+	2.5+	2.5+
4	4	6	6	***
160	160	200	**	***



2	2	4	7
1.8+	2.5+	2.5+	2.5+
10	12	12	16
160	200	**	***

Cluster built out of 2 VMs.
Each VM same config for full cluster scale & node density.

CAN direct upgrade to 15 or 15 SU.

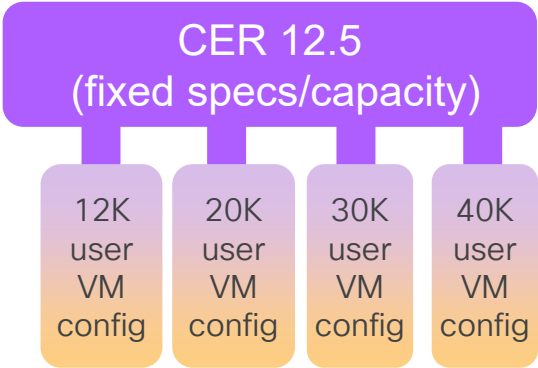
Can NOT direct upgrade to 15 or 15 SU.
Alternatives TBD.

Maybe NOT direct upgrade to 15 or 15 SU.
Check hardware CPU pcores.

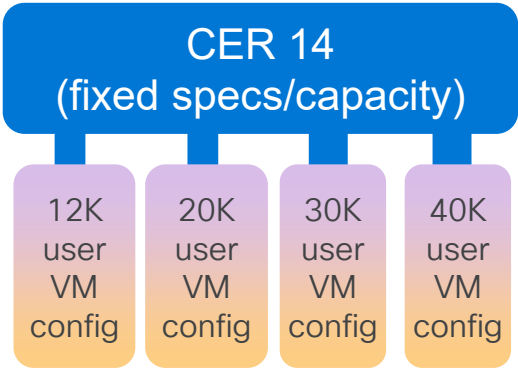
CER 15

VM Configurations / Specs / Footprints

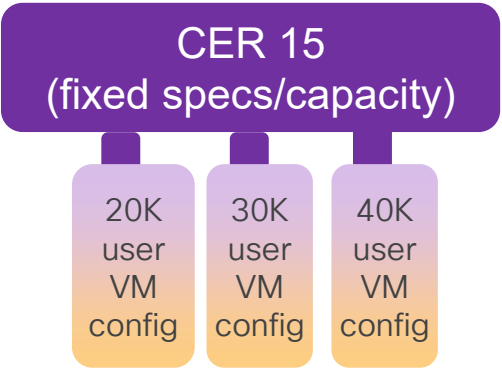
* Min GHz is:
2.0+ for max 12K users.
2.5+ for max 20K users



vcpu	2	1	2	4
Base Freq (GHz), full support	2.0+	2.5+	2.5+	2.5+
vram (GB)	4	4	6	6
vdisk (GB)	1x80	1x80	1x110	1x110



2	1	2	4
2.0+	2.5+	2.5+	2.5+
4	4	6	6
1x80	1x80	1x110	1x110



2	2	4
*	2.5+	2.5+
6	8	8
1x80	1x110	1x110

Cluster built out of 2 VMs.
Each VM same config for full cluster scale & node density.

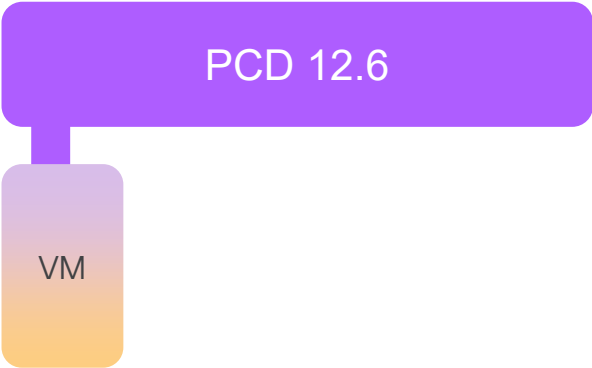
CAN direct upgrade to 15 or 15 SU.

Can NOT direct upgrade to 15 or 15 SU.
Move to another VM config.

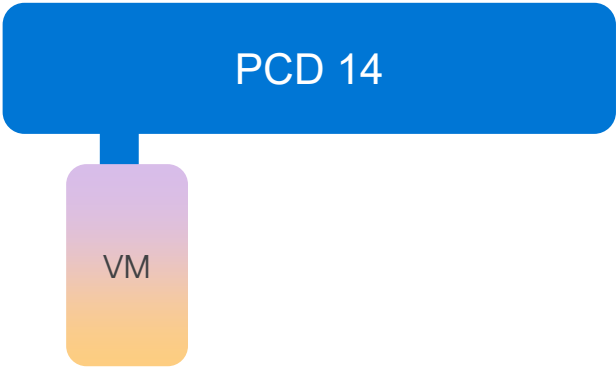
PCD 15

VM Configurations / Specs / Footprints

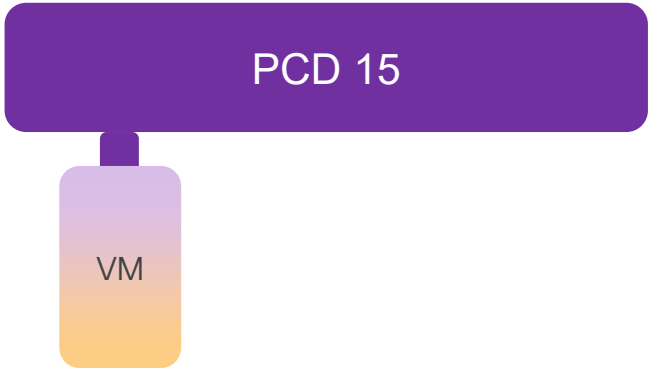
3. Derive VM Count+Specs



vcpu	2
Base Freq (GHz), full support	2.0+
vram (GB)	4
Vdisk (GB), fs, partition ver	1x 80GB ext4, 11.5 (++GB if using PCD as SFTP Server for UCM)



2
2.0+
4
1x 80GB ext4, 11.5



2
2.0+
6
1x 110 ext4, 15

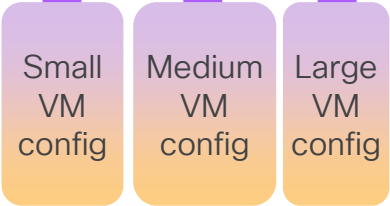
No concerns about vdisk size/qty, ext3 or guest OS partitions as with UCM.

CAN direct upgrade to 15 or 15 SU.

Expressway X15

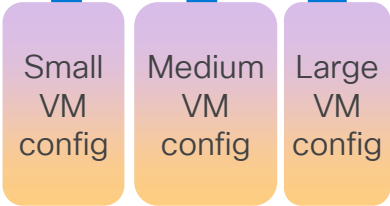
VM Configurations / Specs / Footprints

Expressway X12.7 (fixed specs/capacity)



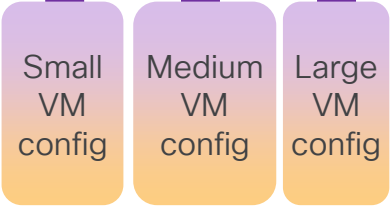
vcpu	2	2	8
Base Freq (GHz), full support	1.8+	2.4+	3.2+
vram (GB)	4	6	8
vdisk (GB)	132	132	132

Expressway X14.3 (fixed specs/capacity)



2	2	8
1.8+	2.4+	3.2+
4	6	8
132	132	132

Expressway X15 (fixed specs/capacity)



2	2	8
1.8+	2.4+	3.2+
4	6	8
132	132	132

Cluster built out of 2 VMs.
Each VM same config for full cluster scale & node density.

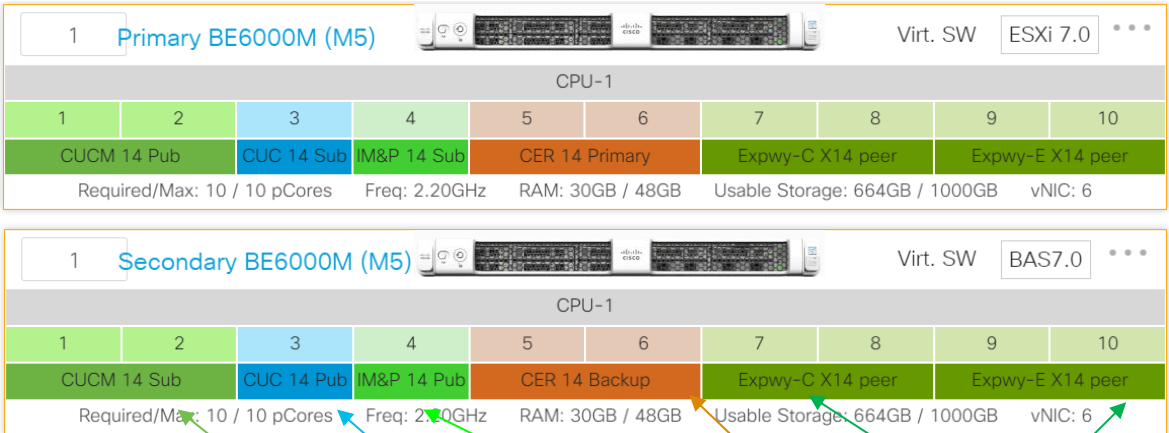
CAN direct upgrade to 15 or 15 SU.

Hardware capacity planning

Example #1 – Small Collaboration

Small Collaboration (<1K users, <1200 devices)
Single-site, redundant BE6000M (M5) appliances
ESXi 7.0 and CSR 14 app versions

From www.cisco.com/go/quotecollab



Available per BE6M-M5-K9 appliance:
1S/10C/2.2GHz CPU
48GB RAM
~1 TB usable storage

Needed for ESXi 7.0 / CSR 14:
10C (fits but no headroom)
30GB RAM (fits with 18GB headroom)
664GB storage (fits with ~336 GB headroom)

Delta for ESX 7.0 > 8.0 & upgrade all VM's 14 > 15:
+1C (won't fit!)
+26GB RAM (if ESXi 8.0) ... won't fit!
or +18 GB RAM (if stay ESXi 7.0) ... barely fits
+60GB storage (fits with ~276 GBheadroom)

Solution on next slide.

	ESXi 7.0>8.0	CUCM 14>15	CUC 14>15	IMP 14>15	CER 14>15	Expressway -C / -E X14.3 >X15
CPU (pcores)	-	-	+1	-	-	-
RAM (GB)	+8	+4	+6	+6	+2	-
Storage (GB)	-	+30	-	+30	-	-
Other	-	-	-	-	-	-

ESXi is per appliance
Rest is per VM



Hardware capacity planning

Example #1 – Small Collaboration (cont'd)

Things explored:

- Is cloud calling an option? **No (airgapped).**
- Would migrating IMP VM's to Webex cloud messaging help? **Not by enough.**
- Can I up-spec the HW? **No (CPU swap not supported, add RAM won't help due to pcore problem).**

Solution:

- Add another appliance. **Provides better redundancy, more expansion headroom, easier change management.**

1	Original BE6000M (M5) #1				Virt. SW		ESXi 8.0		...	
CPU-1										
1	2	3	4	5	6	7	8	9	10	
UCM 15 Pub		CUC 15 Sub		Expwy-C X15 peer		Expwy-E X15 peer				
Required/Max: 8 / 10 pCores Freq: 2.20GHz RAM: 36GB / 48GB Usable Storage: 554GB / 1000GB vNIC: 4										

1	Original BE6000M (M5) #2				Virt. SW		ESXi 8.0		...	
Physical CPU Cores										
1	2	3	4	5	6	7	8	9	10	
UCM 15 Sub		IMP 15 Pub	CER 15 backup		Expwy-C X15 peer					
Required/Max: 7 / 10 pCores (1S/10C) Freq: 2.20GHz RAM: 38GB / 48GB Usable Storage: 432GB / 1000GB vNIC: 4										

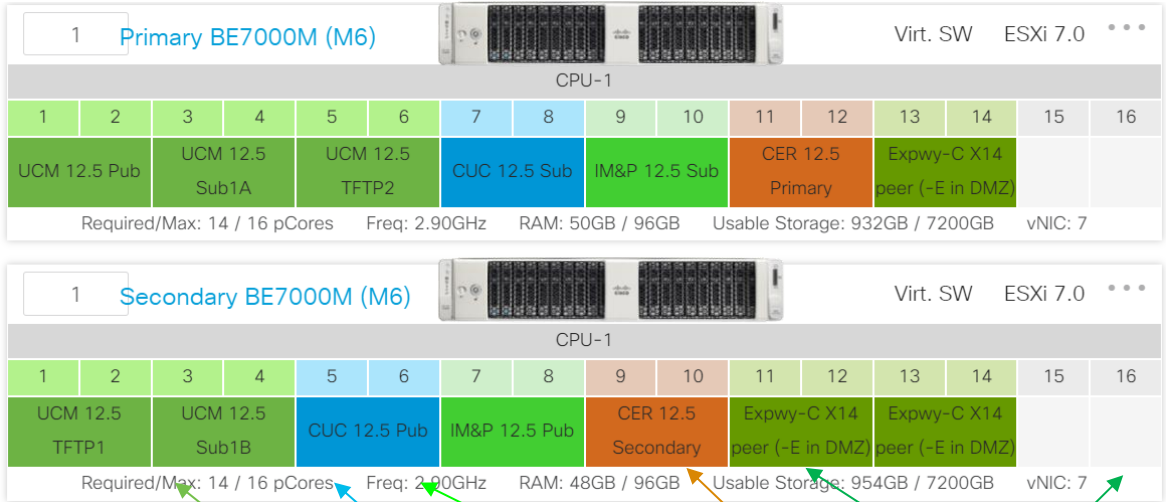
1	Added BE6000M (M6) as #3				Virt. SW		ESXi 8.0		...	
CPU-1										
1	2	3	4	5	6	7	8	9	10	
CUC 15 Pub		IMP 15 Sub	CER 15 primary		Expwy-E X15 peer					
Required/Max: 7 / 10 pCores Freq: 2.30GHz RAM: 38GB / 64GB Usable Storage: 502GB / 3600GB vNIC: 4										

Hardware capacity planning

Example #2 – Medium Collaboration

Medium Collaboration (5K devices)
Single-site, redundant BE7000M (M6) appliances (recent HW refresh).
ESXi 7.0 and CSR 12.7 app versions

From www.cisco.com/go/quotecollab



Available per BE7M-M6-K9 appliance:

1S/16C/2.9GHz CPU
96GB RAM
4x ~1 TB usable storage

Needed for ESXi 7.0 / CSR 14:

14C & 14C (fits with 2C headroom)
50GB & 48GB RAM (fits with 46-48GB headroom)
932GB & 954GB storage (fits with LOTS of headroom)

Delta for ESX 7.0 > 8.0 & upgrade all VM's 14 > 15:

+0C & +0C (fits)
+34GB & +30GB RAM (if ESXi 8.0) ... fits
or +26GB & +22GB RAM (if stay ESXi 7.0) ... fits
+0GB storage (fits)

Can upgrade everything "in place" on existing hardware.

	ESXi 7.0>8.0	CUCM 14>15	CUC 14>15	IMP 14>15	CER 14>15	Expressway -C / -E X14.3 >X15
CPU (pcores)	-	-	-	-	-	-
RAM (GB)	+8	+4	+6	+6	+2	-
Storage (GB)	-	-	-	-	-	-
Other	-	-	-	-	-	-

ESXi is per appliance
Rest is per VM

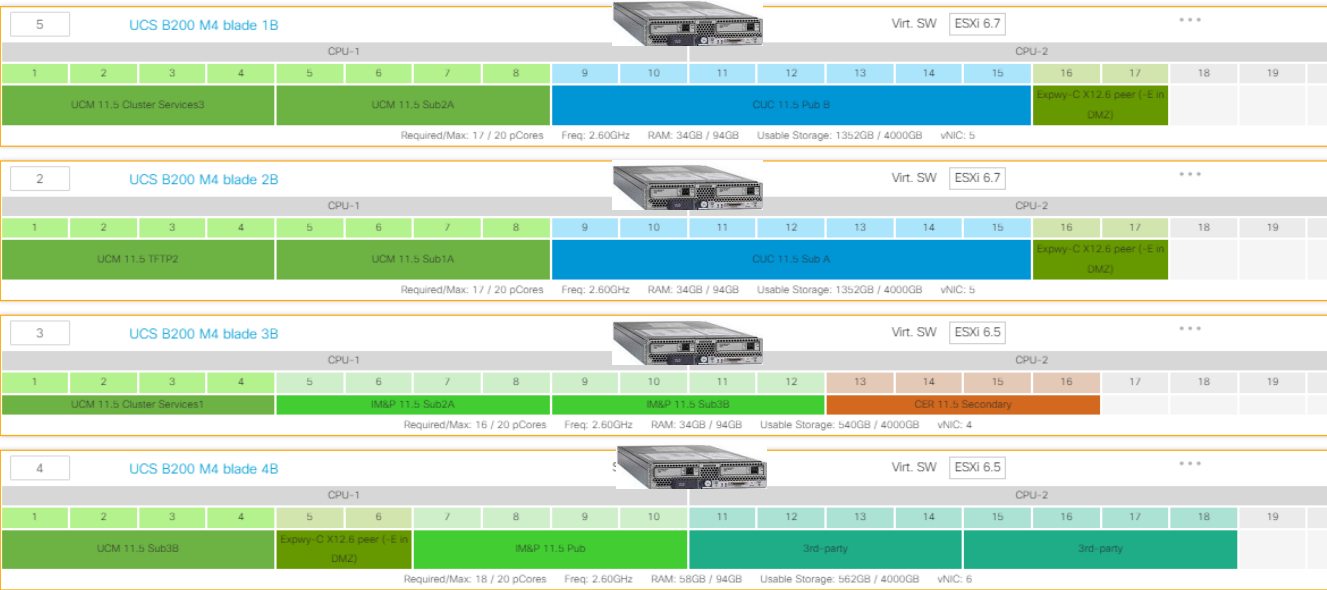
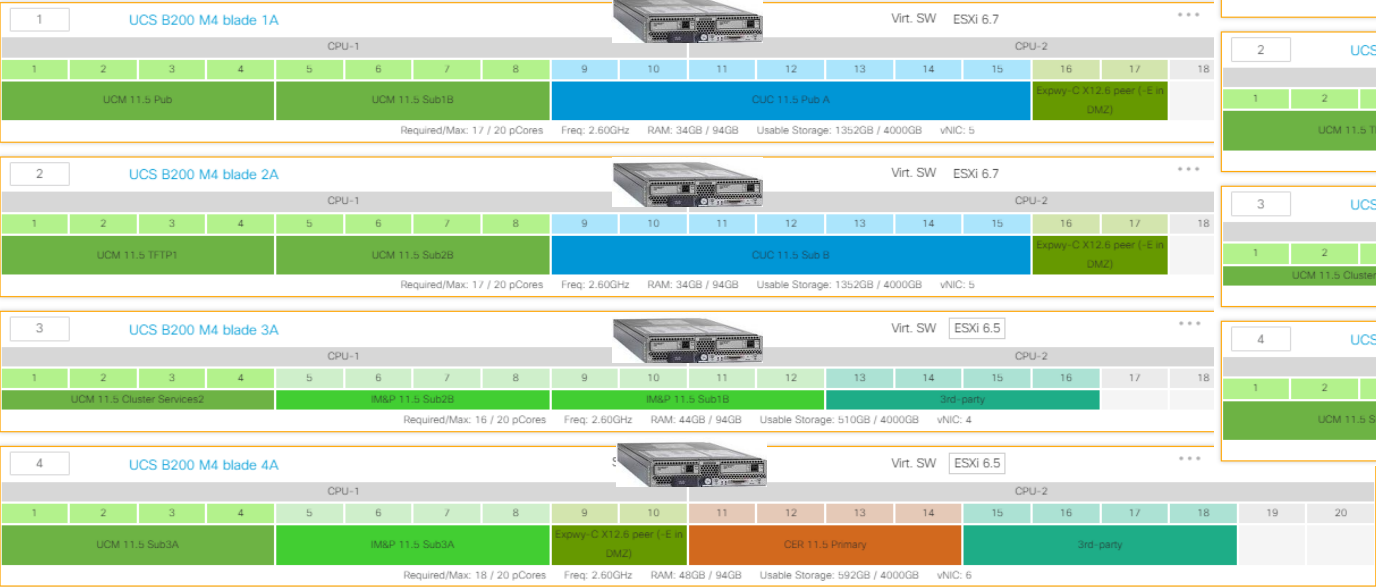


Hardware capacity planning

Example #3 – Large Collaboration

Large Collaboration (10K–20K users with 20K–30K devices)
Dual-DC farms of B200 M4 blade servers (**HW going end of support**).
ESXi 6.x and CSR 11.6 app versions (**SW vers going end of support**).

From www.cisco.com/go/quotecollab



Available per UCSB-B200-M4-U blade server:
2S/10C/2.6GHz CPU
94GB RAM
Diskless / off-blade shared storage

Needed for ESXi 7.0 / CSR 14:
16–18C / blade (fits with 2–4C/blade headroom but HW refresh needed!)
34–58GB RAM / blade (fits with 36–60GB headroom but HW refresh needed!)
Off-box shared storage (fits with LOTS of headroom but HW refresh needed!)

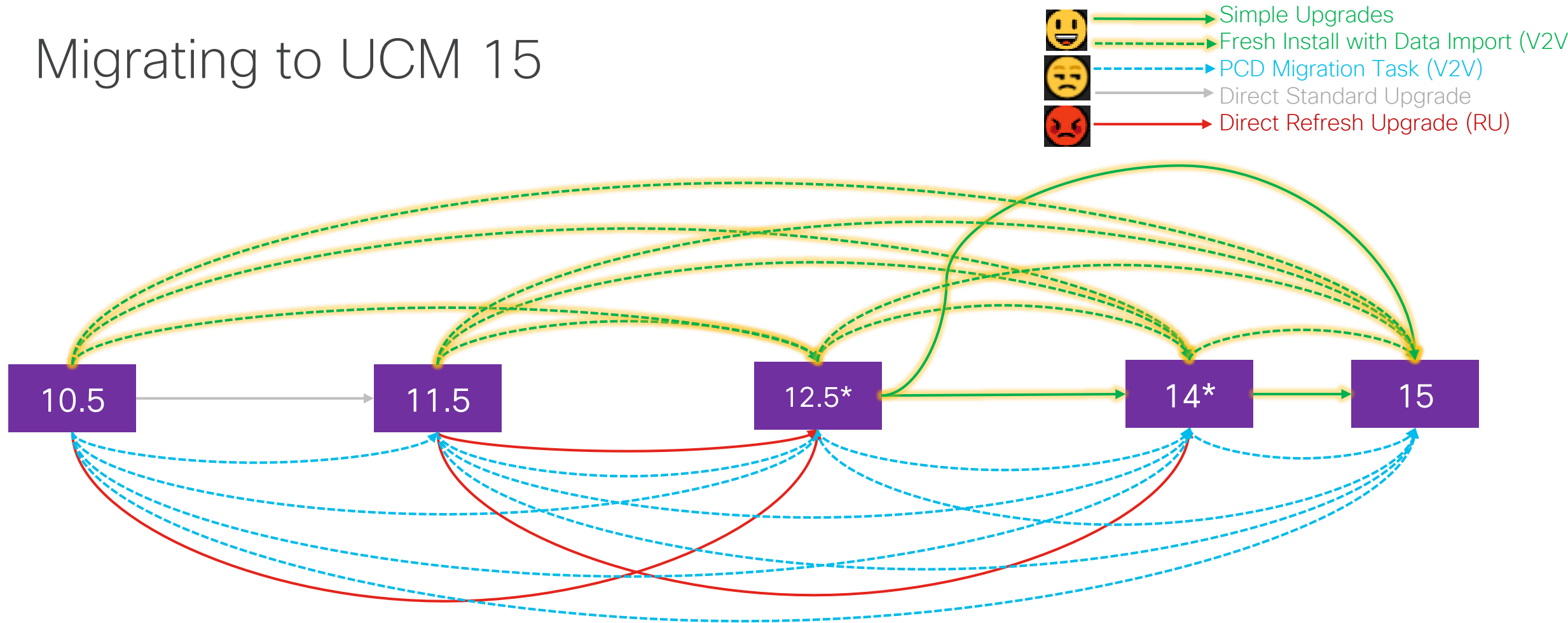
Delta for ESXi 6.x > 8.0 & upgrade all VM's 11.5 > 15:
+0C / blade (fits but HW refresh needed!)
Per blade, +12–38GB RAM (if ESXi 8.0) ... fits if rearrange VMs or +4–30GB RAM (if stay ESXi 7.0) ... fits but HW refresh needed!
+0GB storage (fits)

Refresh HW or adjust VM placement, then can upgrade everything “in place” on existing HW.

	ESXi 6.x>8.0	CUCM 11.5>15	CUC 11.5>15	IMP 11.5>15	CER 11.5>15	Expressway –C / –E X12.6 >X15
CPU (pcores)	–	–	–	–	–	–
RAM (GB)	+8	+6	+6	+6	+2	–
Storage (GB)	–	–	–	–	–	–
Other	–	–	–	–	–	–

ESXi is per blade
Rest is per VM

Migrating to UCM 15



Recommendation:

- Simple Upgrades where you can, Fresh Install with Data Import where you can't.

EDGE

Expressway

Luca Pellegrini – Technical Marketing Engineer

LDAP Support for different port than 636

- LDAP over TCP/UDP uses port 389, while LDAP over TLS uses port 636
- For security reasons, some companies would like to use a different port than 636
- This port is now configurable, while with previous releases it didn't work after TLS handshake
- **STARTTLS**: TLS encryption is used over TCP port.
- **Supported Release**: X15
- **Timeline**: Q4 CY23

The screenshot displays the 'LDAP server configuration' window. The 'Port' field is set to 5389, marked with an orange star. The 'Encryption' dropdown menu is open, showing options: Off, TLS, and STARTTLS. The 'STARTTLS' option is highlighted. Other fields include 'FQDN address resolution' (set to IP address), 'Server address' (10.47.150.115), and 'Certificate revocation list (CRL) checking'.

Field	Value
FQDN address resolution	IP address
Server address	10.47.150.115
Port	5389
Encryption	STARTTLS
Certificate revocation list (CRL) checking	

TLS 1.3 Support

- TLS 1.3 adds security and velocity compared to TLS 1.2
- Faster handshake
- Removes less secure cipher suites and keeps the stronger ones
- Benefits
 - If you set TLS 1.3 as minimum release, Expressway will only talk to systems and devices supporting TLS 1.3
 - In case TLS 1.2 is still present in some scenarios, set the minimum TLS release to 1.2. Fallback to 1.2 will happen if 1.3 can't be negotiated
- Supported Release: X15+
- Timeline: Q2 CY24

The image displays two screenshots of the Cisco TelePresence Video Communication Server Control interface, specifically the 'Logging configuration' and 'Ciphers' sections.

Logging configuration: This section shows various logging options. Under 'Remote syslog servers', a dropdown menu for 'Minimum TLS Version' is open, showing options: TLS v1.0, TLS v1.1, TLS v1.2, and TLS v1.3 (which is selected). Other options include 'Address', 'Port', 'Message Format', and 'Filter by Severity'. An information box on the right states: 'Select the minimum TLS version accepted by this service/client. The service/client allows communication through TLS of all versions greater than or equal to the version selected. Default: TLS v1.2'.

Ciphers: This section shows a list of cipher suites and their minimum TLS versions. A dropdown menu for 'Minimum TLS Version' is open, showing options: TLS v1.0, TLS v1.1, TLS v1.2, and TLS v1.3 (which is selected). The list of cipher suites includes: HTTPS ciphers, HTTPS minimum TLS version, LDAP TLS Ciphers, LDAP minimum TLS version, Reverse proxy TLS ciphers, Reverse proxy minimum TLS version, SIP TLS ciphers, SIP minimum TLS version, SMTP TLS Ciphers, SMTP minimum TLS version, TMS Provisioning Ciphers, TMS Provisioning minimum TLS version, UC server discovery TLS ciphers, UC server discovery minimum TLS version, XMPP TLS ciphers, and XMPP minimum TLS version. An information box on the right states: 'Select the minimum TLS version accepted by this service/client. The service/client allows communication through TLS of all versions greater than or equal to the version selected. Default: TLS v1.2'.

Improved Webex App experience with persistent CUCM login flow

- Webex app registered to CUCM
Registered prompts user to login every 60 days to maintain phone service.
- User may cancel login for phone service and still have access to messaging, meetings, and internal calls as the account is still authenticated with IdP on Cisco Webex cloud
- If the user on Webex app is not authenticated by CUCM:
 - Phone service disconnects which results in Missed calls
 - Mixed calling User Experience. Internal calls may work as connectivity to Cisco Webex cloud is maintained, but PSTN calls don't.

Key Feature

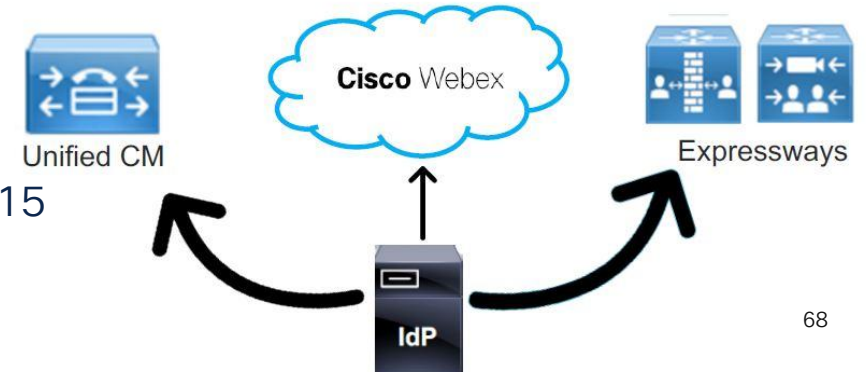
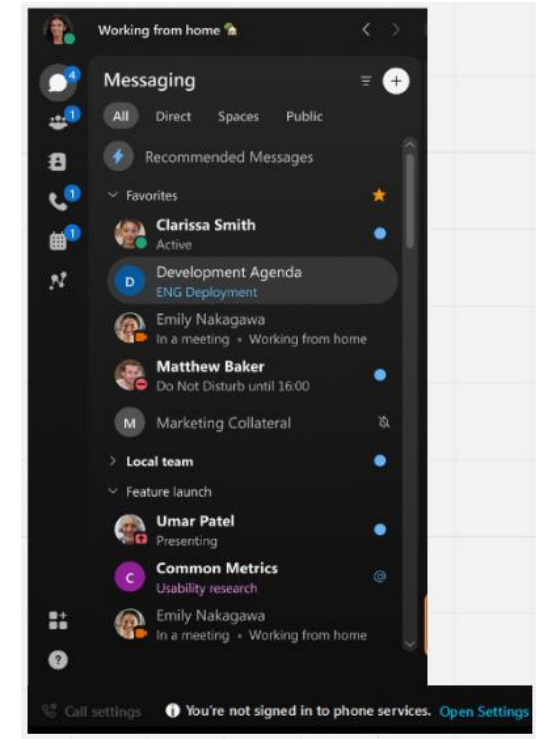
- The refresh token in Webex App is automatically renewed without user intervention so that the user does not get logged out of the phone service.
- Expressway will be able to pass through the auto-renew parameter for refresh token so that the clients will renew the tokens automatically

Benefits

- Improved Webex App user experience resulting in end users not missing calls due to phone service disconnects

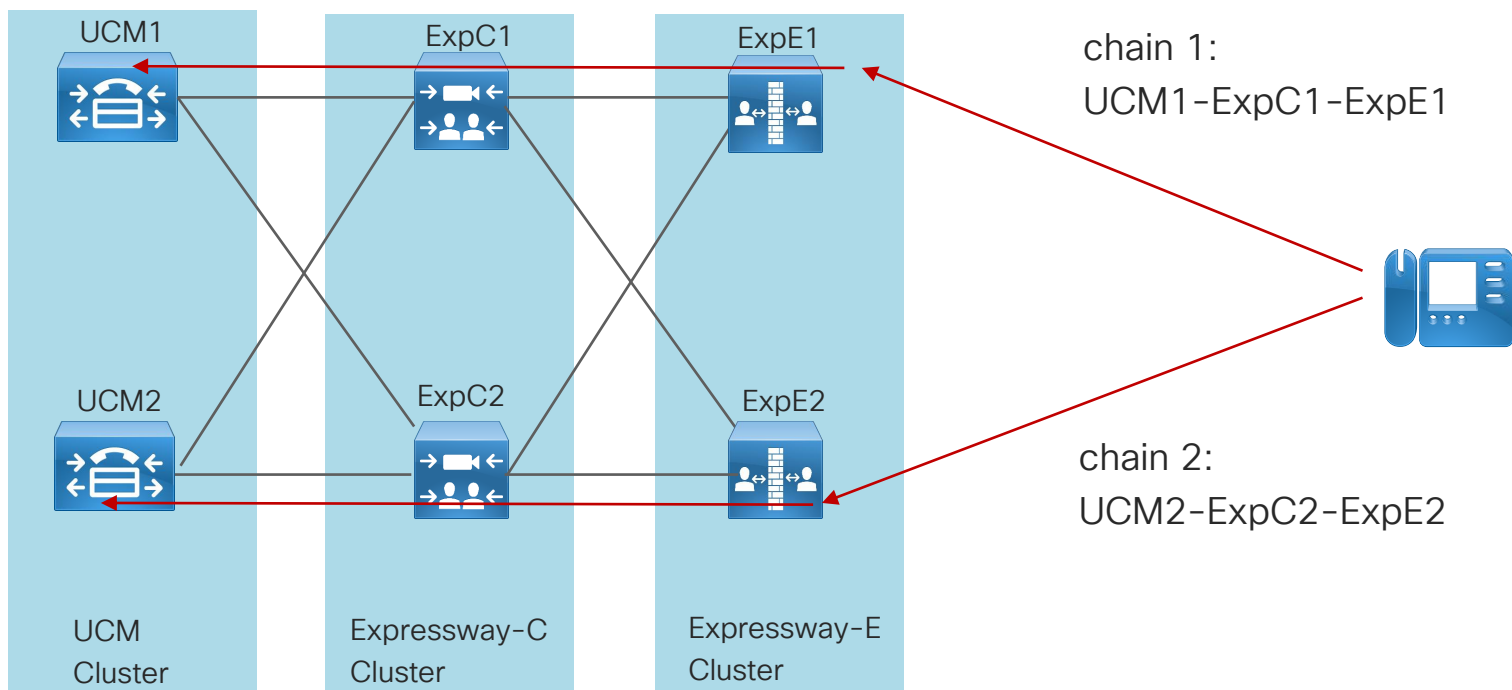
Supported Releases:

- UCM 15, Expressway X15
- Webex App 43.8



88XX and 78XX Series Phone Failover

Actual Failover Scenario



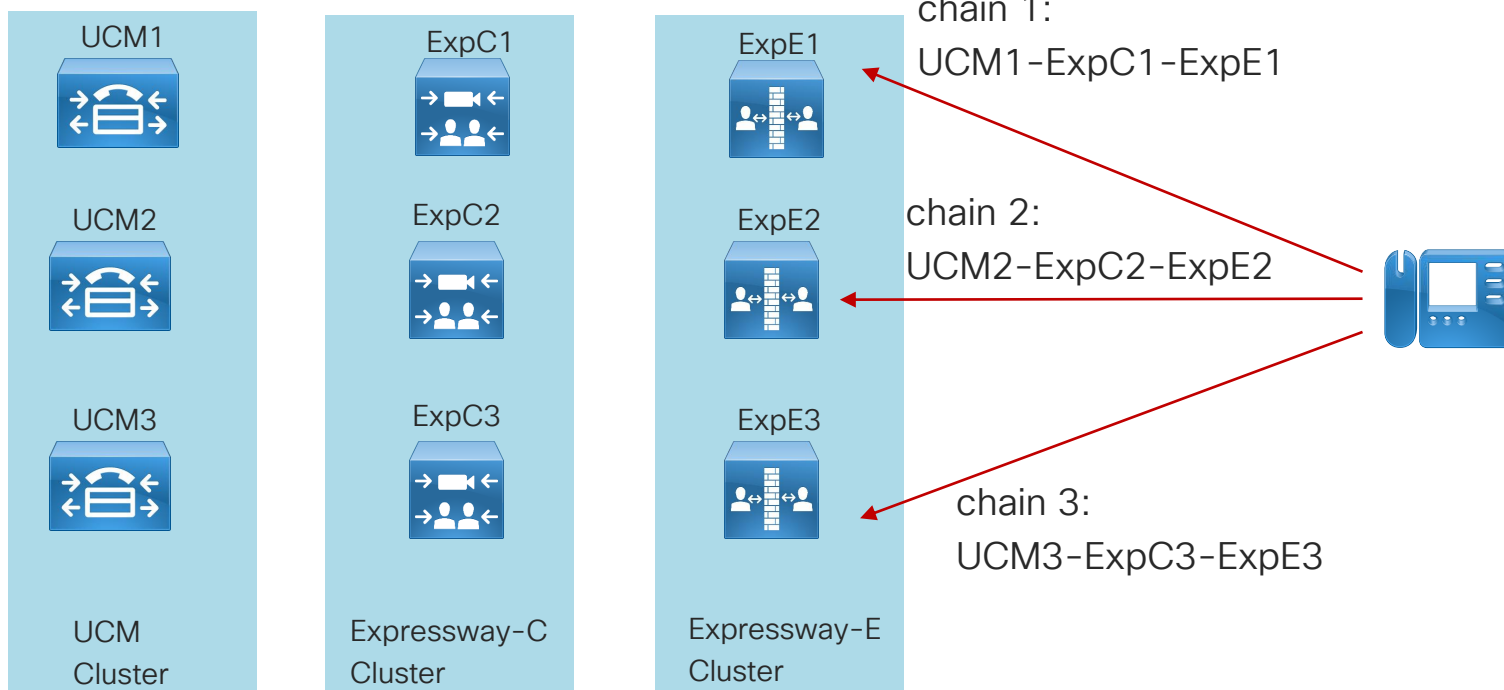
In order to failover to work, multiple servers can fail only inside a chain.

I.e. :

- ExpC1 and ExpE1 fail, then chain 2 will be selected.
- No failover if ExpC1 and ExpE2 fail

88XX and 78XX Series Phone Failover

Actual Failover Scenario: Implications and Limitations



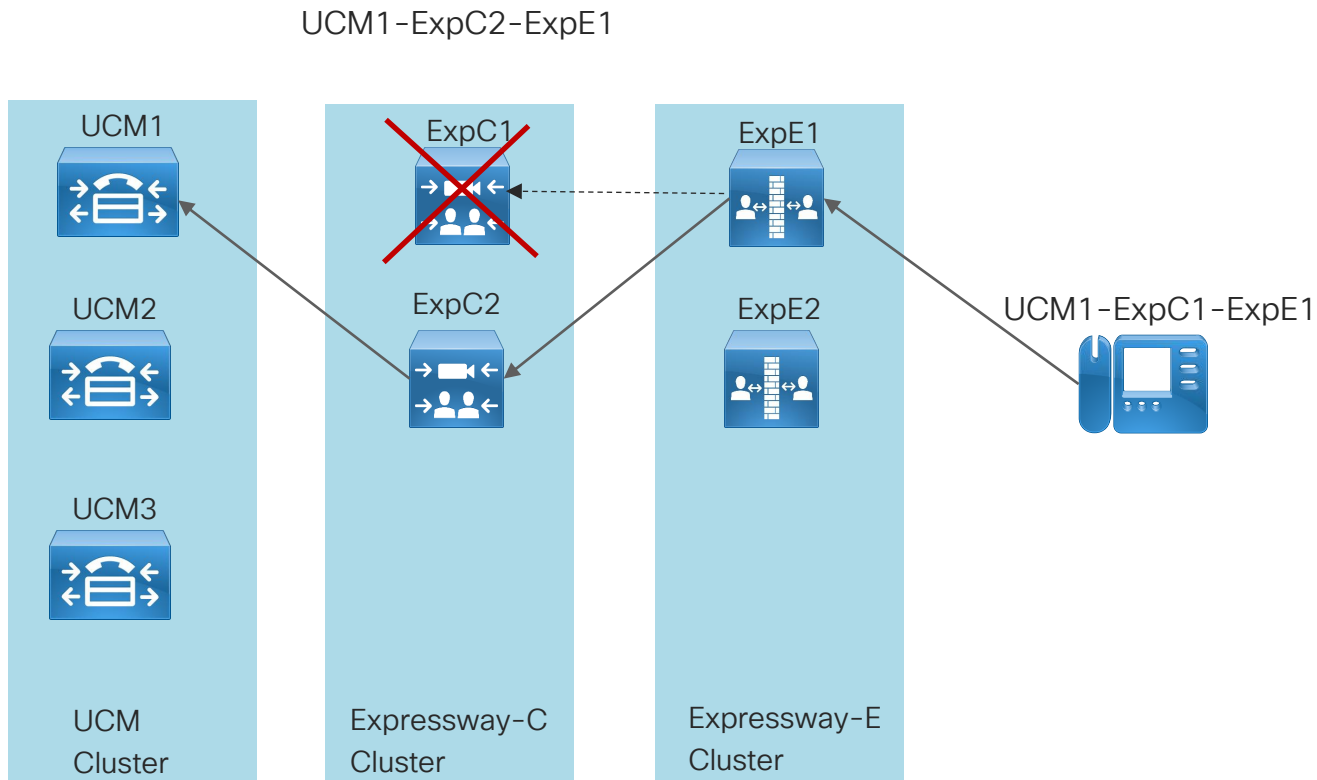
If CallManager group includes UCM1, UCM2, UCM3, then failover might not occur because UCM3 is not associated to any Expressway.

The number of Expressway servers must match the number of servers in the CallManager group.

With multiple CallManager groups, this might become a problem

88XX and 78XX Series Phone Failover

Solution: Adaptive Routing and STUN Based Keepalives



On Registration:

1. Based on get-edge config response, the phone creates one possible route:

ExpE1 - ExpC1 - UCM1

2. ExpC1 is down; ExpC2 is used instead

3. Phone updates the route to:

ExpE1 - ExpC2 - UCM1

After registration:

Phone polls each server through STUN keepalives. If get no response assumes that the server is down and uses another one in the get-edge config response

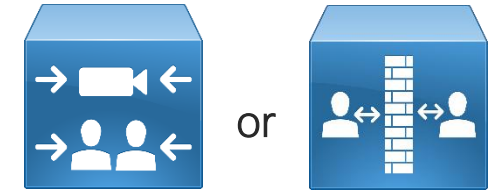
Expressway CE1300 appliance

Latest CE1x00 appliance model

- Refreshes prior CE1200 (M5, LDoS Oct CY28 / EOL14929).
- Provides Edge services to on-prem-Collaboration for Hybrid Work.
- Bare-metal / non-virtualized to satisfy critical customer DMZ security concerns.
- All in an easy to plan/design/quote/deliver *appliance* solution.

CE1300 hardware & software

- Leverages UCS C220 M6S, but is NOT a general-purpose “server”, NOT a UCS convenience quoting bundle.
- SED’s for encryption at rest (factory load defaults to encryption disabled; see Install Guide for how to enable).
- Field Recovery Mechanism updated from SD card to M.2 device-based.
- Only supports Expressway X14.3.1 or later. No other applications. No older versions.
- See [X14.3.1 Release Notes](#) and [CE1300 Install Guide](#) for other details.



Bare-metal install w/ BIOS customizations



EXPWY-1300-K9

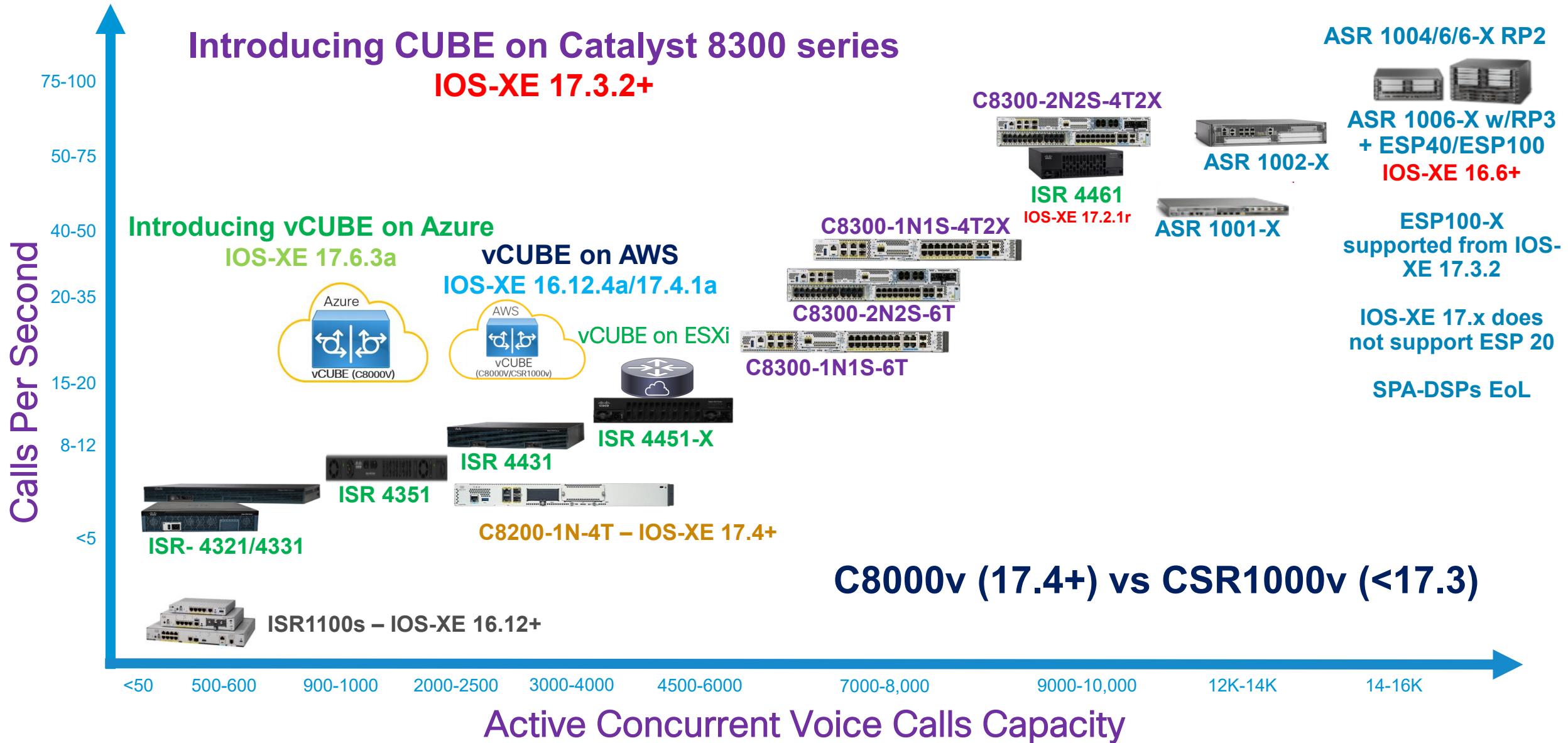
EDGE

CUBE

Hussain Ali – Technical Marketing Engineer

New Platforms for CUBE/vCUBE

CUBE (Enterprise) Product Portfolio [Not to Scale]



CUBE Software Release Mapping

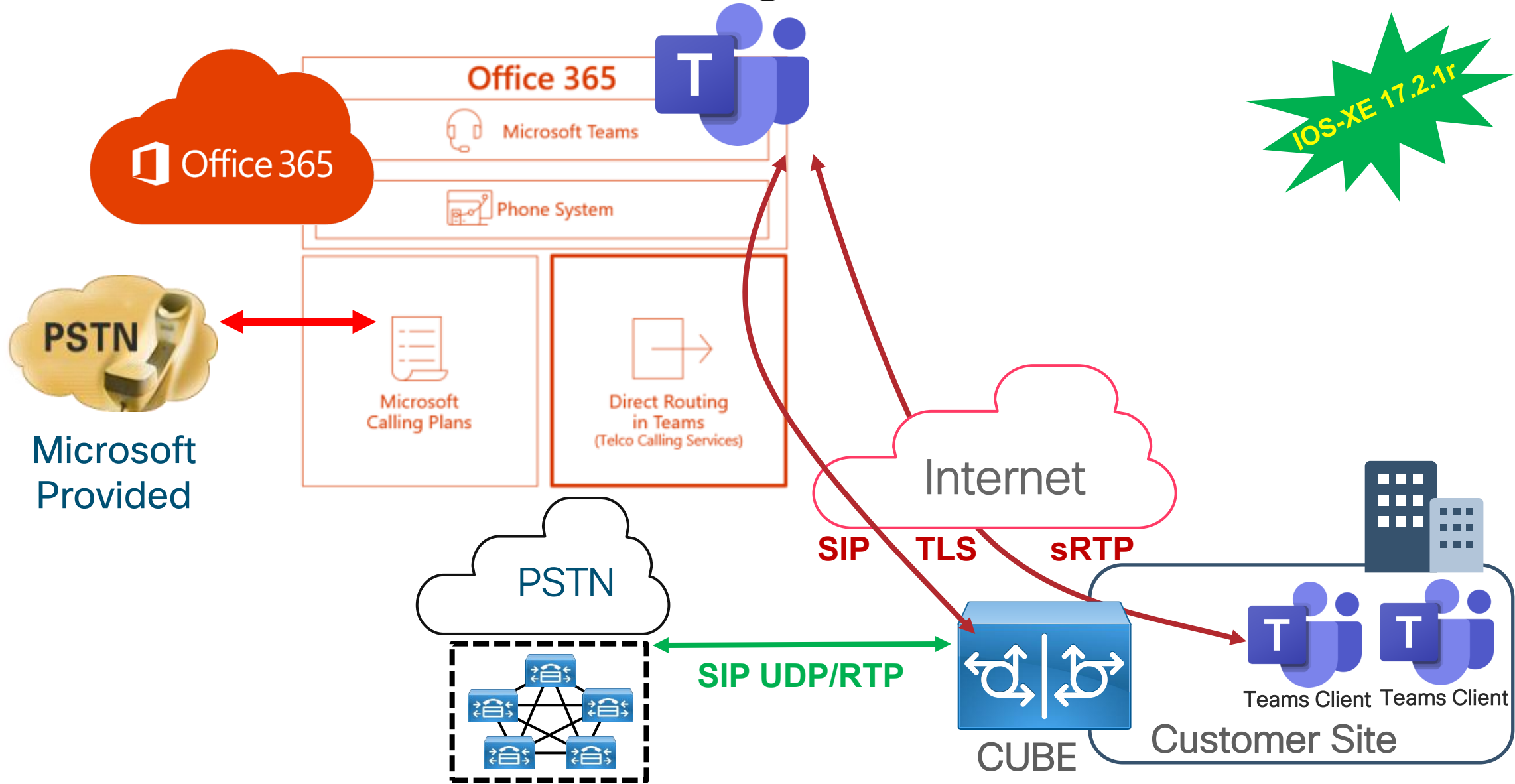
CUBE Software Release Mapping

CUBE Version	Initial IOS-XE Release for this CUBE version and Release date		Subsequent IOS-XE Release for this CUBE version
14.0	17.3.1a	July 2020	17.3.6
14.1	17.3.2*	Oct 2020	17.3.6
14.2	17.4.1a	Nov 2020	17.4.2
14.3	17.5.1	March 2021	17.5.1a
14.4	17.6.1a	July 2021	17.6.6a
14.4	17.7.1a	Nov 2021	17.7.2
14.5	17.8.1a	March 2022	
14.6	17.9.1a	July 2022	17.9.4a
14.6	17.10.1a	Nov 2022	
14.6	17.11.1a	March 2023	
14.7	17.12.1a	July 2023	
14.8	17.13.1a	Nov 2023	

Last release for
ISR4K except
ISR4461

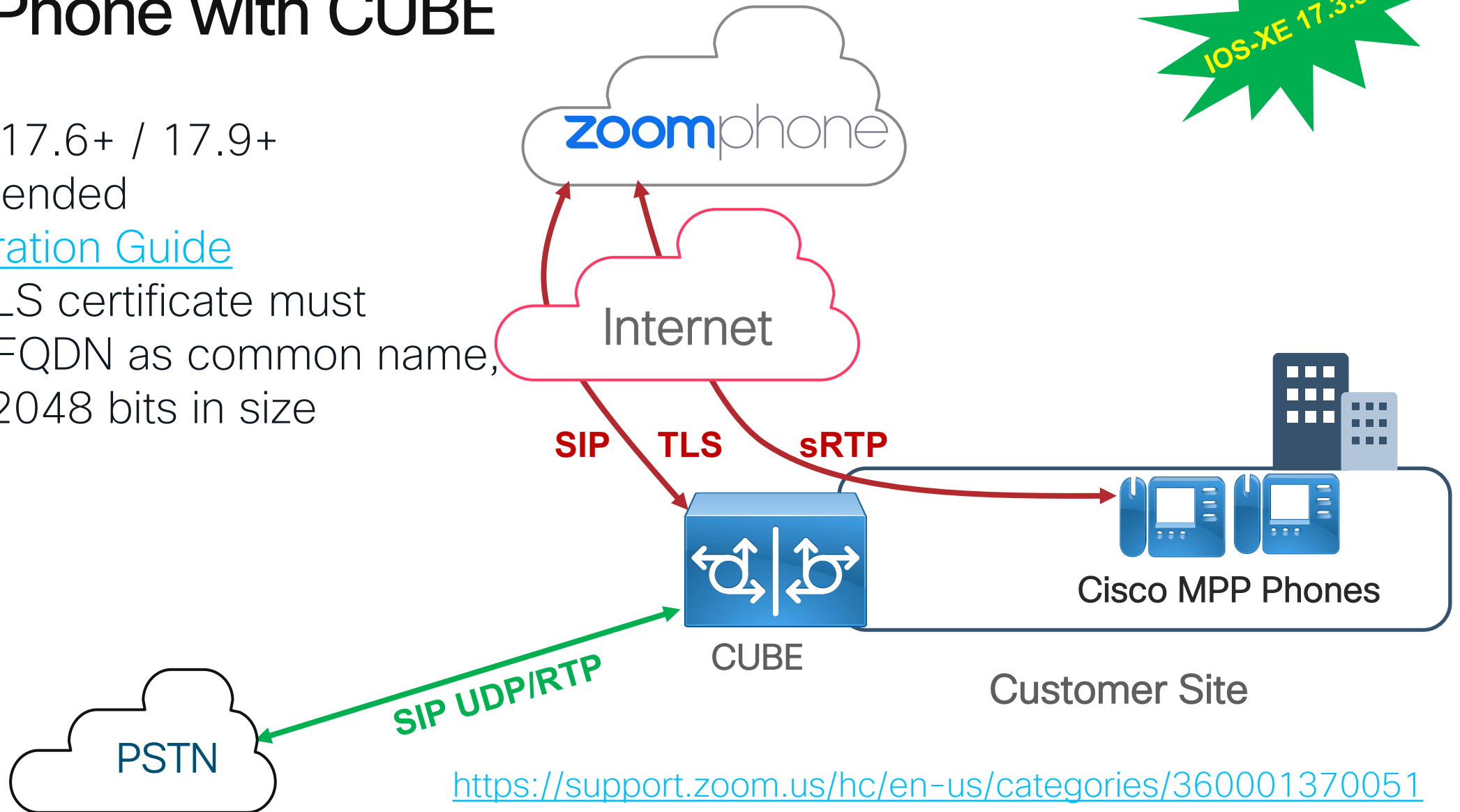
Enabling third-party Cloud Calling with CUBE

Microsoft Teams Direct Routing – Solution Overview



Zoom Phone with CUBE

- IOS-XE 17.6+ / 17.9+ recommended
- [Configuration Guide](#)
- CUBE TLS certificate must contain FQDN as common name, and be 2048 bits in size

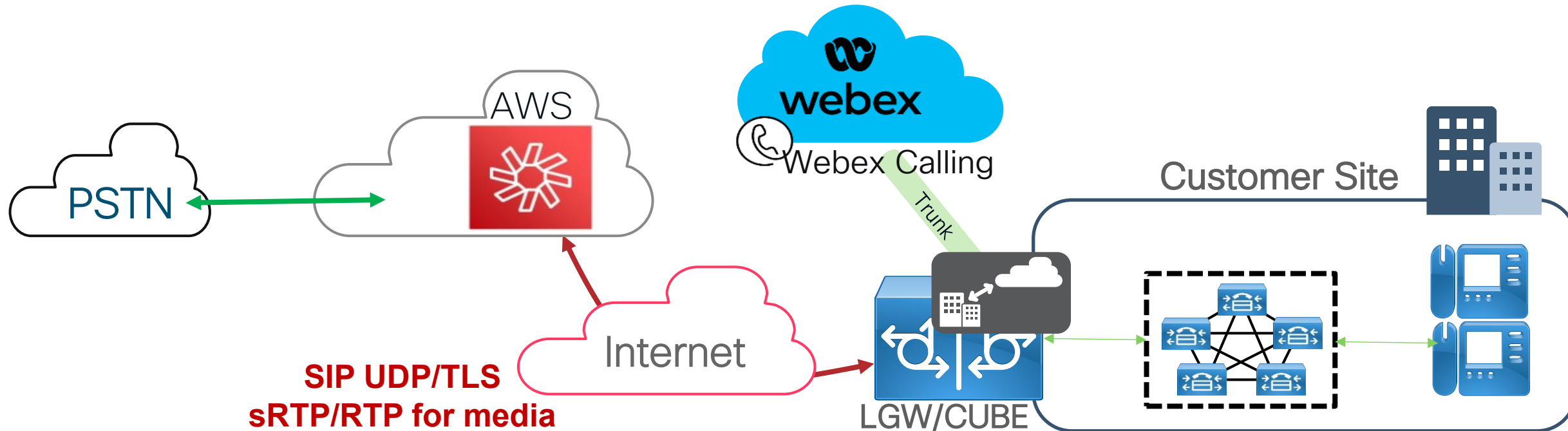


<https://support.zoom.us/hc/en-us/categories/360001370051>

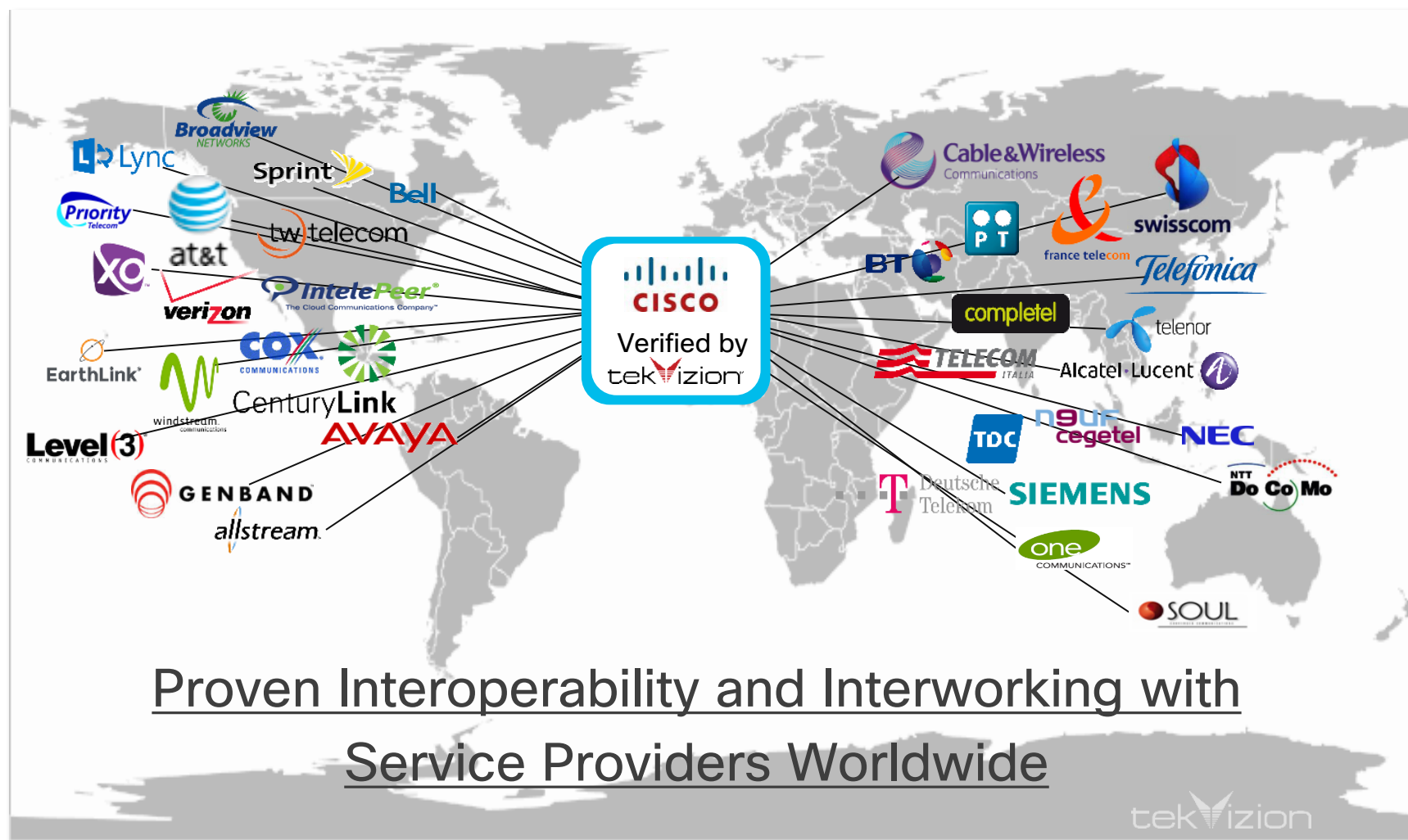
Amazon Chime Voice Connector



- Amazon Chime provides PSTN service to an on-premises or a cloud-based phone system connected via a CUBE
- CUBE can be on-premises or hosted in AWS ([vCUBE in AWS](#))
- IOS-XE 17.9+ recommended, but other IOS-XE releases also supported
- [Configuration Guide](#)
- https://aws.amazon.com/chime/chime-sdk/resources/#Configuration_Guides



CUBE Interoperability Portal for application notes

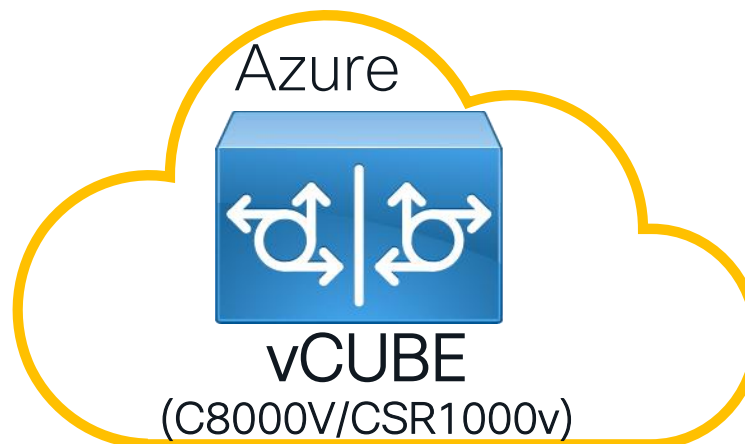


Proven Interoperability and Interworking with
Service Providers Worldwide

- Zoom Phone
- Microsoft Direct Routing
- Amazon Chime Voice Connector
- vCUBE-on-AWS
- vCUBE-on-Azure

Introducing vCUBE on Azure

Introducing vCUBE on Azure



- Platform validated and tested –D2_v2 [2 vCPUs, 7 GB memory]
- IOS-XE versions tested – 17.3.4a (Later IOS-XE releases also supported)
- BYOL – Instance billed only for hosting by Azure and licensing managed through Cisco Smart Licensing
 - CSR1000v required licenses – AX/APPX (No TLS/sRTP), Throughput, and CUBE Session Licenses
 - C8000V required licenses – DNA Network Essentials or above and CUBE Session Licenses
- vCUBE in Azure session counts same as CSR1000v – 2 vCPU

Managing Gateways from the Webex Control Hub

Introducing Gateway Connectors

- Gateway connectors are small applications that run in the gateway Guest Shell to maintain a connection to Control Hub, co-ordinate events and collect status information.
- Guest Shell is independent of IOS-XE running on the platform
- NETCONF and YANG data models are used as opposed to the Command Line (CLI) to manage the gateways, thus, allowing APIs to manage and configure the gateways
- Two types of connectors exist
 - Management Connector - takes care of gateway enrollment to the cloud and lifecycle management of the telemetry connector
 - Telemetry Connector - used for pushing configs and getting command requests from the CH to the gateway

Connector Considerations

- ISR 1100 series are not supported
- CUBE High Availability (HA) mode is not supported
- Controller or SD-WAN mode is not supported (only IOS-XE Autonomous mode is supported)
- Currently two services are supported:
 - Registration-based Local Gateway Configuration Validation
 - Survivability Gateway Configuration template (BRKCOL-2993)
- IOS-XE version required:
 - Local Gateways—Cisco IOS XE 17.3.4a or later
 - Survivability Gateways—Cisco IOS XE 17.9.3 or later

Connector Installation Prerequisites

The platform must have the following configured:

- DNS Server `ip name-server <IP Address>`

- HTTP Proxy

```
ip http client proxy-server <server address> proxy-port <port-number>
```

```
ip http client username <username>
```

```
ip http client password <password>
```

- Default AAA lists to authenticate and authorize NETCONF access

```
aaa new-model
```

```
aaa authentication login default radius local
```

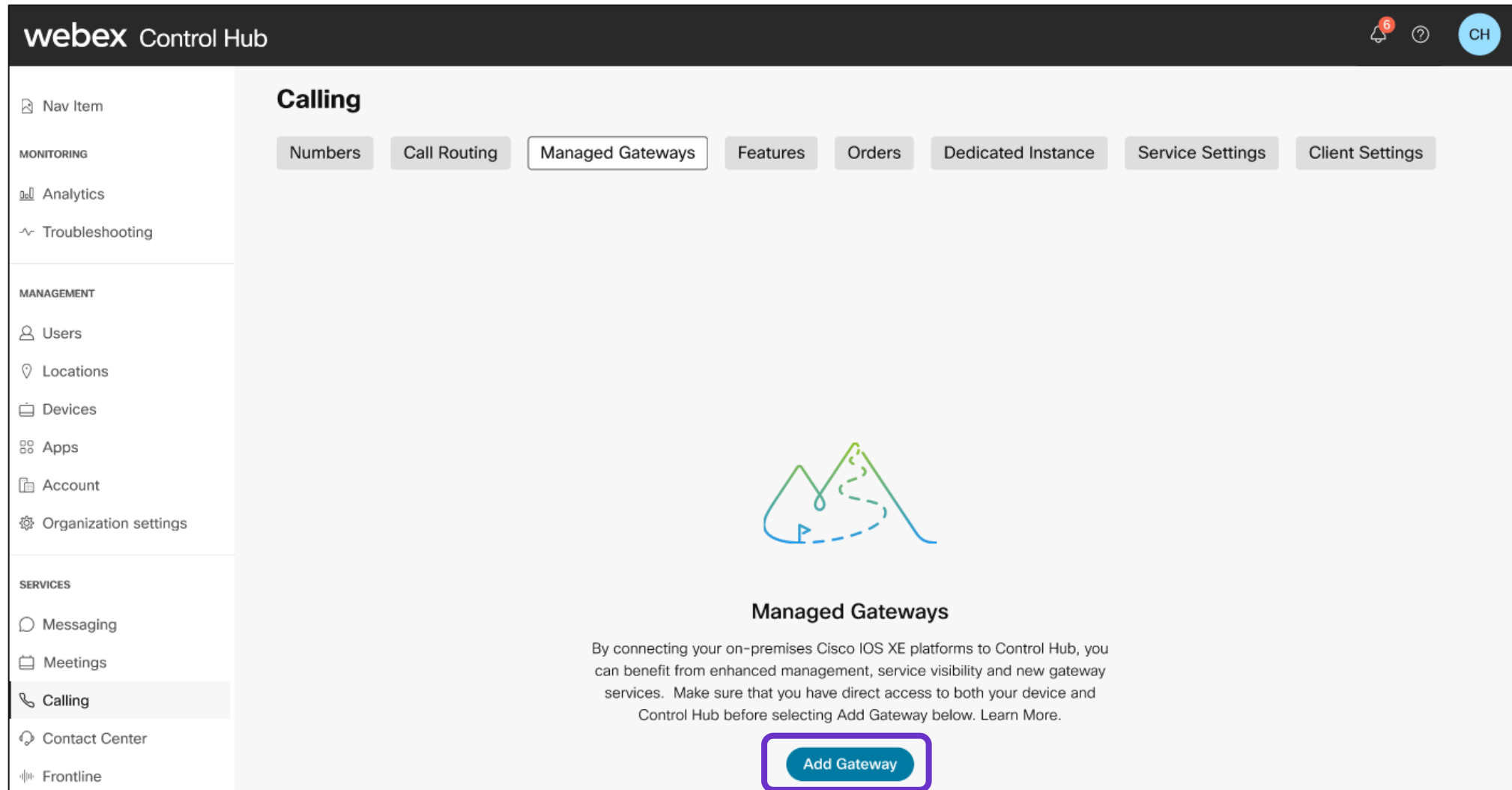
```
aaa authorization exec default radius local if-authenticated
```

```
username test privilege 15 secret <password>
```

- Import the IOS-XE public CA bundle

```
crypto pki trustpool import url http://www.cisco.com/security/pki/trs/ios.p7b
```

Under **Services**, click **Calling** and then click the **Managed Gateways Tab**



The screenshot displays the Webex Control Hub interface. The top navigation bar includes the 'webex Control Hub' logo, a notification bell with a red '6', a help icon, and a user profile icon labeled 'CH'. The left sidebar is organized into three sections: 'MONITORING' with links for 'Nav Item', 'Analytics', and 'Troubleshooting'; 'MANAGEMENT' with links for 'Users', 'Locations', 'Devices', 'Apps', 'Account', and 'Organization settings'; and 'SERVICES' with links for 'Messaging', 'Meetings', 'Calling' (which is highlighted), 'Contact Center', and 'Frontline'. The main content area is titled 'Calling' and features a horizontal tab bar with the following tabs: 'Numbers', 'Call Routing', 'Managed Gateways' (which is selected and highlighted with a white border), 'Features', 'Orders', 'Dedicated Instance', 'Service Settings', and 'Client Settings'. Below the tabs, there is a large graphic of a mountain range with a blue dashed line and an arrow indicating a path. Underneath the graphic, the heading 'Managed Gateways' is followed by a paragraph: 'By connecting your on-premises Cisco IOS XE platforms to Control Hub, you can benefit from enhanced management, service visibility and new gateway services. Make sure that you have direct access to both your device and Control Hub before selecting Add Gateway below. Learn More.' At the bottom of this section, there is a blue button labeled 'Add Gateway' which is highlighted with a purple rectangular border.

In the **Add a Managed Gateway** window, copy the command to install the connector onto the Gateway

Add a Managed Gateway

Before you can add a gateway to Control Hub, you will need to install a connector application on your device. Access the device command line interface and paste the following command in full to start the installation. Once the connector is installed, confirm by checking the box below, then click Next. [Learn more](#)

```
tclsh https://binaries.webex.com/ManagedGatewayScriptProdStable/gateway_onboarding.tcl
```

☐ I have installed the management connector on the gateway.

CancelNext

Run the Management Connector deployment Script

- Run the TCL script
 - tclsh
https://binaries.webex.com/ManagedGatewayScriptProdStable/gateway_onboarding.tcl
- Follow the wizard

```
C8KV-Hussain#  
C8KV-Hussain#$m/ManagedGatewayScriptProdStable/gateway_onboarding.tcl  
Loading https://binaries.webex.com/ManagedGatewayScriptProdStable/gateway_onboarding.tcl !!!  
Cisco IOS XE Software Version: 17.9.20221213  
Script Version: 3.0.3  
Precondition check status: Passed  
Downloading Gateway connector installer package...
```

Select the External Interface to reach Webex Cloud

Webex Gateway Connector Installation

Choose the external-interface from the below list of available interfaces:

Number	Interface	IP-Address	Status
1	GigabitEthernet1	10.52.12.203	up

Enter a number to choose the external interface: 1

Confirm or Edit DNS and Proxy settings

```
These DNS settings were detected in the gateway configuration:  
144.254.71.184 173.38.200.100
```

```
Do you want to use these settings for the connector? [Y/n]: Y
```

```
These proxy settings were detected in the gateway configuration:
```

```
Proxy Server : proxy.esl.cisco.com
```

```
Proxy Port   : 80
```

```
Do you want to use these settings for the connector? [Y/n]: Y
```

Specify the Connector IP Address and Credentials

```
Enter Connector IP address: 10.52.12.216
```

```
Enter Gateway username: hussain
```

```
Enter Gateway password: ****
```

```
Confirm Gateway password: ****
```

```
Enabling guestshell...this may take upto 4 minutes, please wait for completion.
```

Connector Successfully Installed

```
=====
Webex Managed Gateway Connector
=====

*** Cloud connector is installed successfully. ***
-----

*** Interface Status ***
-----

Interface                IP-Address      Status
-----
GigabitEthernet1         10.52.12.203    up
VirtualPortGroup0        10.52.12.203    up
Connector                 10.52.12.216    up
-----

*** App Status ***
-----

Service                  Status
-----
Guestshell                RUNNING
Management Connector      RUNNING
-----
```

Webex Managed Gateway Connector Options

```
=====
Webex Managed Gateway Connector
=====
```

```
Options
```

```
s : Display Status Page
v : View and Modify Cloud Connector Settings
e : Enable Guestshell
d : Disable Guestshell
l : Collect Logs
r : Clear Logs
u : Uninstall Connector
q : Quit
```

```
=====
Select an option from the menu: █
```

In the **Add a Managed Gateway** window, check the **I have installed the management connector on the gateway** check box and click **Next**.

Add a Managed Gateway

Before you can add a gateway to Control Hub, you will need to install a connector application on your device. Access the device command line interface and paste the following command in full to start the installation. Once the connector is installed, confirm by checking the box below, then click Next. [Learn more](#)

`tcsh https://binaries.webex.com/ManagedGatewayScriptProdStable/gateway_onboarding.tcl` 

☒ I have installed the management connector on the gateway.

CancelNext

At the **Add a Managed Gateway** screen, enter the connector IP address that you entered during the connector installation procedure, and a preferred display name for the gateway

Add a Managed Gateway

Enter the following details for your installed connector. Click Next to open the connector web interface where you can complete device enrollment.

Enter the connector IP address

You will need to be able to reach this address directly from your browser.

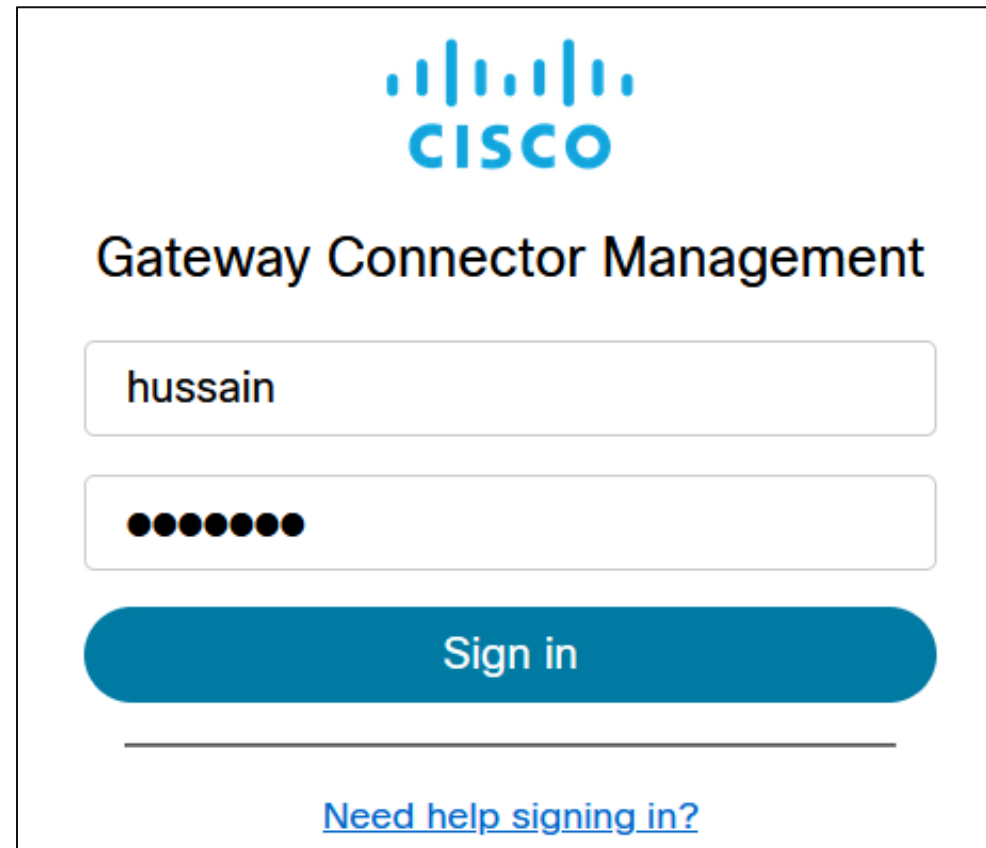
Enter a display name for the gateway

The name is for display purposes only.

Once enrollment is complete, gateways will appear in the Managed Gateway list.

CancelNext ↗

At the Connector Management page, enter the Gateway Admin **Username** and **Password** that you specified during the connector installation procedure





Gateway Connector Management

hussain

●●●●●●●

Sign in

[Need help signing in?](#)

Click the **Enroll Now** button within an hour

Cisco Webex
Gateway Connector Management

Signed in as hussain

Sign out

Enroll Gateway

To complete the enrollment process, a secure connection must be established from this connector to the Cisco Webex cloud.

Use your Webex Calling administrator credentials to authenticate the connection on the next screen.

Enroll Now

Check the **Allow Access to the Gateway Management Connector** check box

Gateway Management Connector

Allow Access to Gateway Management Connector

Permissions are required to allow your Cisco Webex organization to create, read, update, and delete user accounts, as well as read and update information about your organization.

Organization
WxCSA Team Sandbox

FQDN or IP Address
10.52.12.216

☐ **Allow Access to the Gateway Management Connector**
Only allow access to hosts you know and trust

Continue

Enrollment Successful

Gateway Management Connector

Registration Confirmed

You will now be redirected to 10.52.12.216.

Enrollment successful.

You can close this window and proceed to Webex Control Hub to view and associate this gateway with a service.

Managed Gateways

Calling

NumbersLocationsCall RoutingManaged GatewaysFeaturesPSTNService SettingsClient Settings

All Gateways ▾10 Gateway(s)Events HistoryAdd Gateway

Gateway Name	Version	Connector Sta...	Service	Assigned to	Actions
Amsterdam SGW	17.9.3	● Online	Survivability Gateway	Location: Amsterdam Office	...
Hussain-Cat8kv		-	-	-	...
Lisbon SGW	17.9.3	● Online	Survivability Gateway	Location: Lisbon Office	...
London SGW	17.9.3	● Offline	Survivability Gateway	Location: London Branch Office	...
Madrid SGW	17.9.3	● Online	Survivability Gateway	Location: Madrid Office	...
Munich SGW	17.9.3	● Online	Survivability Gateway	Location: Munich Office	...

Managed Gateway now Online

Calling

Numbers

Locations

Call Routing

Managed Gateways

Features

PSTN

Service Settings

Client Settings

Q

Search

All Gateways

▼

10 Gateway(s)

Events History

Add Gateway

Gateway Name	Version	Connector Sta...	Service	Assigned to	Actions
Amsterdam SGW	17.9.3	● Online	Survivability Gateway	Location: Amsterdam Office	...
Hussain-Cat8kv	17.9.20221...	● Online	-	-	...
Lisbon SGW	17.9.3	● Online	Survivability Gateway	Location: Lisbon Office	...
London SGW	17.9.3	● Offline	Survivability Gateway	Location: London Branch Office	...
Madrid SGW	17.9.3	● Online	Survivability Gateway	Location: Madrid Office	...
Munich SGW	17.9.3	● Online	Survivability Gateway	Location: Munich Office	...
Paris SGW	17.9.3	● Online	Survivability Gateway	Location: Paris Office	...

Assign a Service to the Managed Gateway

< Managed Gateways

Hussain-Cat8kv

● Connector Online • Version 17.9.20221213

Actions ▾



Assign Service

Assign the Webex Calling service that you will be using your gateway for.

Assign Service

Select a Service Type

×

Assign Service to Hussain-Cat8kv

Select the Webex Calling service that you will be using your gateway for.

Select service type

▼

Cancel

Assign

Service Type: LGW or SGW

×

Assign Service to Hussain-Cat8kv

Select the Webex Calling service that you will be using your gateway for.

Select service type

Local Gateway

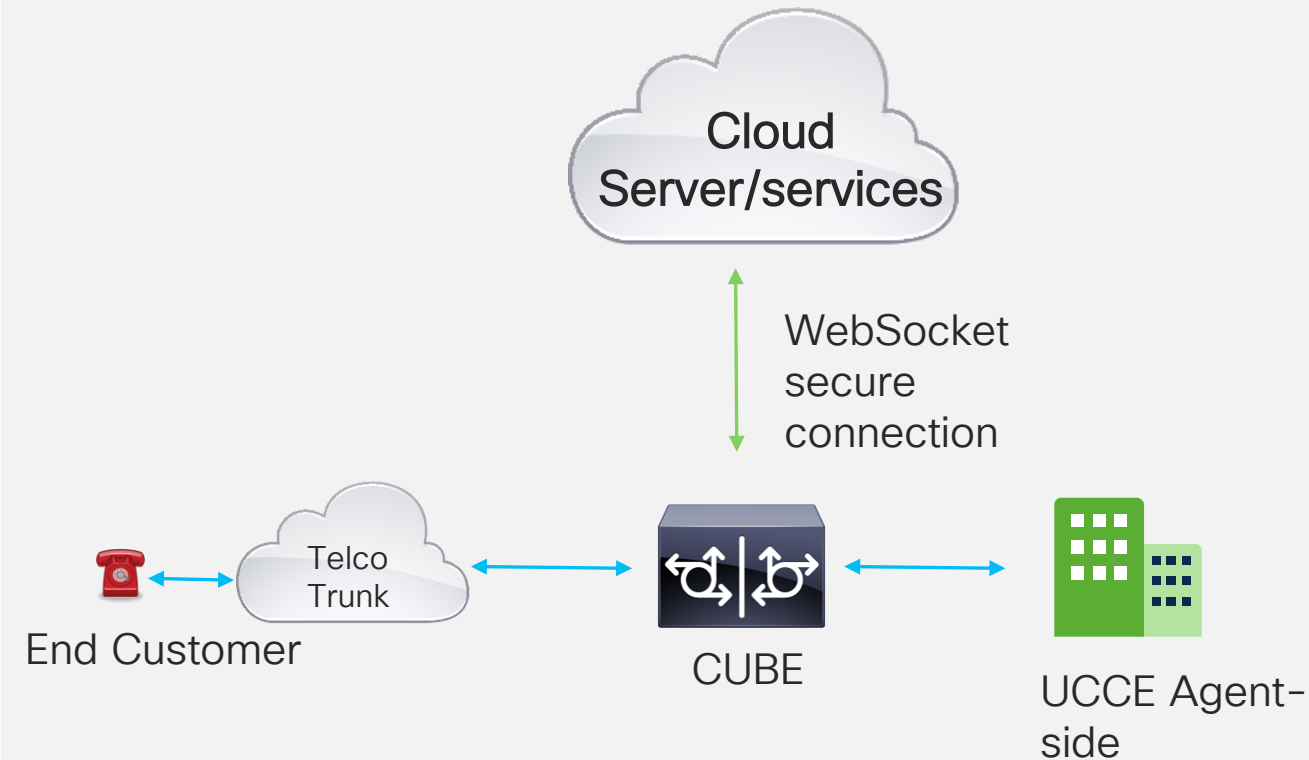
Survivability Gateway

Cancel

Assign

WebSocket-Based Media Forking

Secure media forking for cloud speech services



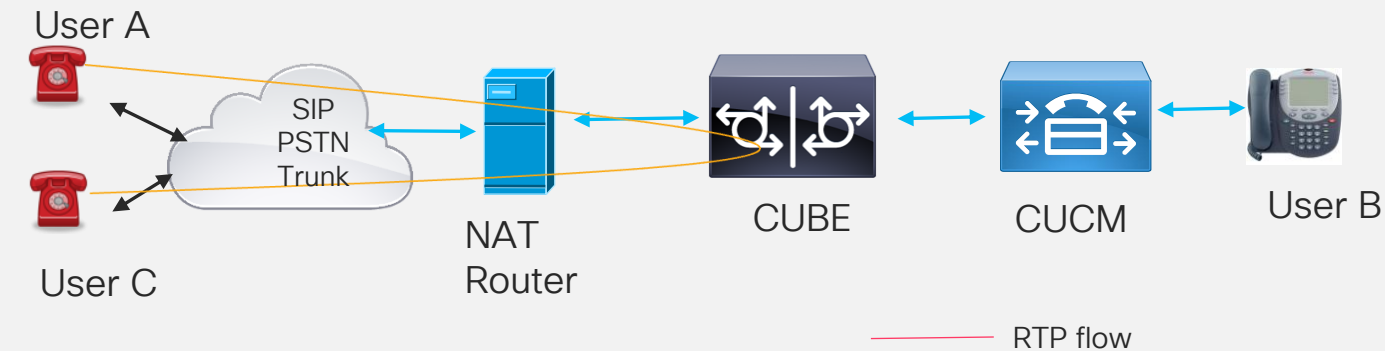
The Feature: CUBE supports GCM Cipher when using WebSocket to fork media in a Cisco Unified Contact Center Enterprise (UCCE) solution deployment with Cloud Speech Service

When the feature is being released: Available with IOS XE 17.12.1a, August 2023.

The value of the feature: More secure TLS connection to Contact Center deployments for Cloud Speech Services.

NAT Traversal using RTP keepalives

NAT Traversal using RTP keepalives



- **The Feature:** Cisco voice solution including CUBE and UCM can be deployed behind a NAT router.
- When a PSTN user A calls an enterprise user B who has setup call forward to an external PSTN user C, the call between A & C will get established, but audio will not flow through.
- A configurable option is provided in CUBE to periodically send the payload-free RTP keepalive packets to keep the pinholes open for media to flow through.

When the feature is being released: Available with IOS XE 17.13.1a, targeted Nov 2023.

- **The value of the feature:** Enhanced calling experience that allows PSTN user A to talk to user C when call is forwarded from enterprise user B.
- Provides an additional option beyond STUN based mechanisms.

References

- [CUBE Performance and Sizing](#)
- [Cisco Unified Border Element Configuration Guide Through Cisco IOS XE 17.5](#)
- [CUBE Config Guide – IOS-XE 17.6 Onwards](#)
- [CUBE Roadmap](#)
- [IOS-UC Sharepoint](#)
- CUBE Support
 - CUBE partner/customer questions: ask-cube@external.cisco.com

MEETINGS

Cisco Meeting Server (CMS/CMM)

Suarabh Khaneja – Product Manager

Hardware Refresh

Overview – CMS1K processor refresh M5v2 to M6



CMS 1000 M5v2
(UCS C220 M5v2)



CMS 1000 M6
(UCS C220 M6)

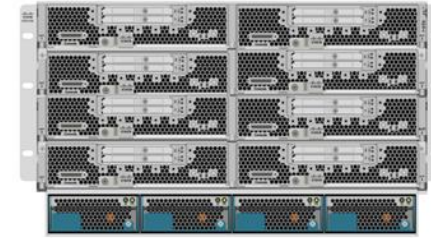
Type of Calls	M5v2	M6
Full HD calls 1080p60 video 720p30 content	30	40
Full HD calls 1080p30 video 1080p30/4K7 content	30	40
Full HD calls 1080p30 video 720p30 content	60	80
HD calls 720p30 video 720p5 content	120	160
SD calls 480p30 video 720p5 content	240	320
Audio calls (G.722)	2200	3000

Overview – CMS2K processor refresh

M5 “Cascade lake” to M6 “Ice lake”



CMS 2K M5v2
(UCS B200 M5v2)



CMS 2K M6
(UCS B200 M6)

Type of Calls	M5v2	M6
Full HD calls 1080p60 video 720p30 content	218	324
Full HD calls 1080p30 video 1080p30/4K7 content	218	324
Full HD calls 1080p30 video 720p30 content	437	648
HD calls 720p30 video 720p5 content	875	1296
SD calls 480p30 video 720p5 content	1250	1875
Audio calls (G.722)	3000	3200

Release: 3.8

August 2023 (GA)



Solution Roadmap (Release 3.8)



Meeting Experience (CMS)



Improving Quality

- Screen Share Quality Improvements

Engaging Experience

- Presence update on Jabber (GA)
- Ability to take Surveys
- Directed chat during meeting
- Czech language support
- Admin control screen sharing resolution

Meeting Management (CMM)



Meeting Control

- Restrict Meeting View for Operators

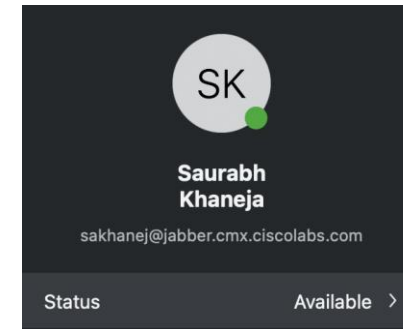
Serviceability

- Smart Licensing Updates

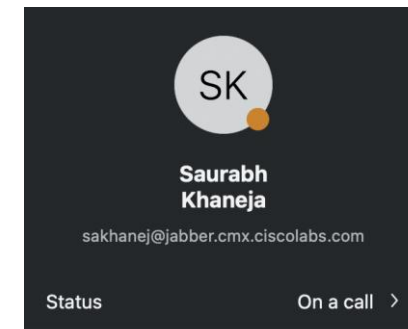
CMS:

Jabber Presence set when in meeting

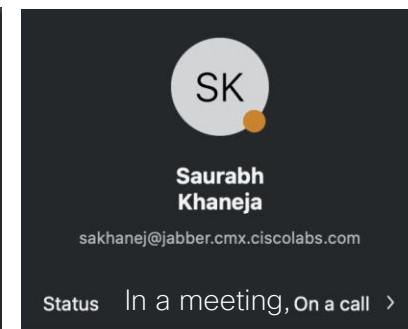
- Integration between Jabber IM&P and CMS to update user presence when in CMS web app meeting
- CMS User id should match Directory URI field in CUCM
- Jabber JID could be CUCM User id or Directory URI
- Implementation Enhancements between IMP and CMS, with TLS support
- Supports only 1 CUCM/IMP cluster
- Presence status set to *`On a call`*
- Jabber reverts to old status post CMS meeting.
- Supported with CUCM/IMP 14.0



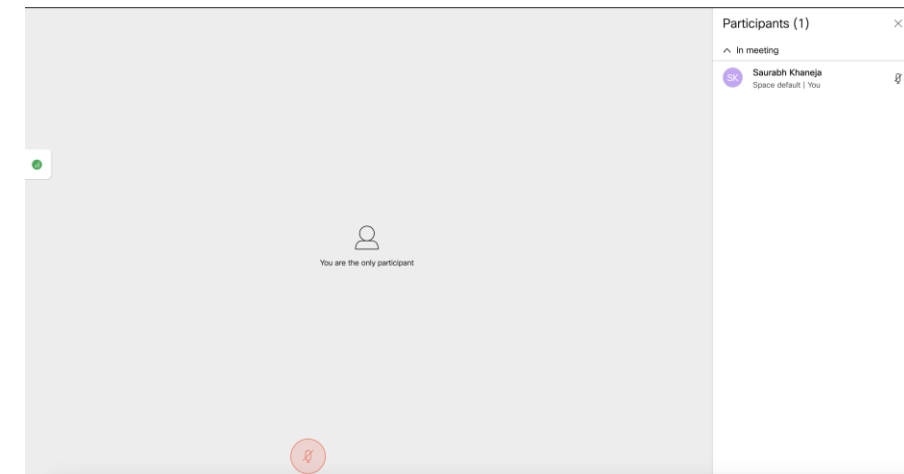
Pre meeting



In CMS meeting
(Non-Scheduled)



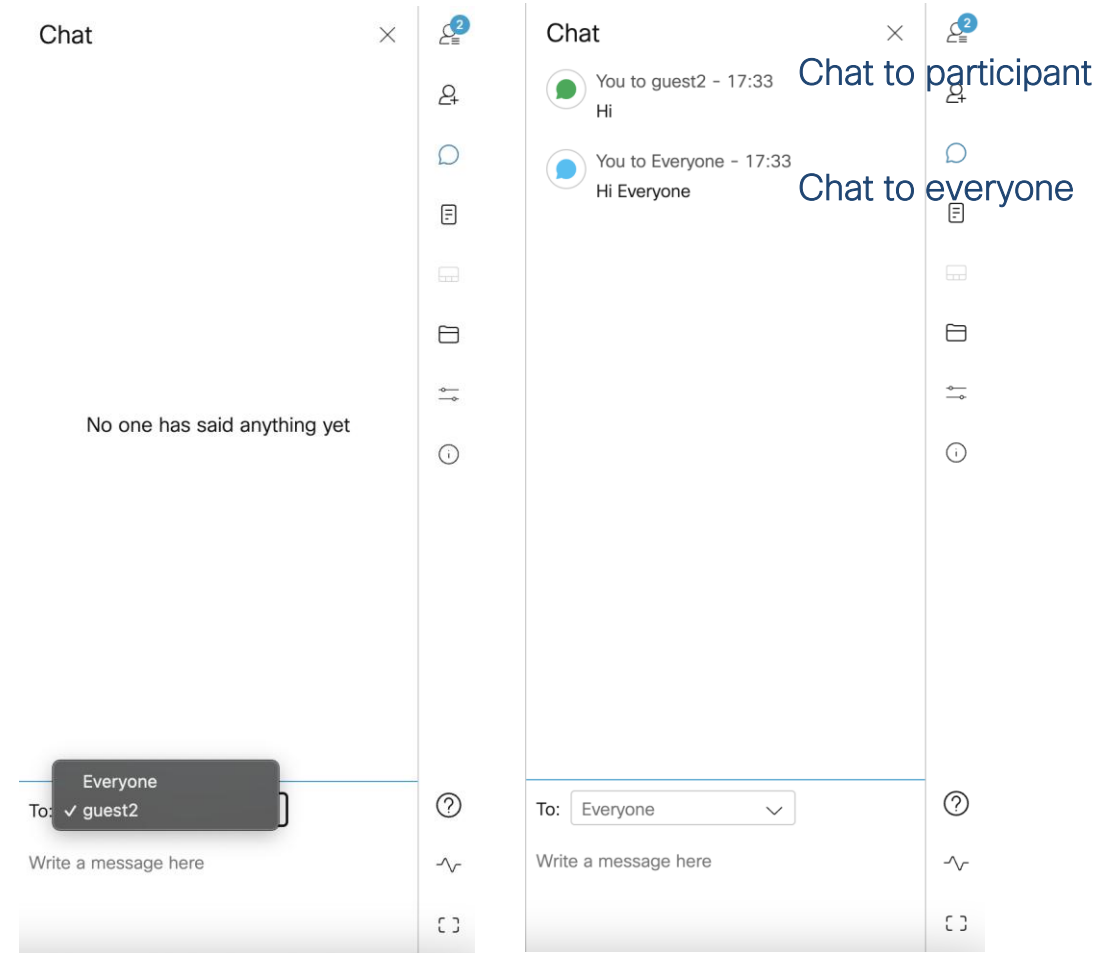
In Scheduled
CMS meeting



CMS web app:

Directed Chat

- In previous versions of CMS, chat was always sent to Everyone in the meeting
- Release 3.8 provides an ability to select a participant to send chat
- Only CMS web app participants are shown in the chat dropdown list
- No chat history is retained if participant drops and joins back again



Expected delivery dates are subject to change

CMS web app:

Ability to take survey

- Web app participants can create and launch surveys
- Multiple surveys can be created but only 1 can be active
- All participants will be notified when survey is available
- Feature introduces 2 config parameters at *CallProfile* (to enable the feature) and *CallLegProfile* (to assign Survey permission)
- Guest users can only participate in the survey
- Surveys are retained till the call is active. Participants disconnecting does not remove their surveys
- Needs `meetingapps` deployment

The image displays two screenshots of the CMS web app's survey interface. The left screenshot shows a 'Test Survey' status 'Yet to Start' at 8/3/2023, 4:20:03 PM. The right screenshot shows the survey configuration screen with two questions: 'Whats your fav animal?' and 'Fav Cuisine?'. The first question has options 'Dog', 'Cat', 'Lion', and 'Deer'. The second question has options 'Japanese', 'Indian', and 'Chinese'. Both questions have 'Add option' and 'Add question' buttons. The interface includes a sidebar with icons for users, messages, documents, and settings, and a top bar with a logo and user profile.

CMS and web app:

Control Screen Share Resolution

- CMS not provides Admin settings to control resolution for screen sharing on web app
- This helps provide the best experience in a lossy network.
- Web app users can always over-ride this setting to get the best experience when sharing text heavy contents
- A restart of ongoing screen sharing is needed for the user selection to take effect

/api/v1/webBridgeProfiles

name	☐	
resourceArchive	☐	
allowPasscodes	☐	<unset>
allowSecrets	☐	<unset>
userPortalEnabled	☐	<unset>
allowUnauthenticatedGuests	☐	<unset>
resolveCoSpaceCallIds	☐	<unset>
resolveCoSpaceUris	☐	<unset>
contentResolution	☐	✓ <unset>
sessionLimit	☐	
sessionLimit	☐	

Admin Configured

Camera

FaceTime HD Camera (3A71:F4B5)

☐ Mirror self-view

☒ Enable HD

Microphone

Default - MacBook Pro Microphone (Built-in)

Sharing

☐ Optimise for Text

Virtual Background

None

Blur

Images

User Config

Expected delivery dates are subject to change



CMM:

Restrict Video Operators view

- Initially Video operators can see all ongoing meetings on CMS through CMM
- This features now controls that experience
- A new coSpace attribute “spaceTag” allows admins to assign a Tag to a space
- On CMM, Video operators can be assigned multiple Tags
- Video Operators with no tags → Can see all spaces with no tags
- Video Operators with tags → Can see only spaces with respective tags
- Tags cannot be assigned to Administrators → Can continue to view all meetings on CMM

/api/v1/coSpaces/7132c6d4-1cf9-4220-9453-ab12946746e0

userProvisionedCoSpace	☐	GUID (n
name	☐	newSpace
uri	☐	
secondaryUri	☐	
callId	☐	
cdrTag	☐	
spaceTag	☒	testSpace
passcode	☐	

Add tags

testS... × Enter tags

You can add up to 10 tags

Expected delivery dates are subject to change

Upcoming Release: 3.9

Feb 2024 (Target mid-Feb)

Solution Roadmap (Release 3.9)

Tentative



Meeting Experience (CMS web app)



Improving Quality

- AV1 Codec for Screen Sharing (*Beta*)

Engaging Experience

- CUCM Multi-cluster support for presence update
- Ability to select Audio Output device on web app
- Web app connection resiliency (*Beta*)
- User search improvements

Security

- Support for ECDSA Certificates

Meeting Management



Management

- Create CMM Spaces

ENDPOINTS

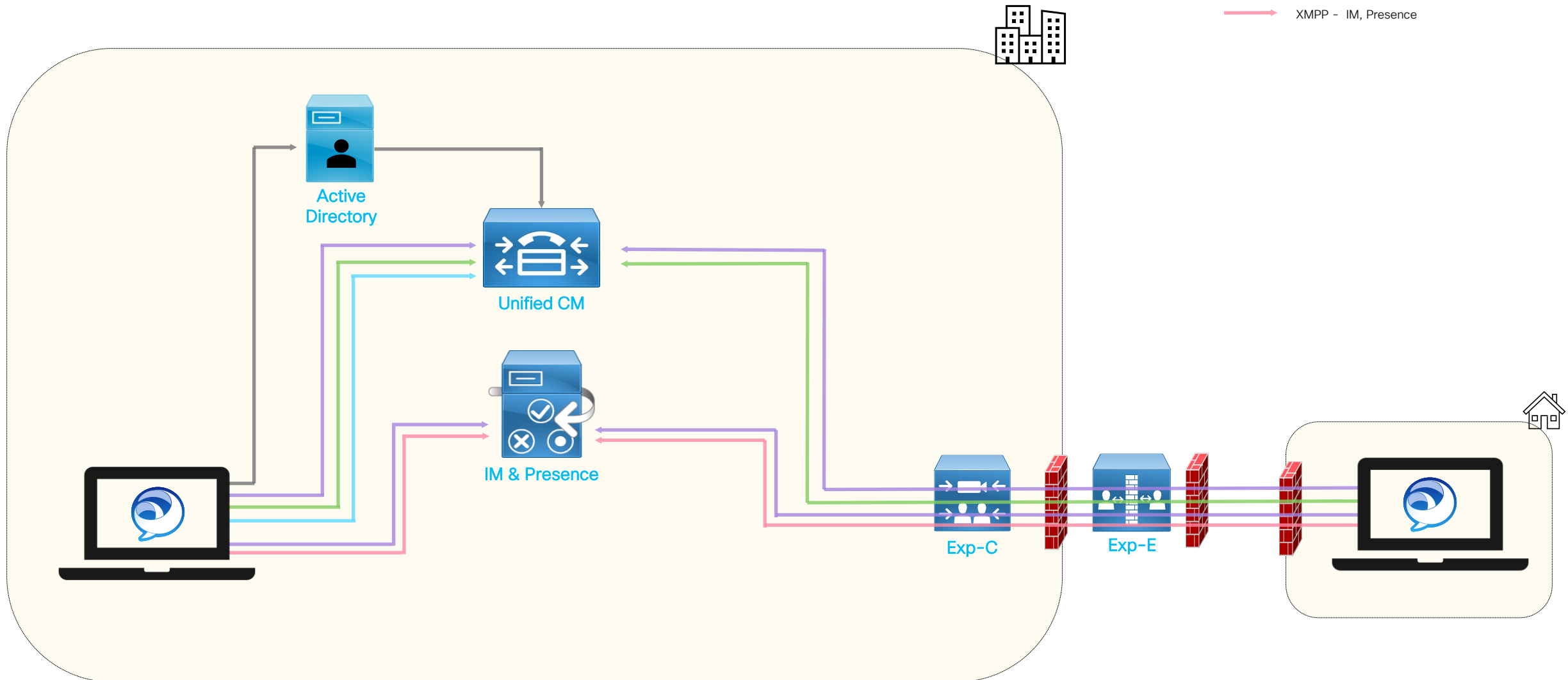
Jabber

Shane Long – Technical Marketing Engineer

Agenda

- Jabber Deployment Modes
- Recent Jabber Updates
- Jabber Roadmap
- Webex App with Unified CM Calling Architecture
- Webex App Unified CM Calling Features
- MS Teams Calling Integration

Jabber Architecture



Jabber Deployment Models



Full UC Mode

- Calling (softphone mode)
- Calling (deskphone mode)
- Contacts
- Presence
- Instant Messaging
- Requires IM & Presence



Phone Only Mode

- Calling (softphone mode)
- Calling (deskphone mode)
- IM & Presence not required



Phone Only with Contacts

- Calling (softphone mode)
- Calling (deskphone mode)
- Contacts
- Presence (optional)
- Requires IM & Presence



Windows



Mac
iOS



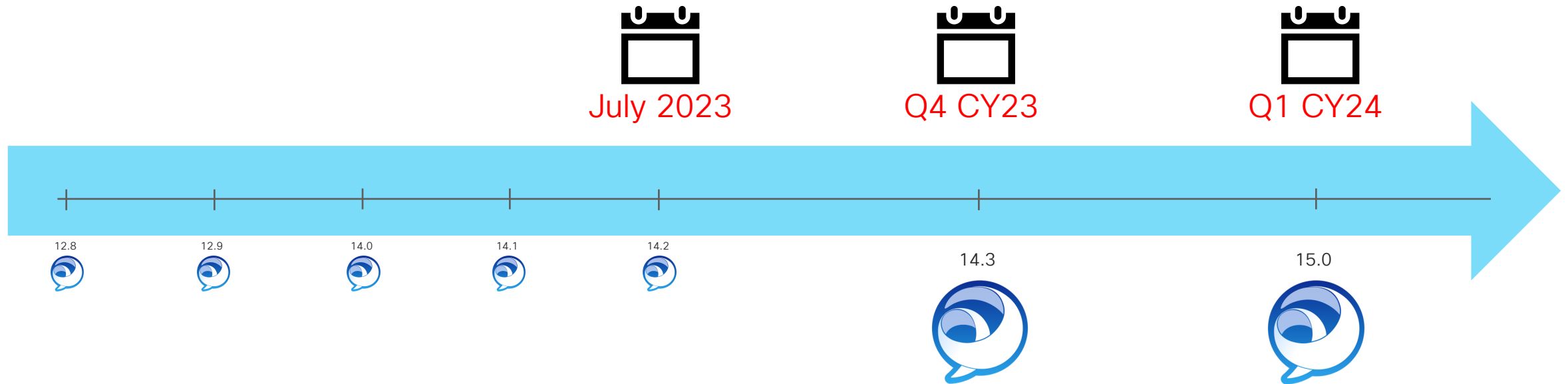
Android



Virtual Desktop

vmware®

Jabber Releases



Jabber 14.2 Release Updates

New Device Support

- Apple M1 chip native
- Samsung Xcover6 Pro
- Samsung Galaxy A53
- iPhone 15 / 15 Plus / 15 Pro / 15 Pro Max
- Apple local push notifications for iOS over Wi-Fi
- Admin settings to allow ignoring call using “x” button in call notification toast
- Android dual VPN support

New OS Support

- iOS 17 and iPadOS 17
- Android 13

Cisco Jabber priorities



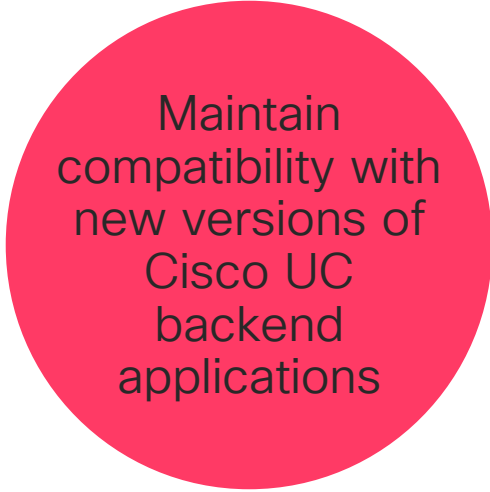
Provide critical security fixes



Deliver selective regulatory compliance features



Maintain compatibility with new versions of device hardware and OS



Maintain compatibility with new versions of Cisco UC backend applications

Cisco Jabber future release plan

Cisco Jabber 14.3 (Q4CY23)

- macOS Sonoma support
- Android 14 support
- Replace IE with Microsoft WebView2
- Outlook Calendar integration for CMS meeting
- Jabber for Android: Jabber presence status goes to "Away" when Jabber is in the background

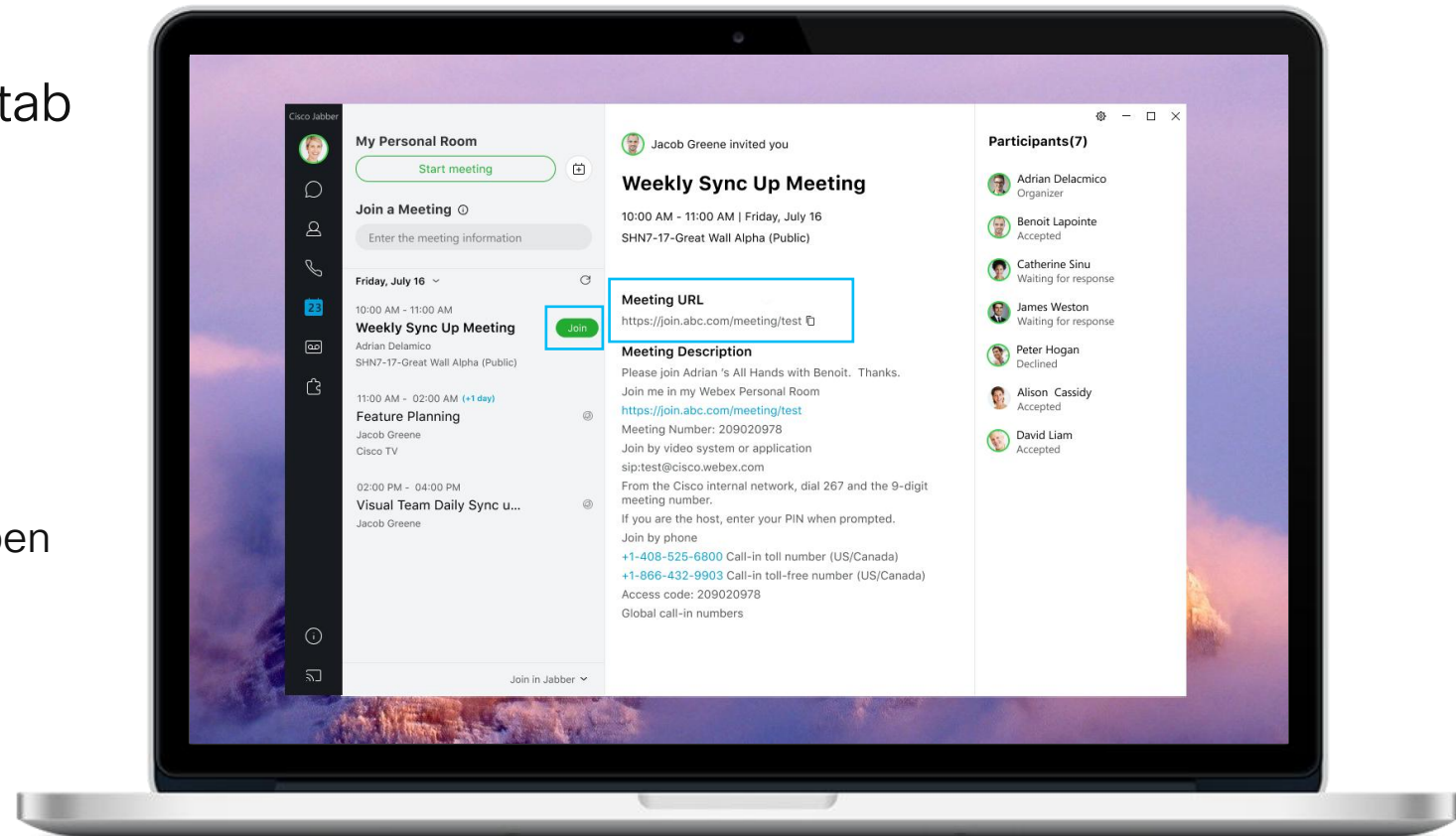
Cisco Jabber 15 (Q1CY24)

- Background blur for Windows (not virtual background)
- TLS 1.3 support
- Apple local push notifications over private LTE/5G network
- Enhancing security for Cisco Jabber Chrome extension

Outlook Calendar Integration for CMS Meetings

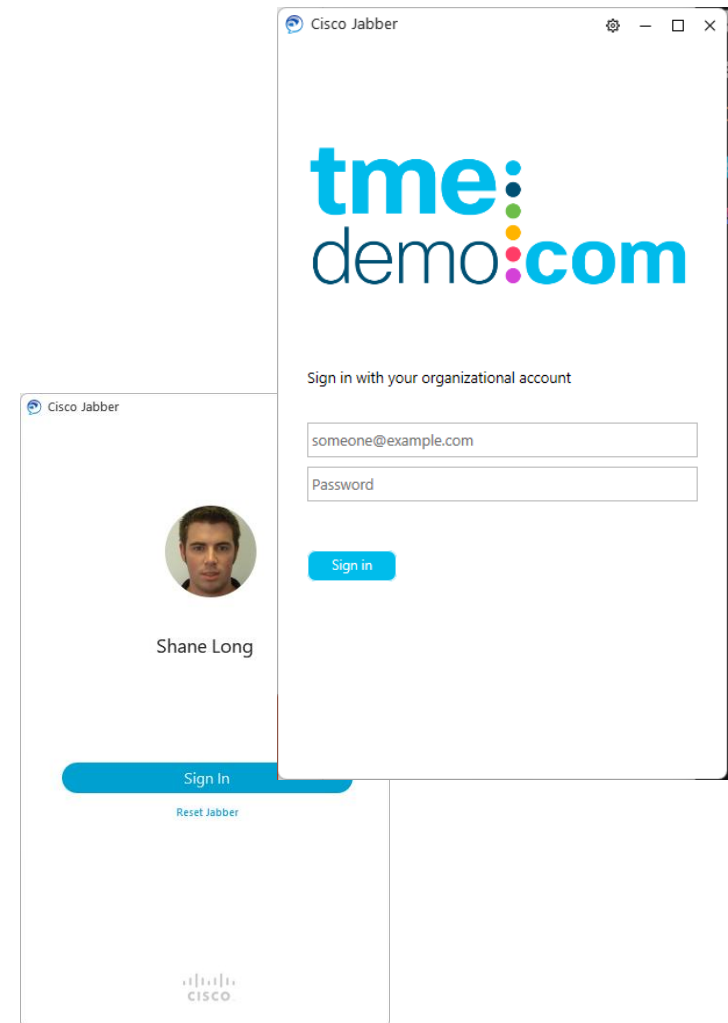
Allows for launching of the CMS Web App direct from the Jabber meetings tab or meeting notification

- To configure the organization CMS meeting domain:
<CMSMeetingDomains>cms.abc.com, join.abc.com
<CMSMeetingDomains>
- Click 'Join' button, launch **CMS Web app**, open the Meeting URL: <https://join.abc.com/meeting/test>



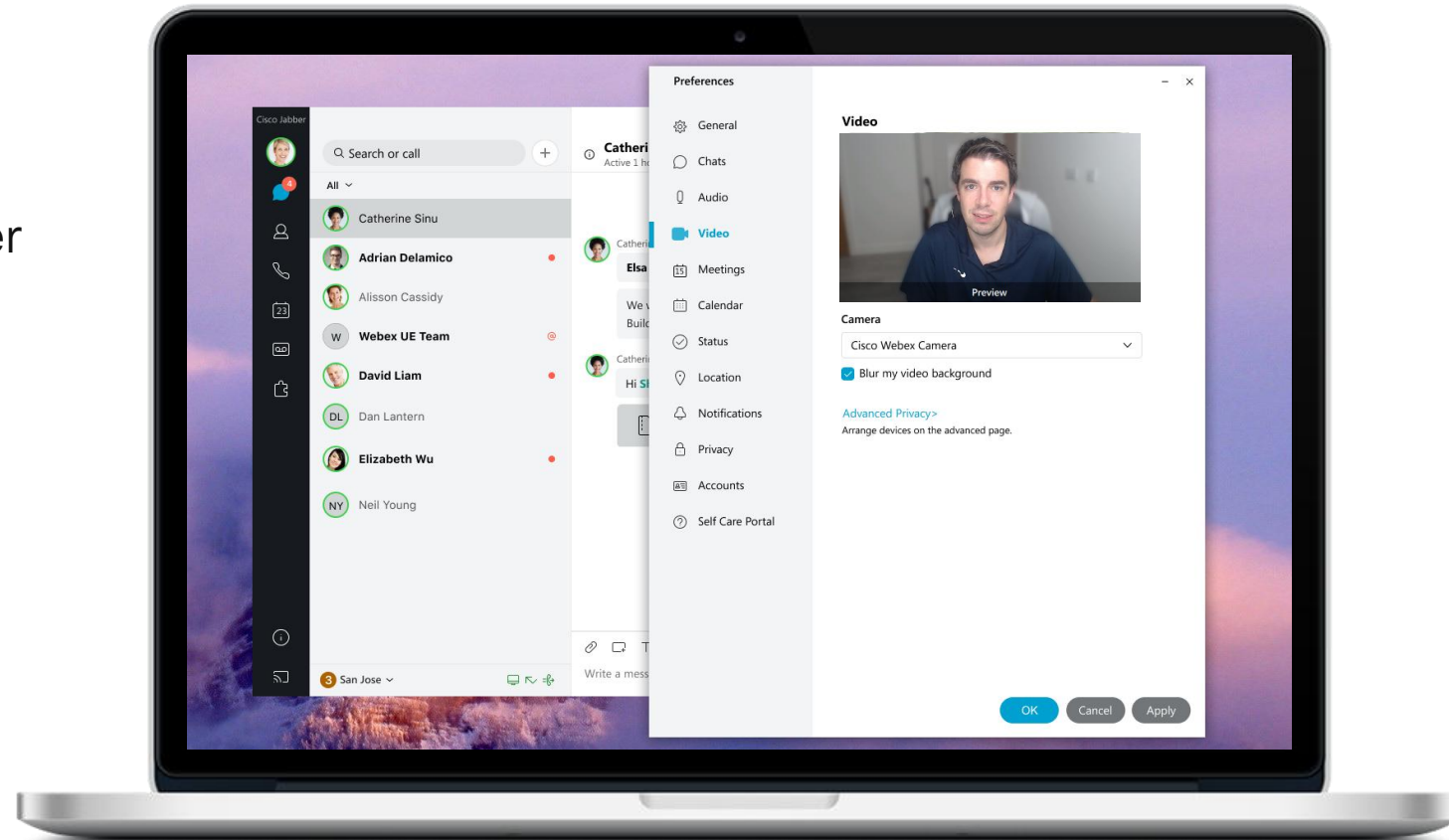
Replace IE with Webview2

- Jabber for Windows uses Internet Explorer as the engine for displaying web content during sign in and Chrome for displaying content in custom tabs
- Default web engine will be updating to Webview2 from 14.3
- This is controllable via a jabber-config key
- The following will be the default:
 <BrowserEngineForLogin>EDGE</BrowserEngineForLogin>
 <BrowserEngineForCustomTab>EDGE</BrowserEngineForCustomTab>

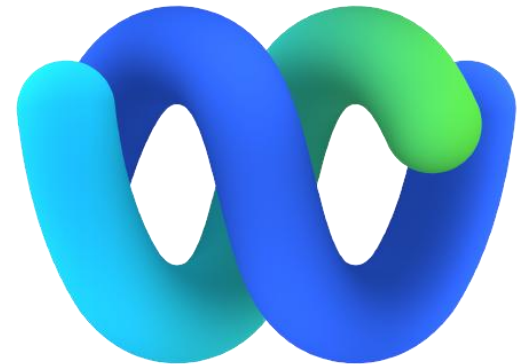


Blur Background

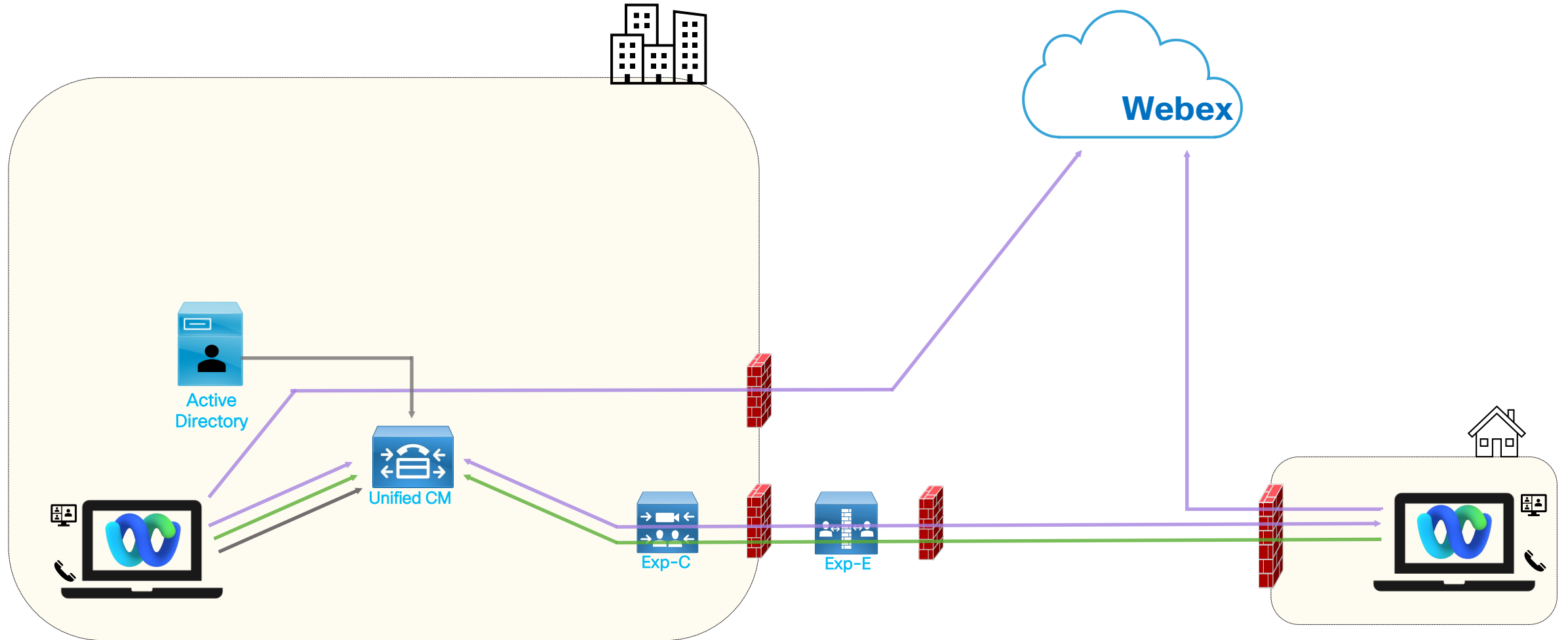
- Blur Background will be available on Jabber for Windows only
- Virtual Backgrounds are not planned



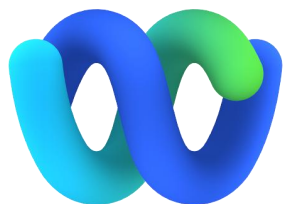
Are you ready to move from Jabber to Webex App?



Webex App Unified CM Calling Architecture

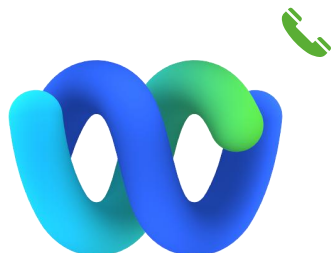


Deploy Webex to meet your needs



Webex Suite

- Calling (softphone mode)
- Calling (deskphone mode)
- Contacts
- Presence
- Messaging
- Meetings



Calling Only Mode

- Calling (softphone mode)
- Calling (deskphone mode)
- Presence
- Contacts



Windows



Mac
iOS



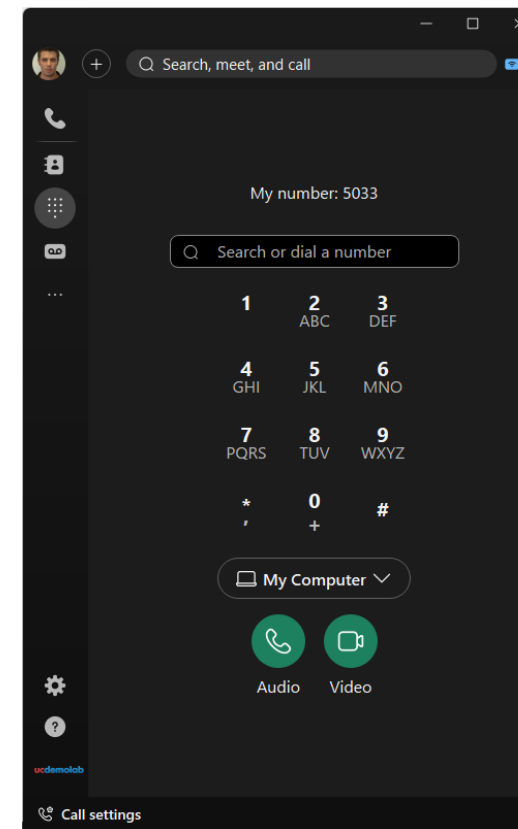
Android



Linux



vmware Virtual Desktop



Calling Only Mode



Webex App
reached feature
parity with Cisco
Jabber for most
knowledge workers

		Webex App (Unified CM)	Cisco Jabber
Messaging	IM-only Share	✓	✓
	Remote Desktop Control (Mouse & Keyboard)	✓	✓
Device	Desk-phone Control	✓	✓
Calling	HD Video & Audio Softphone	✓	✓
	Contacts (Corporate, Personal, Local Search)	✓	✓
	In-a-call Presence, DND	✓	✓
	Hold/Resume, Consultative Transfer, Conference	✓	✓
	Call History	✓	✓
	Voicemail (Visual & MWI & Call VM)	✓	✓
	Hunt Group	✓	✓
	Call Pickup	✓	✓
	Call Park	✓	✓
	Extend & Connect & Dial-via Office	✓	✓
	In-Call Sharing	✓	✓
	Call Recording	✓	✓
	Multi-Line	✓	✓
	FECC (Far End Camera Control)	✓	✓
	ICE Media Optimization	✓	✓
	3rd Party Headset Control	✓	✓
	CTI-Control by Other Application	✓	✓
	Enterprise Contact Center (UCCE/X) Support	✓	✓
	Virtual Desktop Integration (VDI)	✓	✓

Incremental features in Webex App

Webex App delivers many more features that are not supported by Cisco Jabber

Calling

- Virtual & custom background
- Noise removal and music mode
- Remote side (PSTN) noise removal
- Local push notifications
- Business texting
- Multi call window
- Seamless call handover between networks (Wi-Fi, LTE/4G)
- B2B2C app-to-app calling with annotation and whiteboard
- Move call to meeting
- E911 RedSky native support
- App share with desk phone control
- Desk phone control over MRA
- Whisper coaching
- Barge in for shared line
- Realtime closed captions and translation

Messaging

- Group chat escalation to meeting

Admin/Deployment

- Cisco headset management
- App auto update
- Auto device provisioning
- Epic Rover integration

Device Integration

- Proximity pair with Joining Meeting and 1:1 calling
- Seamless move between device and Webex App
- Wireless share & whiteboard with cloud devices
- 3rd party headset control on Mac
- Cisco headset control

End User Experience

- Single unified app for calling and meetings

Contact center features in Webex App

Webex App supports key calling features for contact center solutions

- Make/receive call
- Hold/resume
- Transfer
- Conference
- DTMF
- Recording (BIB)
- Multi-line
- Silent monitoring
- Call notification control for admin
- Call windows control for user and admin
- Zip tone for auto pickup
- Whisper announcements
- cBarge
- Agent greetings
- Whisper coaching

Further Information

<https://www.cisco.com/go/ct>

Clients

Table 3. Collaboration Transitions: Clients

Workload	START		TARGET		Key Considerations & Benefits	Transition Documentation	Product / Solution Documentation
	Type	Product(s)	Type	Product(s)			
Clients	On-Premises	Jabber	Hybrid	Webex	§ Make calls in Webex through on-premises Unified CM and control Unified CM registered desk phones directly from Webex . § Use the configuration that's in place for your Unified CM deployment (such as location, bandwidth settings, point-to-point media, and so on) § Reduce on-premises infrastructure by moving messaging to Webex.	Transition Map Deployment Guide	Jabber Data Sheet (Desktop) Jabber Data Sheet (Mobile) Unified CM Data Sheet Unified CM IM&P Service Jabber Support Docs Unified CM Support Docs Unified CM IM&P Support Docs Webex App Support Docs

Contact Center

Table 4. Collaboration Workload Transitions: Contact Center

Workload	START		TARGET		Key Considerations & Benefits	Transition Documentation	Product / Solution Documentation
	Type	Product(s)	Type	Product(s)			
Contact Center	On-Premises	Unified Contact Center Express (CCX)	Cloud	Webex Contact Center (CC)	§ Quickly transform your contact center from a cost center to a strategic business asset. § Webex Contact Center in the cloud provides flexibility and scalability. § Webex Contact Center provides high levels of security and data protection.	Transition Map Deployment Guide	Unified CCX Data Sheet Webex Contact Center Data Sheet Unified CCX Support Docs

https://www.ciscolive.com/on-demand/on-demand-library.html?search=BRKCOL-2062#



The bridge to possible

Best Practices for Migration

From Jabber to the Webex App

Steven Fly, Technical Marketing Engineer Leader

BRKCOL-2062

CISCO Live!

#CiscoLive

Microsoft Teams Integration

Call Tab

Access dial pad, contact search and speed dials

Dialpad

Access to dialpad to key in numbers

Add Speed dial

Click to add up to 25 speed dial cards

Contact Search

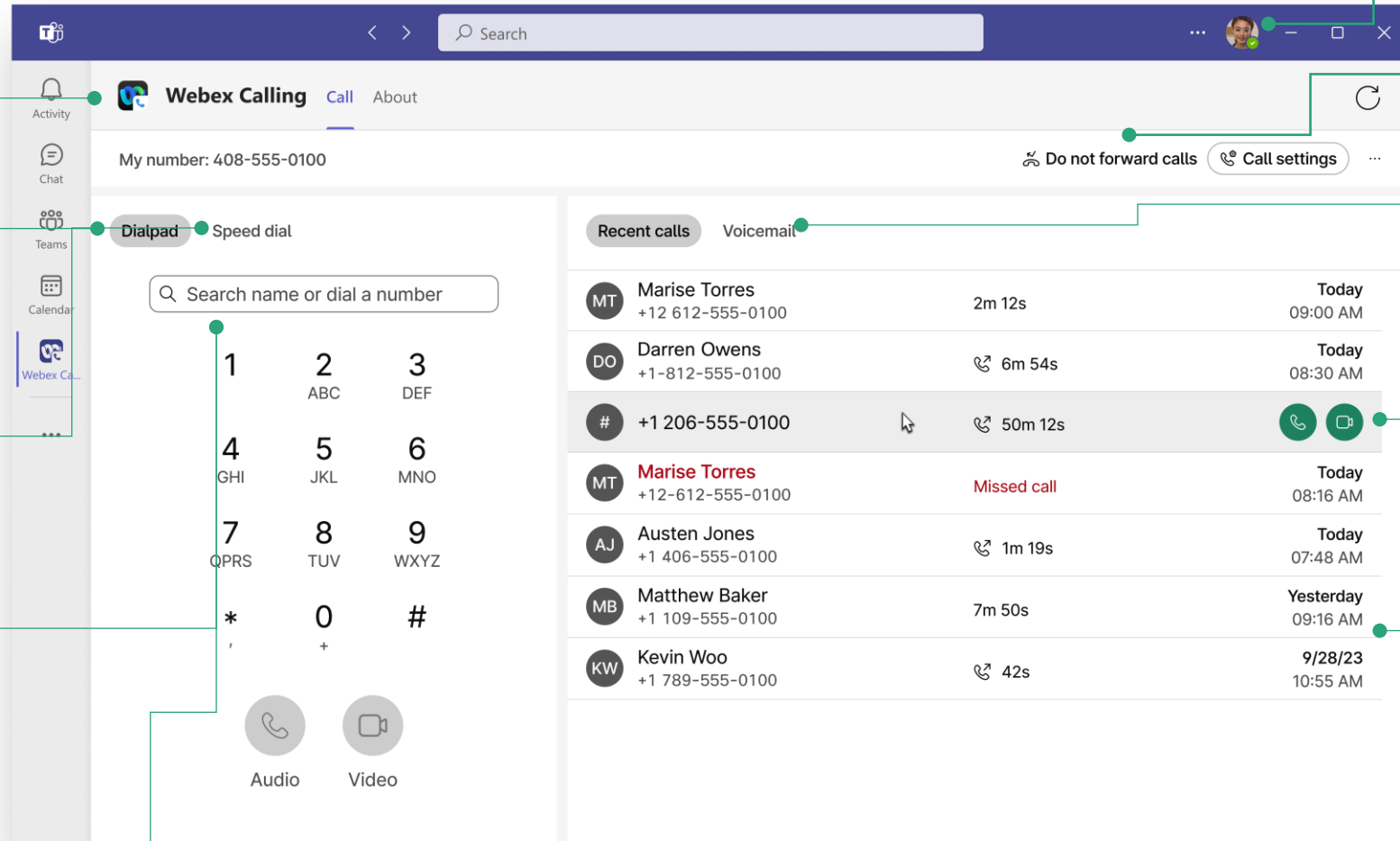
Search for personal and corporate contacts

Azure AD
Contacts

Outlook 365
Contacts

Contact Resolution

Resolves the incoming call identifier to Azure AD



Bi-directional Presence

Display your on UCM call presence to your contacts and see their call presence too

Call Forwarding

See your call forwarding status at a single glance

Voicemail

Access and play back the voicemail

Call

Select number and call contact

Call History

See history of incoming/outgoing/missed calls

Call/Search Input

Enter a number, SIP address or type a name to search for Azure AD/Outlook contact

UCM - Pre-Req:

UCM 15, 12.5 SU7 or 14 SU3

CUC - 15 / 12.5 / 14 SU3

On-premises: Webex Cloud-Connected UC (CCUC)

DEMO

Webex Calling for MS Teams

ENDPOINTS

Devices

David Scott – Product Manager

The first desk phone built for hybrid work.

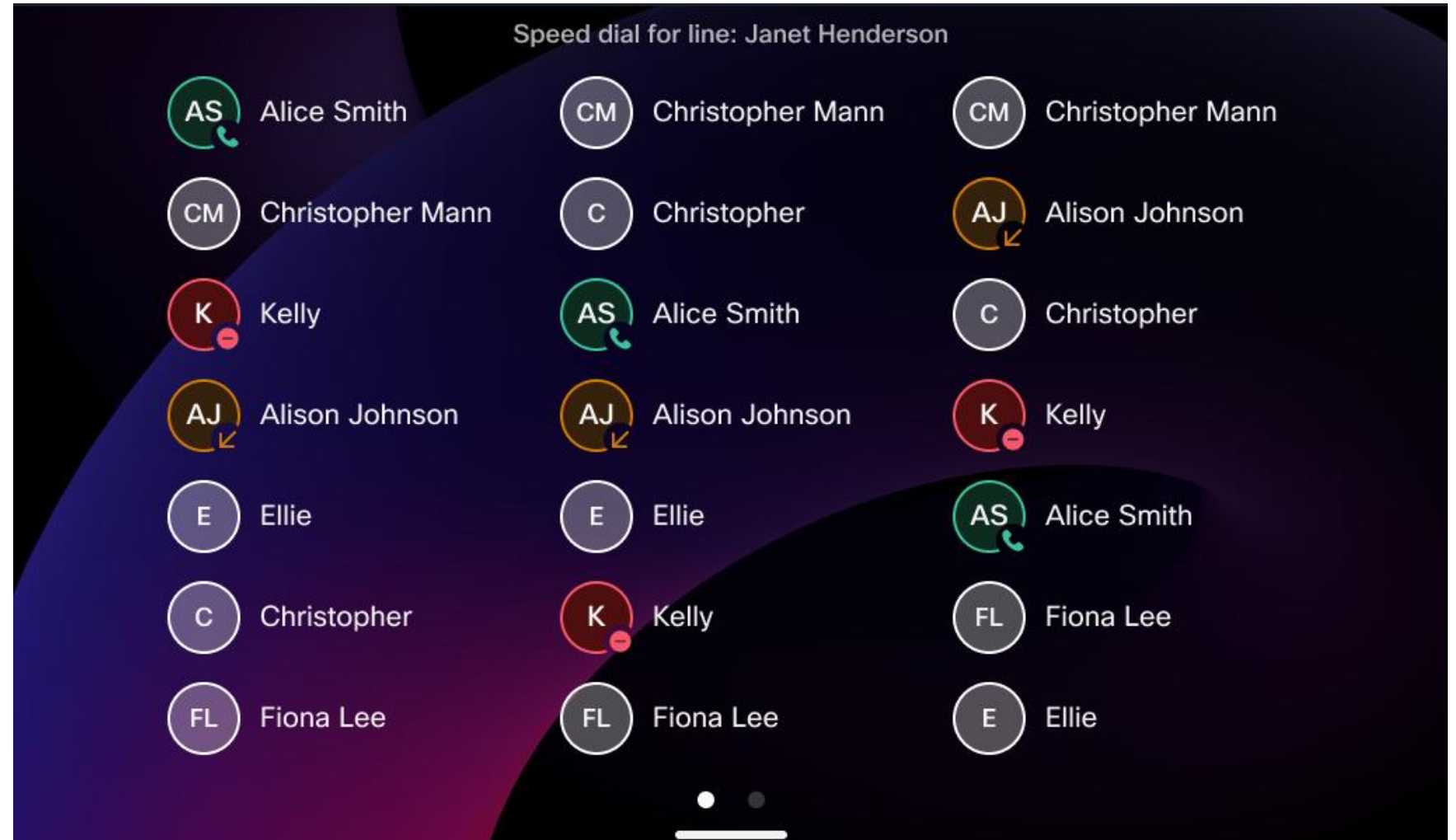


Carbon Black



Swipe Down for Speed Dial

- Access Speed Dials by swiping down from the top of the touch screen
- Scroll horizontally to access more
- BLF status will indicate if they are in a call or on DND
- Up to 108 speed dials supported on CUCM



Active Bar

- When navigating away from a call, the call appears as an active bar at the bottom of the screen
- Tap the bar to return to the call
- An active meeting will also appear as an active bar when you navigate away from the meeting



8875 Custom Wallpaper & Logo

- You can now have a custom logo and custom wallpaper simultaneously
- These require a List.xml file
- The admin can remotely push a custom wallpaper
- For details, view the video below:
<http://cs.co/8875Tutorial>



8875 Training Resources

- Useful for End Users or Admins
- <http://cs.co/8875Tutorial>



Cisco Video Phone 8875 Training

Cisco Phones, Headsets, and Peripherals

Public ▾

13 videos 1 view Updated today

▶ Play all ⌵ Shuffle

No description



1:20



0:46



1:08



1:02



0:49



3:09

Cisco Video Phone 8875 Line Level Call Features
Cisco Phones, Headsets, and Peripherals • 1 view • 13 hours ago

Cisco Video Phone 8875 Security
Cisco Phones, Headsets, and Peripherals • 2 views • 13 hours ago

Cisco Video Phone 8875 IP Phone Services
Cisco Phones, Headsets, and Peripherals • 2 views • 13 hours ago

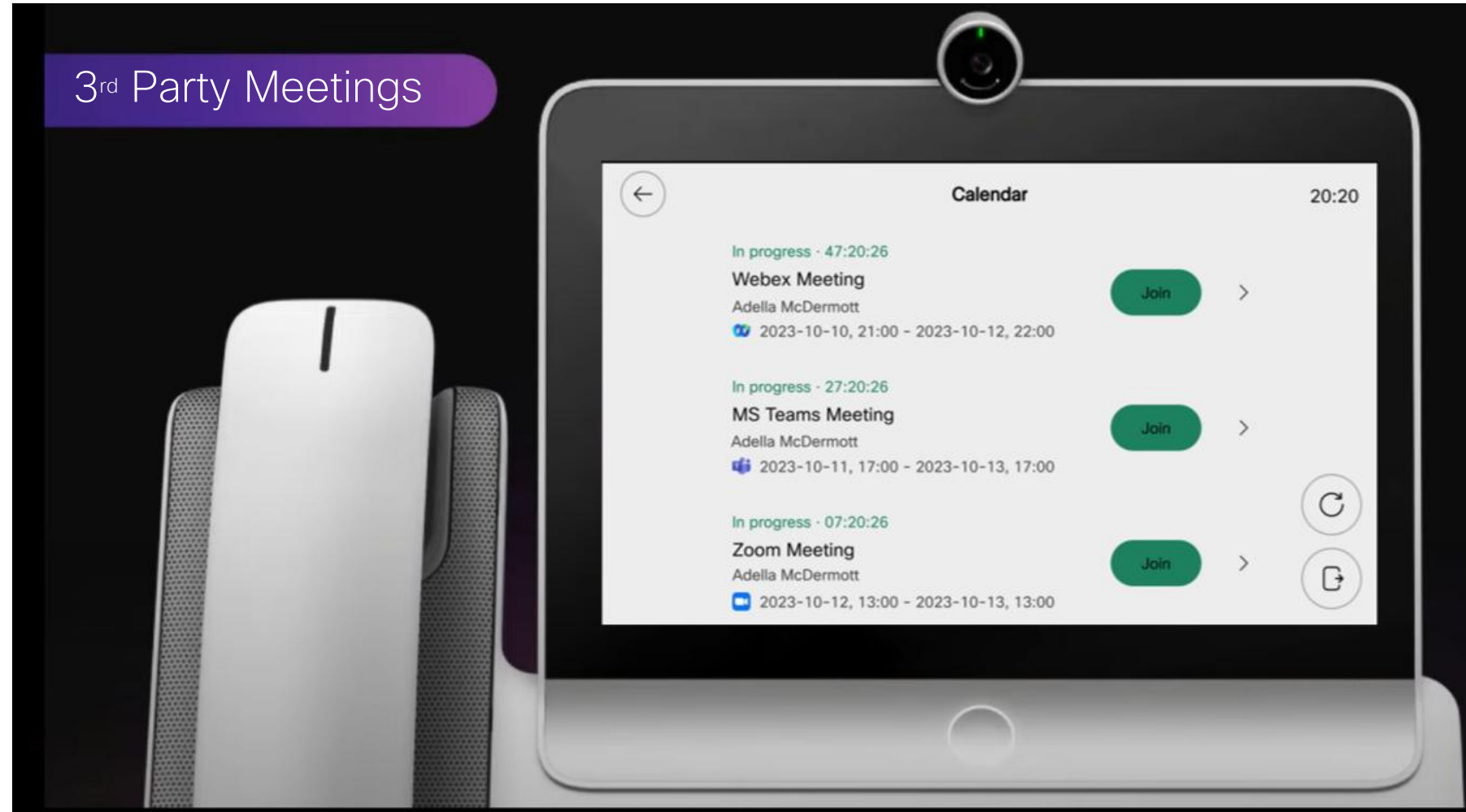
Cisco Video Phone 8875 Hot desking and OBTJ
Cisco Phones, Headsets, and Peripherals • 2 views • 13 hours ago

Cisco Video Phone 8875 Customized Display
Cisco Phones, Headsets, and Peripherals • 1 view • 13 hours ago

Cisco Video Phone 8875 Admin Guide for Custom Logo & Background
Cisco Phones, Headsets, and Peripherals • No views • 13 hours ago

Early Preview: 8875 3rd Party Meetings using OBTJ

- Join MS Teams, Zoom, or Webex meetings from your 8875
- Turn the 8875 video phone into a meeting companion device
- For more information, contact us or your Cisco representative



Enterprise Phone Headsets Enhancements

Enhancements:

- Simplified Extension Mobility Login with Expressway MRA for 7800/8800
- Simplified Extension Mobility Login with 730 USB Adaptor for 8851/61/65
- Bluetooth Mute Sync for 730 for 8851/61/65
- Cisco Headset 720 support for 8851/61/65
- Cisco Headset 980 support for 8851/61/65
- 560 Auto Answer on/off dock support for 7800/8800
 - By default, you can answer calls by simply lifting your 560 headset off the base and ending them when you put it back down. This firmware release enables you to toggle this feature on and off through Webex App or on Cisco IP Phones with firmware 14.2 or later.

Enterprise Phone Other Enhancements

- External Phone Number Mask Display for Non-primary Line
- SIP OAuth Support on SRST

Cisco Wireless Phones

Cisco Wireless Phone 840

IP65 ruggedized, water and dust resistant

4" Dragontail™ glass touch screen

8MP rear camera, 5MP front camera



Version 1.9
May 2023

- SRST Support
- Call Pickup on CUCM
- Call Pickup for Txfer on WxC
- Audible Caller ID
- Ringtone/Alert Sound and Wallpaper management (CUCM)
- FW + application updates (CUCM)
- Diagnostics Application
- Security updates
- Bug Fixes

Version 1.10
September 2023

- InformaCast XML Support
- Hoteling login (WxC)
- Japanese-localized applications
- UX Improvements
- Security updates
- Bug Fixes

Cisco Wireless Phone 860

IP68 ruggedized, waterproof and dustproof

5.2" Gorilla® glass touch screen

13MP rear camera, 8MP front camera

Hot swappable battery for 100% up-time



Android 10 OS
Google Play Store

Webex app installed
CUCM Registration

EMM/MDM Control
802.11 Wi-Fi

Scanner HW Option
Optional charging station

Customizable buttons
Vertical Applications

Cisco Wireless Phones Key Resources

Wireless Phone Overviews

- Technical Overview: Hardware & Features
<http://cs.co/wireless-phone-overview>
- Technical Overview2: Core Telephony
<http://cs.co/wireless-phone-overview2>

Core Documentation

- Wireless LAN Deployment Guide
<http://cs.co/webex-wireless-dg>



Free EMM Alternative

- Video Demo
<http://cs.co/emma-demo>
- Tool URL
<https://configure.cisco.com>

MDM 840/860 Tutorials

- Microsoft Intune
<https://youtu.be/i5NcAGJI3e4>
- VMWARE Workspace One (AirWatch)
<https://youtu.be/23jC15zInlc>
- Meraki SM
<https://youtu.be/ysmW6bcOjT0&t=863s>

Android 10 OS
Google Play Store

Webex app installed
CUCM Registration

EMM/MDM Control
802.11 Wi-Fi

Scanner HW Option
Optional charging station

Customizable buttons
Vertical Applications

Firmware Release 5.1(2) for 6800 Series IP DECT

Delivers customer requested features to the 6800 series IP DECT devices:

- Cisco IP DECT Phone on CUCM
- Configuration Changes Logging
- Configuration Changes Reporting
- EDOS Profile Supports all XML Parameters
- Stateful Firewall to Control Inbound Ethernet Traffic
- Voice-VLAN support (Network-Related Information Extraction using CDP/LLDP)



6800 DECT on CUCM

- The 6800 DECT is now supported on CUCM. It must be added as a Third-Party SIP (Advanced) endpoint.
- The MAC address in CUCM should be arbitrary and any value other than the actual DBS MAC.
- You must configure the DSB for TCP and not UDP.
- Video on how to install:
<https://youtu.be/75018CuYzKc>



Find the perfect peripherals

Integrated with Webex app and Cisco devices.

Microsoft Teams certified versions.

Dedicated Headset and Camera apps

Manage & view insights on your fleet of peripherals in Webex Control Hub

900 Series

Executive Bluetooth® Wireless headsets with premium audio



Bang & Olufsen Cisco 980



Bang & Olufsen Cisco 950

700 Series

Work everywhere with Bluetooth® wireless headsets for hybrid work featuring intelligent noise removal



Headset 730

Updated ergonomics!



720 Series

Certified for Microsoft Teams

500 Series

Professional wired and wireless headsets for everyday calling and contact center applications



560 Series



530 Series



520 Series

300 Series

Professional value line with convenient controls



320 Series

Certified for Microsoft Teams

Desk Camera

Intelligent USB webcams for use on external displays / tripods



Desk Camera 4K

Certified for Microsoft Teams



Desk Camera 1080p



Cisco and Bang & Olufsen joined forces to deliver an audio experience that is equipped for business communications, excels in immersive sound, and designed with elegance.

The Bang & Olufsen Cisco 900 Series

The new Bang & Olufsen Cisco 950 true wireless earbuds combine the B&O craftsmanship, design, and legendary sound with purpose built software for business users. It's your new discreet and powerful all-day audio companion.



Bang & Olufsen Cisco 950

Take great audio anywhere

-  Bluetooth 5.2
-  Adaptive active noise cancellation and transparency mode
-  Immerse yourself in B&O legendary sound
-  Six built-in microphones for crystal clear communications

All-day comfort

-  Tailored fit with 4 sizes of silicon tips or COMPLY™ memory foam tips
-  Stereo or mono use on left or right earbud

User centric experiences

-  Only earbuds optimised for Webex app and Cisco devices with on-ear call controls over native BT
-  Touch or motion controls to easily navigate calls and music
-  In-ear voice or tone notifications
-  Customize settings in the B&O mobile app

Peace of mind for IT

-  Enterprise grade hardware and software security
-  Centralized inventory management in Control Hub
-  Built to last with premium materials including aluminum and tempered glass

True wireless earbuds designed for professionals.



B&O | cisco

Conclusions & References

Release 15 and On-Premises Update

Ongoing and continued investment in on-premises products and solutions

- Release 15 (Unified CM, IM&P, Unity Connection, Emergency Responder + Platform)
- Expressway
- CUBE
- Cisco Meeting Server (CMS)
- Jabber
- Devices

References – Release 15 / Platform

- Unified CM 15 Beta: <https://gobeta.webex.com/key/cucm>
- Partner webinar 'Cisco UCM 15 Update - Get Current or Go Cloud Now': <http://cs.co/cvoe>
- Unified CM Release EoL mailers: ask-ucm-release-eol@cisco.com or ask-ucm-release-eol-partner@cisco.com
- Data Center Migration for Push REST Service: <https://www.cisco.com/c/en/us/products/collateral/unified-communications/unified-communications-manager-callmanager/uni-cm-im-presence-service-bulletin.html>
- Phone Deprectation Release 14:
https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/trouble/14_0_1/fieldNotices/cucm_b_deprecated_phones-14.html
- Quote Collab Tool: <http://www.cisco.com/go/quotecollab>
- Unified Communications Manager Support Docs: <https://www.cisco.com/c/en/us/support/unified-communications/unified-communications-manager-callmanager/series.html>

References - Expressway

- Expressway X14.3 Release Notes:
https://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/expressway/release_note/X14-3-x/Cisco-Expressway-and-Cisco-Expressway-Select-Release-Note-X14-3-x.pdf
- Expressway CE1300 Appliance Install Guide:
https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/expressway/install_guide/X14-3-1/CE1300/exwy_b_cisco-expressway-ce1300-appliance-installation-guide-x1431.html
- Expressway Support Docs: <https://www.cisco.com/content/en/us/support/unified-communications/expressway-series/series.html#~tab-documents>

References – CUBE (1 of 2)

- CUBE Interop Portal: <http://www.cisco.com/go/interoperability>
- Zoom Phone with CUBE: <https://assets.zoom.us/docs/zoom-phone/Zoom-Phone-Config-Guide-Cisco-UBE-ISR4331.pdf>
- Amazon Chime with CUBE: [https://d1.awsstatic.com/Products/product-name/Chime/Amazon Chime Voice Connector SIP Trunking Configuration with Cisco UCM and Cisco UBE v2.1.pdf](https://d1.awsstatic.com/Products/product-name/Chime/Amazon_Chime_Voice_Connector_SIP_Trunking_Configuration_with_Cisco_UCM_and_Cisco_UBE_v2.1.pdf)
- Management Connector Deployment Script: https://binaries.webex.com/ManagedGatewayScriptProdStable/gateway_onboarding.tcl
- CUBE Box: <https://cisco.box.com/CUBE-Enterprise> (Internal Only – external folks can be added if they request access via email)
- [CUBE Performance and Sizing](#)
- [Cisco Unified Border Element Configuration Guide Through Cisco IOS XE 17.5](#)
- [CUBE Config Guide – IOS-XE 17.6 Onwards](#)

References – CUBE (2 of 2)

- [CUBE Roadmap](#)
- [IOS-UC Sharepoint](#)
- CUBE Support

CUBE partner/customer questions: ask-cube@external.cisco.com

References – Cisco Meeting Server

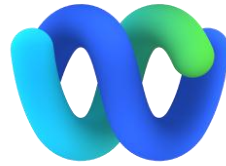
- Cisco Meeting Server Support Docs: <https://www.cisco.com/content/en/us/support/conferencing/meeting-server/tsd-products-support-series-home.html>

References - Jabber

- Collaboration Transitions: <https://www.cisco.com/go/ct>
- Jabber Support Docs: <https://www.cisco.com/content/en/us/support/unified-communications/jabber/series.html#~tab=documents>

References - Devices

- 8875 Tutorial: <http://cs.co/8875Tutorial>
- Wireless Technical overviews: <http://cs.co/wireless-phone-overview> and <http://cs.co/wireless-phone-overview2>
- Wireless LAN Deployment Guide: <http://cs.co/webex-wireless-dg>
- Mobility Management: <http://cs.co/emma-demo> | <https://configure.cisco.com/> (Tool)
- MDM Tutorials - MS Intune: <https://youtu.be/i5NcAGJI3e4>
- MDM Tutorials - VMWare Workspace One (AirWatch): <https://youtu.be/23jC15zInlc>
- MDM Tutorials - Meraki SM: <https://youtu.be/ysmW6bcOjT0&t=863s>
- 6800 DECT: <https://youtu.be/75018CuYzKc>



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